Shu Exam Questions

G1s

Fall 2026

Preface

Compiled in preparation for the Shu Exam on November 13, 2026.



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Chapter 1

General Physics

1 State Kepler's Laws and discuss two applications or implications of the laws. Be prepared to talk about how these laws relate to the energy and angular momentum of a planet in its orbit.

1. Planetary orbits are ellipses with the Sun/star at one of the foci. In a bound orbit, the total energy of the system must be negative, which results from the fact that gravity is an inverse-square law (parabolic and hyperbolic trajectories will occur when the energy is 0 or positive, respectively).

2. A planet in its orbit will sleep out an equal area over an equal amount of time. For a non-circular orbit, the planet will move faster closer to the star (fastest at the point of closest approach, the periapsis) and slower further from the star (slowest at the point of furthest approach, apoapsis). This is a natural consequence of conservation of angular momentum for a body experiencing a central force.

   Proof. We work in polar coordinates, $\vec{r} = (r, \theta)$. During a time interval dt, a planet moves an angle $d\theta$, sweeping out a thin triangle of area

   $$dA = \frac{1}{2} r^2 d\theta \implies \frac{dA}{dt} = \frac{1}{2} r^2 \underbrace{\frac{d\theta}{dt}}_{=\omega} = \frac{1}{2} r^2 \omega. \tag{1.1}$$

   The angular momentum magnitude is

   $$L = |\vec{r} \times (m\vec{v})| = mr \underbrace{v_\perp}_{=r\omega} = mr^2\omega \implies r^2\omega = \frac{L}{m}. \tag{1.2}$$

   Combining Eqs. 1.1 and 1.2 gives

   $$\frac{dA}{dt} = \frac{L}{2m}. \tag{1.3}$$

   Since gravity acts radially, the net torque on the system is zero, so angular momentum is conserved, and thus $\frac{dA}{dt}$ is constant. This assumes a body orbiting a fixed center; in the generalized two-body system, the total angular momentum L belongs to the entire system, and we recover that the area swept out by the line connecting the two bodies is

   $$\frac{dA}{dt} = \frac{L}{2\mu}, \tag{1.4}$$

   where $L = \mu r^2 \omega$ and $\mu = \frac{m_1 m_2}{m_1 + m_2}$ is the reduced mass. □

3. The orbital period squared is proportional to the semi-major axis cubed. This results from a balance of centripetal force and gravity. For the Solar System, the form is $P^2 = a^3$ for P measured in Earth years and a measured in AU. The general form is

$$(m_1 + m_2)P^2 = \frac{4\pi^2}{G}a^3. \tag{1.5}$$

Proof. We outline the simpler derivation from balancing centripetal and gravitational forces assuming circular orbits, $a = r$. Consider two masses, m_1 and m_2, separated by a distance r and each separated from the center of mass of the system (which they orbit) by r_1 and r_2, respectively. Then, $r = r_1 + r_2$ and $m_1 r_1 = m_2 r_2$, so that

$$r_1 = r\left(\frac{m_2}{m_1 + m_2}\right). \tag{1.6}$$

Focus on m_1; let us balance the centripetal and gravitational forces acting on it. The centripetal force is $m_1\omega^2 r_1$ and the gravitational force is Gm_1m_2/r^2. So,

$$\frac{Gm_2}{r^2} = \omega^2 r_1 \overset{(1.6)}{=} \omega^2\left[r\left(\frac{m_2}{m_1 + m_2}\right)\right] \implies \omega^2 = \frac{G(m_1 + m_2)}{r^3}. \tag{1.7}$$

The angular velocity is $\omega = 2\pi/P$ so that we recover Eq. 1.5

$$\frac{4\pi^2}{P^2} = \frac{G(m_1 + m_2)}{r^3} \implies (m_1 + m_2)P^2 = \frac{4\pi^2}{G}r^3. \tag{1.8}$$

The general derivation for an elliptical orbit involves integrating Eq. 1.4 over the full area of the orbit and using the angular momentum of the system. □

Applications of Kepler's laws include:

- Exoplanet orbits
- Indirect mass measurements of black holes
- Jean's mass derivation (?)

2 Describe the physics of tidal forces, and discuss the meaning of the Roche Limit.

Gravitational forces depend on distance as $1/r^2$, so a non-point source object will have one side closer to a mass experiencing a slightly stronger gravitational force than the side further away from the source, as in Fig. 1.1. These are **tidal forces**.

The **Roche Limit** is defined as the critical distance at which the tidal forces from a mass overcome an object's self-gravity and the object is ripped apart. For example, Saturn's moons lie outside Saturn's Roche Limit and can form large objects, but the rings are within the limit, preventing clumping and accretion into a single mass. For a larger object of mass M and satellite of mass m, the Roche limit is

$$r_{\mathrm{R}} = R_m\left(\frac{2M}{m}\right)^{1/3} = R_M\left(\frac{2\rho_M}{\rho_m}\right)^{1/3} \qquad \text{(Rigid body)}, \tag{1.9}$$

$$r_{\mathrm{R}} \approx 2.44\, R_m\left(\frac{M}{m}\right)^{1/3} = 2.44\, R_M\left(\frac{\rho_M}{\rho_m}\right)^{1/3} \qquad \text{(Fluid body)}, \tag{1.10}$$

where R_M and R_m are the radius of larger object and satellite, respectively, and ρ_M and ρ_m are their densities. The fluid body equation accounts for deformation of the satellite.

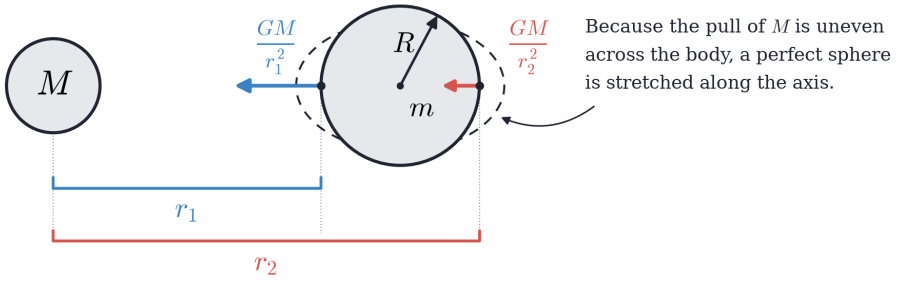


Figure 1.1: Tidal forces.

3 What is the "free fall" time? What physical variables does it depend on? Describe its relevance in at least two areas of astronomy.

The **free fall time** is the characteristic time for a (spherical) body to collapse under its own gravity:

$$\tau_{\text{ff}} = \left(\frac{3\pi}{32G\bar{\rho}}\right)^{1/2}. \tag{1.11}$$

Note this only depends on the mean interior density, not on the physical size of the body.

Proof. We start by deriving the free fall time for a test particle of mass m falling from an initial rest position at R_0 to a much larger central mass $M \gg m$. From conservation of energy, we have that

$$\overbrace{\frac{1}{2}\left(\frac{dr}{dt}\right)^2 - \frac{GM}{r}}^{=E(t)} = \overbrace{-\frac{GM}{R_0}}^{=E_0} \implies \frac{dr}{dt} = -\sqrt{2GM\left(\frac{1}{r} - \frac{1}{R_0}\right)}. \tag{1.12}$$

We integrate to $r = 0$ to find

$$\int_0^{\tau_{\text{ff}}} dt = -\frac{1}{\sqrt{2GM}} \int_{R_0}^{0} \frac{dr}{\sqrt{\frac{1}{r} - \frac{1}{R_0}}} \implies \tau_{\text{ff}} = \frac{\pi}{2}\sqrt{\frac{R_0^3}{2GM}}. \tag{1.13}$$

For a uniform, spherically symmetric body with initial radius R_0 and initial density ρ_0, the mass within the cloud is $M = \frac{4}{3}\pi R_0^3 \rho_0$, so we recover Eq. 1.11:

$$\tau_{\text{ff}} = \frac{\pi}{2}\sqrt{\frac{R_0^3}{2G\left(\frac{4}{3}\pi R_0^3 \rho_0\right)}} = \sqrt{\frac{3\pi}{32G\rho_0}}. \tag{1.14}$$

Why can we substitute the mass of a uniform sphere into the point-mass derivation? For any spherically symmetric mass distribution, a gash shell feels a gravitational pull from the mass inside its radius as if all that interior mass were concentrated at the center; any test particle at R_0 is falling toward a point mass M at the center. And, in this situation, we have **homologous collapse**: every layer of the cloud reaches the center at the same time. Shells at smaller initial radii have both less interior mass and a shorter distance to travel. If the cloud is denser at its center, we get **inside-out collapse**: the inner layers collapse faster than the outer layers. □

Applications of the free fall time include

- One can approximate the timescale of core collapse in core-collapse SNe as the free fall time of the core
- Homologous collapse happens on a free-fall timescale.

4 State the Virial Theorem and its assumptions, and give at least two applications.

The **virial theorem** relates the time-averaged kinetic and potential energies for a bound system in equilibrium. In the version of the theorem here, we also assume that the system is bound by an inverse square force like gravity:

$$2\langle K\rangle + \langle U\rangle = 0, \qquad E = \frac{\langle U\rangle}{2}. \tag{1.15}$$

The general form is for any stable, bound system where the potential energy follows a power law of the form $U(r) \propto r^n$, in which case we have

$$2\langle K\rangle = n\langle U\rangle. \tag{1.16}$$

Applications of the virial theorem include

- Mismatch between the virial theorem and observations led to a theory for dark matter. From observations of the Coma Cluster, in particular galaxy velocities, Zwicky derived kinetic energies and used the virial theorem to estimate the gravitational potential energy. The mismatch between the mass derived from this method and the luminous matter mass led him to conclude the existence of dark matter.
- One can derive the Jeans instability by setting the condition for gas cloud collapse at $2\langle K\rangle < -\langle U\rangle$, which gives the Jeans mass $M_{\mathrm{J}} \propto T^{3/2}\rho^{-1/2}$. See Question 28.
- The virial theorem can be used to relate the mass, size, and velocity dispersion of galaxies. The virial mass estimator is

$$M \propto \frac{\sigma^2 R}{G}, \tag{1.17}$$

using $\langle K\rangle = \frac{1}{2}M\sigma^2$ and $\langle U\rangle = -\frac{GM^2}{R}$.

5 How does the Planck function for blackbody radiation scale with frequency in the Rayleigh-Jeans limit and the Wien's limit? What is Wien's Law, and how is it related to the Planck function? Sketch Planck functions for two temperatures.

The **Planck function** is

$$B_\nu(T) = \frac{2h\nu^3}{c^2}\frac{1}{\exp(h\nu/kT)-1}. \tag{1.18}$$

The **Rayleigh-Jeans limit** is the case when $h\nu \ll kT$, in which case we can expand $\frac{1}{e^x-1}$ as $\frac{1}{x}+\ldots$ to get

$$B_\nu(T) \approx \frac{2\nu^2 kT}{c^2}. \tag{1.19}$$

Wien's limit is the case when $h\nu \gg kT$, and we get

$$B_\nu(T) \approx \frac{2h\nu^3}{c^2}\exp(-h\nu/kT). \tag{1.20}$$

The Planck function for a few temperatures is given in Fig. 1.2. Note $I_\nu = B_\nu(T)$ for an ideal blackbody radiator.

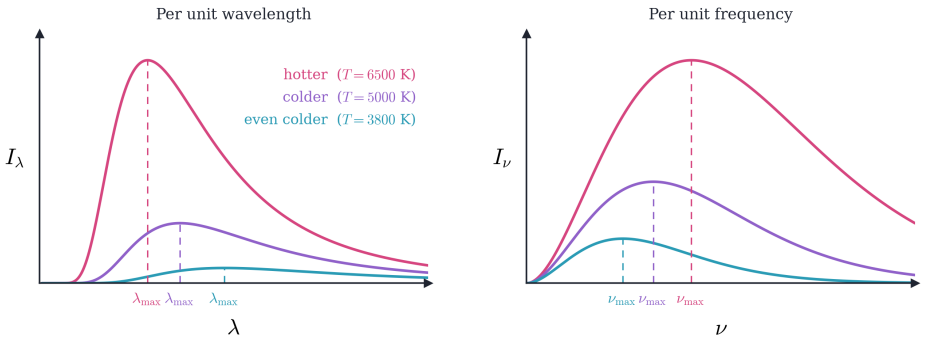


Figure 1.2: The Planck function at several temperatures.

Wien's Law states that a blackbody spectrum peaks at

$$\lambda_{\text{peak}} = \frac{2.9 \times 10^{-3}\,\text{m} \cdot \text{K}}{T}, \qquad \nu_{\text{max}} = 2.82\frac{kT}{h}. \tag{1.21}$$

At higher temperatures, the peak shifts to shorter wavelengths and higher frequencies. Also, when sketching the Planck function at different temperature, it is important that $B_\nu(T_1) > B_\nu(T_2)$ for $T_1 > T_2$ for all frequencies.

6 Describe the formation of spectral lines (emission and absorption) of both atoms and molecules. What kind of energy levels or states are atoms and molecules characterized by? Why are spectral lines dark?

Spectral lines in atoms arise from the absorption/emission of photons via the excitation/de-excitation of electrons. These spectral lines typically lie at UV, optical, or IR energies. They can exhibit fine or hyperfine structure caused by electron spin states and nucleus spin states, respectively, which split the emission lines in spectra.

Spectral lines in molecules are more complicated. They arise from electronic transitions, but also vibrational and rotational modes. Electronic transitions still cause UV, optical, and IR lines. Vibrational oscillations are typically at IR energies and rotational transitions are typically in the sub-millimeter or microwave ranges.

Spectral lines in stars are dark because they are caused by absorption of the intrinsically blackbody emission from the hot dense core by atoms and molecules in the cooler outer layers of the star (photosphere, chromosphere). Although these particles re-emit the absorbed photons, they do so isotropically rather than directionally (i.e., radially), which reduces the amount of photons at those frequencies along our line of sight, making them dark compared to the continuum.

7 Describe the physics governing the shape of spectral lines. Describe at least three effects.

- **Natural ("lifetime") line width/broadening**: This is a result of the fact that an excited state has a finite lifetime, so emission of photons also takes a finite time. In QM, energy is the Fourier transform of a time-varying signal; when you take the Fourier transform of a finite-time signal, the result is not a delta function (perfectly sharp energy), but a **Lorentzian (Cauchy) distribution** centered around the energy difference between the states. This is tied to the (Heisenberg) Uncertainty Principle.

- **Doppler/thermal broadening**: Atoms and molecules are always moving. The line emission of each atom is then randomly redshifted or blueshifted, leading the overall shape to follow a **Gaussian distribution**. In general, this effect is much stronger than natural broadening. At the macro scale, bulk motions (such as turbulence) can also cause Doppler broadening even if a gas is "cool."
- **Collisional/pressure broadening**: This is arguably the most complicated effect, with active research for specific cases. The basic idea is that if there are many collisions, photon emissions gets "interrupted" early (shorter lifetime), so there is another Lorentzian distribution.

When there is both Doppler and pressure broadening, the spectral line shape is the convolution of the Gaussian and Lorentzian distributions: a **Voigt profile**.

8 What is the Doppler shift? Name three applications to astronomical measurements. How does the Doppler shift relate to the redshift? How do we measure redshifts?

The **Doppler shift** is the change in wavelength or frequency of a wave in relation to an observer due to relative motion between the emitter and the observer. Redshift, specifically **cosmological redshift**, is caused by the expansion of space itself, stretching light waves as they travel across the universe (hence why there cannot be a cosmological blueshift for our expanding universe). Both are quantified by the same parameter,

$$z = \frac{v}{c} = \frac{\lambda_{\text{obs}} - \lambda_{\text{rest}}}{\lambda_{\text{rest}}}. \tag{1.22}$$

Applications of the Doppler shift and cosmological redshift to astronomical measurements include:

- Exoplanet detections, for the radial velocity technique which measures the Doppler shift induced by the orbit of the star around the center of mass of the star-exoplanet system (see Question 31).
- Lyman-α forest cosmology, which uses a series (a "forest") of Lyman-α absorption lines from neutral hydrogen at different redshifts along a line of sight to a distant galaxy or quasar to map the distribution of baryonic matter in the universe.
- Rotation curves of galaxies, obtained by measuring the Doppler shift of spectral lines at different points in a galaxy or even in the Milky Way.
- P Cygni profiles, characterized by a red(Doppler-)shifted emission line paired immediately with a blue(Doppler-)shifted absorption trough at the same rest wavelength, caused by stellar wind or gas shells expanding toward the observer (blueshifted absorption) and broader winds or circumstellar envelopes expanding in all directions (redshifted or standard emission). These features are common in Luminous Blue Variables (LBVs), which the prototype star P Cygni is classified as; hot supergiants; or SNe. The width and depth of the peak and trough can be used to measure speed and mass-loss rate of stellar winds; the P Cygni profiles of SNe are extremely broad and deep, reflecting high velocities, but over weeks and months, they gradually transition into pure emission lines as the ejecta thins and slows down.

To measure redshifts, we ideally use spectra of objects, in which we can either look for the shifts of known spectral lines relative to their rest wavelengths or for wavelengths at which the flux drops sharply. The latter includes the Lyman break at 912 Å, below which hydrogen gas absorbs any photon, or the 4000 Åbreak, below which the Balmer series and metal lines become dominant; the observed location of these breaks is redshifted in the same way as spectral lines. We can also measure **photometric redshifts** using multiple bandpass filters: if an object's flux drops to zero in filters sensitive to wavelengths below a certain value, we can deduce a spectral break has been redshifted to a wavelength in the window between the first filter for which the object disappears and the last filter it was present in.

9 Describe the principle of hydrostatic equilibrium and sketch out its derivation. State its relevance to at least two areas of astronomy.

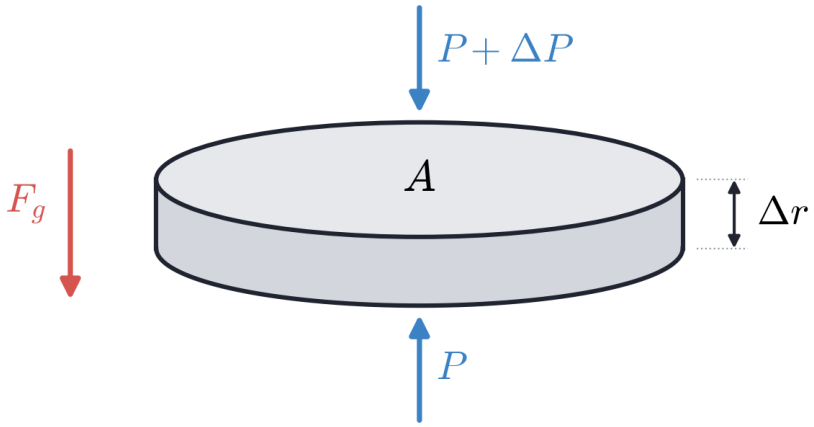


Figure 1.3: The setup for hydrostatic equilibrium.

Hydrostatic equilibrium is when outward pressure balances gravity. The condition in a vertical or radial direction is

$$\frac{dP}{dr} = -\rho g, \tag{1.23}$$

where P is pressure, r is height or radius, ρ is the fluid density, and g is acceleration due to gravity. The spherical condition is

$$\frac{dP}{dr} = -\frac{G\rho(r)M(r)}{r^2}, \tag{1.24}$$

where $M(r)$ is the enclosed mass at radius r.

Proof. First consider the vertical case. Take a slice of height Δr and area A from a fluid of density ρ, as in Fig. 1.3. The pressure pushing up on the slice is P and down on the slice is $P + \Delta P$. So, balancing forces in the two vertical directions, we have

$$AP = A(P + \Delta P) + Mg = A(P + \Delta P) + (\rho A \Delta r)g \implies \frac{dP}{dr} = -\rho g, \tag{1.25}$$

where ρ and g are allowed to vary with r. Using that, for a spherical distribution, $g(r) = \frac{GM(r)}{r^2}$, immediately gives us the spherical condition,

$$\frac{dP}{dr} = -\frac{G\rho M(r)}{r^2}, \tag{1.26}$$

□

Applications to astronomy include:

- Planets, for which the balance between internal pressure and gravity imparts their spherical shape.
- Stars, which maintain equilibrium through gravity and outward thermal and radiative pressure. The balance can be disrupted in either direction. If fusion drops, gravity overpowers pressure, causing the core to contract and heat up. If fusion spikes, outward pressure overpowers gravity, causing the star to expand.

- Core collapse for clouds in the ISM stars when gravity overpowers pressure, which can happen when too much mass has accumulated or external forces (SN shocks, cloud collisions) compress the cloud.

10 Describe the EM spectrum, including approximate energy, wavelength, and frequency scales, regions that can be observed from Earth, regions that cannot (and why not), and several common astrophysical emission mechanisms in each region.

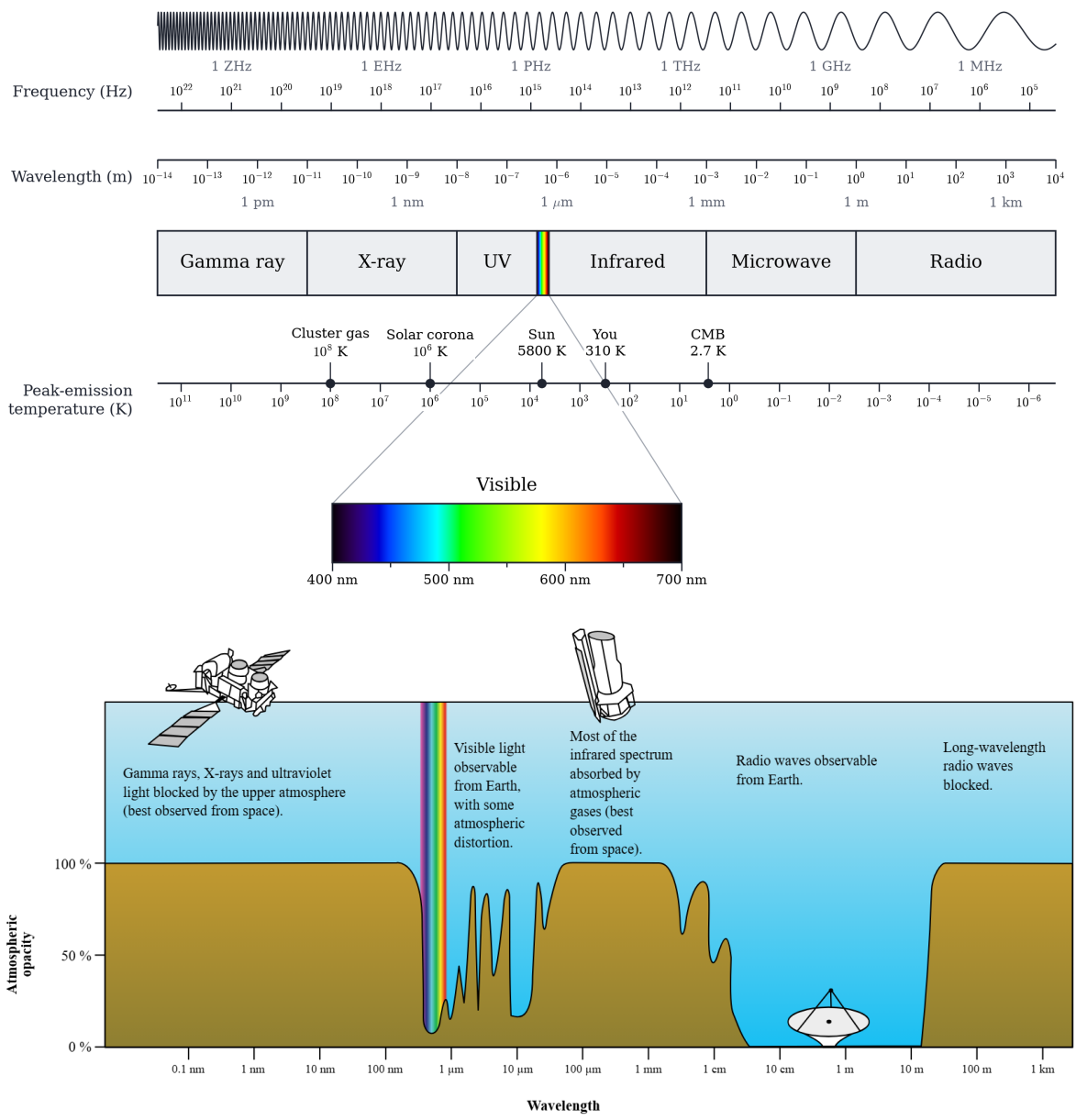


Figure 1.4: The electromagnetic spectrum.

See Fig. 1.4 for the EM spectrum. Astrophysical sources of each region of the electromagnetic spectrum include:

- Radio: 21 cm line emission from H I, AGN jets and blazars (synchrotron radiation), pulsars, FRBs, radio recombination lines (recombination, cascade down through $n \gtrsim 30 - 100$ energy levels, though these can also be in the microwave, e.g., H30α).
- Microwave: CMB, molecular rotational lines (e.g., CO $J = 1 \to 0$ transition).
- Infrared: planets, protoplanetary disks, M dwarfs, brown dwarfs, dust emission (absorb starlight and re-emit in the far IR), redshifted galaxies.
- Optical: stars, SNe, AGN (especially quasars), recombination radiation (e.g., from H II regions, Balmer series lines)
- UV: O/B stars, AGN
- X-ray: X-ray binaries, hot ISM (thermal bremsstrahlung), intracluster medium (ICM).
- Gamma ray: GRBs (core-collapse SNe, neutron star mergers), nuclear transitions (decay of radioactive elements produced, for example, in SNe).

Why are there regions of the spectrum that cannot be observed from Earth, as in Fig. 1.4?

- Long wavelength radio: for $\lambda \gtrsim 10$ m, there is ionospheric absorption and reflection, as this is below the ionosphere's characteristic plasma frequency.
- Infrared, sub-millimeter, microwave, and radio up to ~ 3 cm: mainly molecular transitions from H_2O, CO_2, O_2, etc. Rotational transitions are more towards the sub-millimter, microwave, and radio regions, while vibrational transitions are more towards the mid- or far-infrared.
- UV: ozone (O_3).
- UV and soft X-ray: photoelectric absorption (photoionization of, e.g., O_2 and N_2 in the atmosphere; note that this is what creates the ionosphere that causes absorption/reflection for long wavelength radio.
- Mid/hard X-ray and Gamma ray (keV–MeV range): Compton scattering with free electrons in the ionosphere
- Gamma ray (> 1.022 MeV): pair production, which occurs when high-energy gamma rays interact with the electric field of atomic nuclei or electrons ($\gamma + Z \to Z + e^- + e^+$). This requires creating both an electron and a positron, so the threshold energy is twice the rest-mass energy of a single electron (511 keV).

Chapter 2

Measurements in Astronomy

11 What is parallax, and how does it provide a measurement of distance? What is the parallax of an object one parsec away? Approximately how far can the Gaia satellite determine good parallaxes?

Parallax is the apparent shift of an object viewed along different lines of sight; nearby objects show a larger parallax than farther objects. See Fig. 2.1. In the astronomical setting, parallax is the apparent shift of a nearby star against much more distant background objects. As the Earth orbits the Sun, a nearby star traces a small ellipse on the sky over the course of a year. The parallax angle p is half of the total shift, which corresponds to a baseline of 1 AU. Since these are small angles, we have

$$\tan p \approx p = \frac{1\,\text{AU}}{d} \implies d\,[\text{pc}] = \frac{1}{p\,[''\,]}. \tag{2.1}$$

Hence, one parsec is defined as the distance at which 1 AU subtends an angle of one arcsecond. The nearest star, Proxima Centauri, has $p = 0.77''$, which means that every star has $p < 1''$.

Gaia can determine the parallax of stars up to 30,000 lyr (~ 10 kpc) away to an accuracy of about 20%.

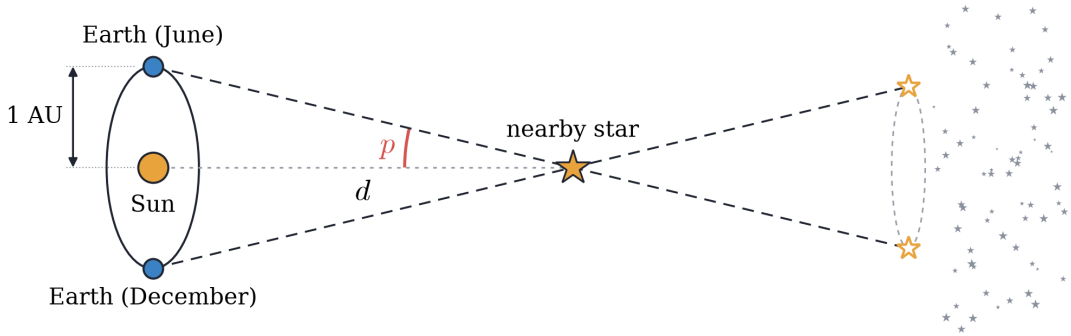


Figure 2.1: Parallax.

12 Write down the relation between apparent and absolute magnitudes and flux. What is the difference between monochromatic and bolometric flux? How does one measure each?

Flux is the amount of light passing through an area per unit time. It is related to the distance from the emitter r as $F \propto r^{-2}$, the **inverse-square law**, because the same luminosity is spread out over the surface of a sphere with surface area going as r^2. **Apparent magnitude** is a logarithmic rescaling of the flux, and thus is a measure of how bright an object is to the observer relative to some astronomical standard:

$$m = -2.5 \log_{10} \left(\frac{F}{F_0} \right) + m_0, \tag{2.2}$$

where F_0 and m_0 are the standard reference, often the star Vega. **Absolute magnitude** is a logarithmic rescaling of the luminosity, and thus is a measure of the intrinsic brightness of an object:

$$M = -2.5 \log_{10} \left(\frac{L}{L_0} \right) + M_0, \tag{2.3}$$

where again L_0 and M_0 are the standard reference and are defined such that the absolute magnitude of an object is equal to the apparent magnitude of the object if it were placed 10 pc away. Thus, we can define the **distance modulus** as $\mu = m - M$ so that

$$\mu = 5 \log_{10}(d\,[\mathrm{pc}]) - 5, \qquad d\,[\mathrm{pc}] = 10^{\frac{\mu}{5}+1}. \tag{2.4}$$

Monochromatic flux is the flux measured for one wavelength or frequency. This is measured by using a dispersive element to get a spectrum of monochromatic flux versus wavelength. The (cgs) units are erg s^{-1} cm^{-2} Hz^{-1}. The **bolometric flux** is the flux integrated over all wavelengths or frequencies. In practice, this cannot be measured directly; one could integrate across observations from different bands (a pseudo-bolometric flux, since it only measures the flux within the outermost filters), or fit and integrate a model spectrum or **spectral energy distribution (SED)**. Between monochromatic and bolometric fluxes are **photometric or integrated bandpass fluxes**, which is what is measured in photometry with bandpass filters and is equal to the integrated product of the true spectrum and the transmission curve of a physical filter.

13 What is a K-correction? What is needed to determine the K-correction for a source?

K-corrections are used to approximate the rest frame measurements of an object's photometry. Since we only measure the part of the spectrum that passes through our bandpass filter, we cannot directly compare measurements of objects at different redshifts observed through the same filter without such corrections. Redshift causes both a bandpass shift and a bandwidth broadening by a factor of $1 + z$. Given a source's redshift and its approximate SED (often using SED templates), one can infer the true photometric measurement of a source in its rest frame. The equation for absolute magnitude with K-corrections accounted for is

$$M = m - 5 \log_{10}(D_L\,[\mathrm{pc}]) + 5 - K_{\mathrm{Corr}}, \tag{2.5}$$

or equivalently using the distance modulus defined in Question 12, $\mu = m - M - K_{\mathrm{Corr}}$, and $K_{\mathrm{Corr}} = -2.5 \log_{10}(C/(1+z))$ for some complicated factor C.

14 Describe the factors affecting the total throughput between an astronomical source and the counts recorded on a CCD.

A **CCD**, or **charge-coupled device**, works through the internal photoelectric effect, whereby photons strike the silicon substrate of a pixel, promoting electrons in the valence band to the conduction band of the silicon crystal, which creates an electron-hole pair. An electric field created by the pixel's metal-oxide-semiconductor (MOS) capacitor and depletion region separates the electron and the hole, trapping the photoelectron in a potential well (the "bucket"). The accumulated charge in each pixel, which is proportional to the intensity of light that hit it, is shifted and read out sequentially into a digital signal. We want to cool CCDs to reduce dark current, through which thermal energy kicks electrons from the valence band to the conduction band without any photon absorption, and thermal noise, which is generated by dark current following a Poisson distribution.

The **throughput** between an astronomical source and the counts recorded on a CCD is the relationship between a perfect observation of the source and the image produced. This is affected by astrophysical factors, such as interstellar dust absorption and scattering; atmospheric factors, such as extinction from Rayleigh scattering by air molecules or aerosol scattering by dust, water droplets, or smoke; optical factors, such as absorption and reflection losses by lenses, filters and mirrors, or obstruction by secondary mirrors or internal support structures; detector factors, such as the quantum efficiency of the silicon, which is the efficiency of photon transmission and conversion to electrons; and electronic factors, such as the system gain, which maps electrons to recorded counts.

15 Describe two advantages of a large telescope aperture over a small aperture.

A large telescope aperture gives better resolution by decreasing the diffraction limit (see Question 16) and better sensitivity, because the light-collecting area scales with the square of the aperture diameter.

16 What is the equation for angular resolution, absent effects from the Earth's atmosphere? What is the physical origin of this limit?

Angular resolution, also known as the **Rayleigh criterion**, follows

$$\theta_{\text{min}} \approx 1.22 \frac{\lambda}{D}, \tag{2.6}$$

where θ_{min} is the minimum angular separation required for two point sources of light to be distinguished as separate objects, λ is the wavelength of the observation, and D is the aperture diameter.

The physical origin of this limit is from diffraction. Light from a point source passing through a circular aperture of finite size creates an **Airy disk**, as in Fig. 2.2; the first minimum of the Airy pattern is at $\sin\theta_{\text{min}} \approx 1.22\frac{\lambda}{D}$, and we approximate $\sin\theta \approx \theta$ for our small angles. Two point sources can be separated or resolved when the center of one Airy disk lies outside the first minimum of the other Airy disk.

17 What causes a telescope's point spread function (PSF)? How can one improve (decrease) the PSF?

A telescope's **point spread function** is the shape of a point source as seen on the detector. This is fundamentally an interference pattern and ideally is just an Airy disk (see Question 16), the central peak of which can be approximated as a two-dimensional circular Gaussian. In reality, the atmosphere distorts the PSF, as do detector geometry and pixel-pixel interactions.

Turbulence in the atmosphere causes rapidly changing variations in the refractive index along the light path to the detector, which degrades the PSF. This effect, called atmospheric **seeing**, limits the angular

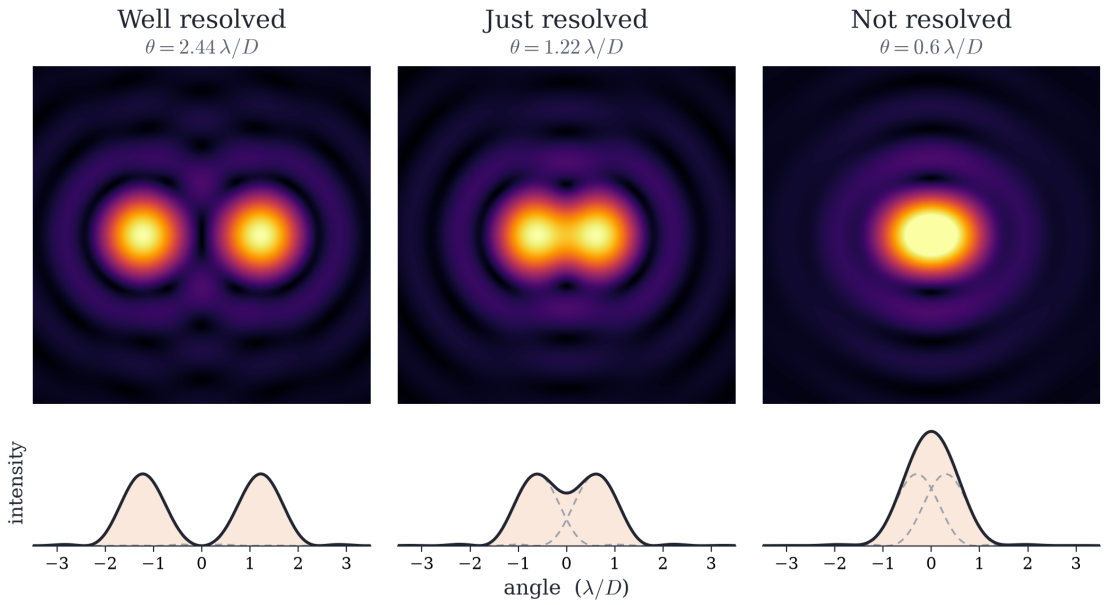


Figure 2.2: Airy disks and angular resolution.

resolution of ground-based telescopes to have $\theta_{\min}$ be above its diffraction limit. The solution for this is to use a space-based telescope or include adaptive optics, which measures (through a wavefront sensor) and mitigates (through a deformable mirror) light distortions in real time through observations of a bright, stable reference source. This source can either be a reference star or an artificial laser guide star, created by laser excitation of sodium atoms in the atmosphere. If we are truly diffraction limited, we can only improve the PSF by increasing the aperture diameter.

Quantization, or rather pixelization, also affects a telescope's PSF, since we are discretizing a continuous PSF over some number of pixels. The natural solution is to increase the number, or reduce the size, of detector pixels. Aberrations caused by optical elements, such as chromatic aberration (the wavelength dependence of refraction through lenses) or wavelength-dependent reflectivity off mirrors can all change the PSF. These can be mitigated by precise optical alignment and telescope maintenance.

18 What are gravitational waves? How do we measure them?

Gravitational waves are gravitational radiation propagating through spacetime. These waves are ripples in spacetime caused by a changing mass quadrupole moment, which is motion that alters the mass distribution in a non-spherical or non-cylindrical manner. The types of astronomical phenomenon that produce gravitational waves include

- All binary systems. The orbital speed of the system controls the wave frequency. Compact binary systems are detectable; they can orbit very close together and have high mass, so they produce strong, high-frequency gravitational waves. As they lose energy to gravitational radiation, they inspiral and eventually merge. Ordinary systems, such as stellar binaries or planetary systems, still emit gravitational waves, but have small masses and large orbital separations, so the amplitudes are minimal and the orbits take trillions of years to decay noticeably.

- Non-spherically symmetric SNe. In particular, core-collapse SNe have several asymmetrical mechanisms that can lead to gravitational waves. These are detectable with current observatories only if they occur within or nearby the MW, because they are relatively faint.

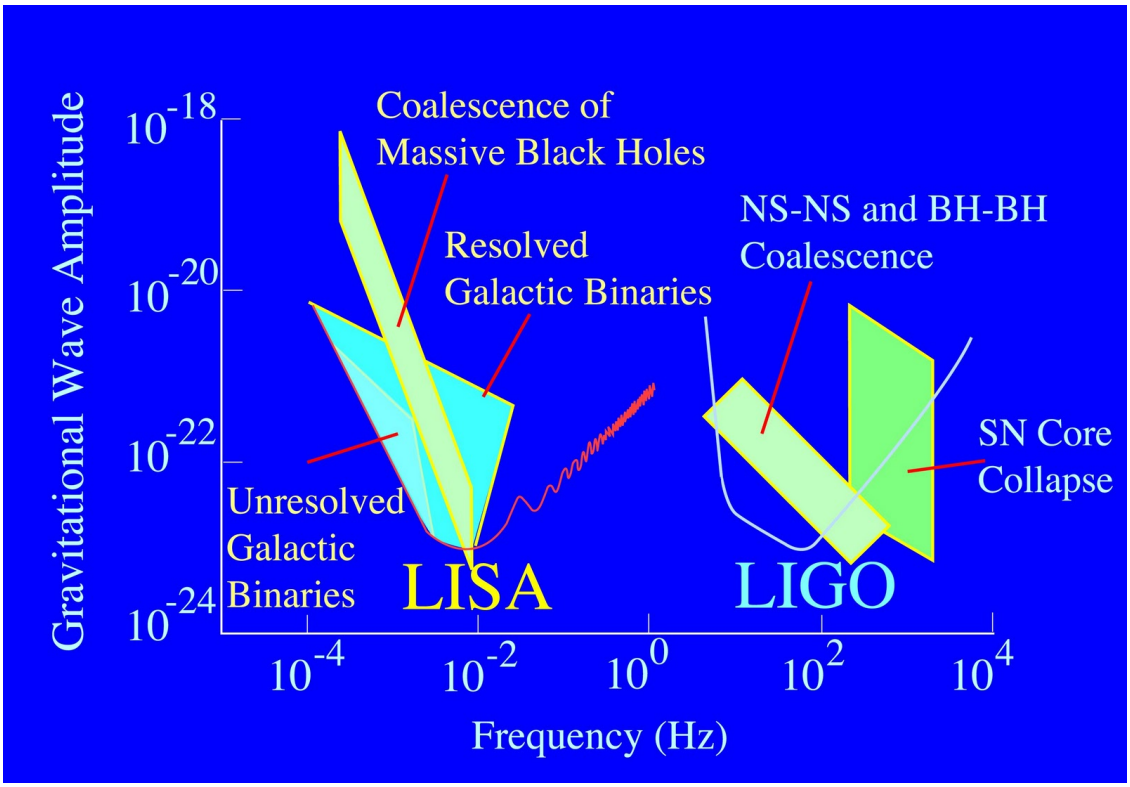


Figure 2.3: Gravitational wave measurement regimes for LISA and LIGO.

- Asymmetric neutron star rotation. A 10 cm asymmetry on a neutron star is effectively a large mountain, so a millisecond pulsar (MSP) with such a bump would produce continuous gravitational waves at a frequency equal to twice the spin frequency of the star, unlike the short, explosive "chirp" signals from merging binaries. This gravitational radiation would cause the spin to gradually slow down over time. In principle, this could be detectable, even though the **strain amplitude**, defined by $h = \Delta L/L$, is extremely faint, and in fact, for the fastest MSPs, the lack of a detection thus far suggests that these stars are incredibly smooth.

We measure gravitational waves using laser interferometers and, hopefully, pulsar timing arrays. Interferometers measure the strain amplitude, the difference in path length caused by the stretching and compressing of spacetime. For LIGO, the arms are 4 km long and they reflect light back and forth several times; typically, ΔL is on the order of a proton width.

LIGO/VIRGO/KAGRA can measure frequencies of 10 Hz to 10 kHz, corresponding to stellar mass mergers. Pulsar timing arrays, such as NANOGrav, can measure frequencies of order nanohertz, targeting the stochastic gravitational wave background, which is dominated by supermassive black hole (SMBH) binaries. LISA, with baselines of 2.5×10^6 km, will measure frequencies of order millihertz, targeting SMBH mergers.

19 What are the sources of atmospheric absorption/scattering? Why is the night sky not perfectly dark? Describe techniques for mitigating these effects in (ground-based) astronomical observations.

Molecules such as water absorb and scatter light, the latter through Rayleigh scattering, which is much stronger at shorter wavelengths. See Questions 10 and 14

The night sky is not perfectly dark because of reflected sunlight on the moon, satellites in orbit, light

pollution from the ground, and airglow, which is driven by recombination of atoms and molecules that were photoionized or energized by solar radiation during the day. There is also zodiacal light, which is sunlight scattered off interplanetary dust in the inner solar system, causing a faint glow along the ecliptic plane.

To mitigate these effects in ground-based astronomical observations, we build our telescopes in high, dry, and remote locations (see Question 20), schedule observations around lunar phases to avoid the full moon, and aim to observe objects at high elevations and away from the moon. We also can use narrowband filters to isolate specific emission lines while blocking some ambient background light, and, in data processing, we perform sky subtraction, where we measure the background light near a target and subtract it from an image.

20 Why do we build telescopes in Hawaii, Chile, Arizona, etc., but not in Boston? Why are some radio telescopes placed at the South Pole?

Three factors make for an ideal observing location:

1. Dryness, since water vapor and suspended water droplets in the atmosphere scatter light, create haze, and absorb photons.

2. High altitude, since there is less atmosphere or airmass to observe through.

3. Remote locations, to minimize light pollution from heavily populated areas.

Thus remote mountains (Mauna Kea, Hawaii), desert mountains (Arizona), or, best of all, remote desert mountains (Chile) are best for ground-based observations, but Boston, which is at sea level, humid, and highly populated, is not.

The South Pole is extremely dry, because nearly all the moisture is frozen out of the atmosphere, so high-frequency radio waves and infrared light are not absorbed. The atmosphere is also very stable, because the sun only rises and sets once per year, so there are minimal temperature fluctuations. It is also at very high altitude and very remote. Thus, some radio telescopes are placed at the South Pole. However, the conditions are poor for optical telescopes. The upper atmosphere is not calm, but near the ground, where air touches the frozen ice sheets, there is a highly turbulent layer of wind and temperature variation that causes poor seeing. Also, the South Pole is in continuous daylight conditions for six months of the year, so that optical telescopes are unable to observe for half the calendar, whereas radio telescopes are uninterrupted by daylight.

21 How does the signal-to-noise ratio of an observed source depend on integration time?

We assume that the number of photons hitting our detector in a given integration time follows a **Poisson distribution**. The standard deviation of a Poisson distribution scales with $\mu^{1/2}$, where μ is the average or expected number of photons in the integration time. Note μ scales with time, since we assume the average rate of photons hitting the detector is constant. The signal-to-noise ratio follows as

$$S/N = \frac{\mu}{\sigma} \propto \frac{\mu}{\mu^{1/2}} = \mu^{1/2} \propto t^{1/2}. \tag{2.7}$$

This is the photon noise limited case; in reality, there is also background sky noise and detector read noise. The former also scales with $t^{1/2}$ and the latter is constant, so at long exposures, we recover the $t^{1/2}$ scaling for the signal-to-noise ratio.

Let us also consider radio observations, for which the statistics are Gaussian rather than Poisson. This is because radio telescopes measure continuous EM waves and voltages rather than counting discrete photons.

The signal from a radio telescope remains stable over time, and the noise scales as $1/\sqrt{t}$, since noise fluctuates randomly and averages out statistically. We recover the same scaling as the optical case:

$$S/N \propto \frac{1}{1/\sqrt{t}} = t^{1/2}. \tag{2.8}$$

The actual radiometer equation is

$$S/N = \frac{T_A}{T_{\text{sys}}}\sqrt{\Delta\nu \cdot t}, \tag{2.9}$$

where T_A is the antenna temperature (proportional to the power received from the source), T_{sys} is the system noise temperature (noise from the receiver, atmosphere, and sky), and $\Delta\nu$ is the observation bandwidth.

22 You have collected a two dimensional set of data points. From a scatter plot you observe a correlation between the two dimensions. How can you determine if the correlation is statistically significant? Describe at least two techniques.

One can compute a correlation coefficient and test the null hypothesis to determine if a correlation is statistically significant. This yields a p-value, the probability of obtaining a test statistic at least as extreme as the observed value from uncorrelated data. Some tests include:

- **Pearson's coefficient (linear):** This tests the strength of a linear relationship, with correlation coefficient $-1 \leq r \leq 1$,

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}} = \frac{\text{Cov}(X,Y)}{s_X s_Y}, \tag{2.10}$$

  where $\text{Cov}(X,Y)$ is the sample covariance and s_X, s_Y are the standard deviations of the variables. To determine if r is significantly different from zero, we convert it into a Student's t-statistic with $\nu = n-2$ degrees of freedom, assuming that the joint distribution is approximately bivariate normal, as

$$t = r\sqrt{\frac{n-2}{1-r^2}}. \tag{2.11}$$

  Under the null hypothesis, this statistic follows a Student's t-distribution with $n-2$ degrees of freedom. The two-tailed p-value is:

$$p = 2 \cdot P(T \geq |t|), \quad T \sim t_{n-2}. \tag{2.12}$$

  If $p < \alpha$ (typically $\alpha = 0.05$), we reject the null hypothesis and conclude that the linear correlation is statistically significant.

- **Spearman's coefficient (rank):** This measures monotonic relationships and is robust to outliers. It finds ranks then computes Pearson's coefficient on the ranks, r_S, for which we can then apply the Student's t-test to measure statistical significance.

- **Permutation test:** When the distribution is non-normal, contains outliers, or when evaluating non-linear monotonic relationships, a permutation test provides a distribution-free measure of significance. After calculating r or r_S on the original paired dataset, we simulate the null distribution by randomly shuffling the y values among the x values many times, recomputing the correlation coefficient r^* or r_S^*. This is valid because under the null hypothesis, X and Y are independent, so any y_i is equally likely to pair with any x_j. The p-value is the fraction of shuffles that give a correlation at least as extreme as the observed value.

- **Bootstrapping (significance):** Rather than permuting labels, we can resample paired data (x_i, y_i) with replacement B times to construct an empirical sampling distribution for r. This preserves the covariance structure of each pair, and it outputs the confidence intervals of r. If the 95 % bootstrap confidence interval does not encompass zero, the correlation is statistically significant at $\alpha = 0.05$.

Often, to fit empirical relations (say, the Cepheid period-luminosity relation, the X_{CO} conversion factor, Tully-Fisher relation, etc.) , we do something like generalized linear regression (allowing for uncertainties in both axes) and assess the fit quality using a generalized χ^2 goodness-of-fit:

$$\chi^2 = \sum_{i=1}^{N} \frac{(y_i - (\alpha + \beta x_i))^2}{\sigma_{y,i}^2 + \beta^2 \sigma_{x,i}^2 + \sigma_{\mathrm{int}}^2}, \tag{2.13}$$

where $\alpha + \beta x$ is the linear fit, σ_{int} is the intrinsic scatter and $\sigma_{x,i}, \sigma_{y,i}$ are the uncertainties in the individual measurements. For a good model, where the intrinsic scatter accounts for the spread, the reduced chi-squared, $\chi^2_\nu \equiv \frac{\chi^2}{N-k} \approx 1$, where $N - k$ is the degrees of freedom. To show whether the correlation is statistically significant, we must demonstrate that the slope β is distinguishable from zero ($|\beta| / \sigma_\beta \geq 3 - 5\sigma$ or $P(\beta \leq 0 \mid \text{data}) \approx 0$ via MCMC sampling).

Chapter 3

ISM and Star Formation

23 What is interstellar dust? How is it observed? Why is it important (state at least two reasons)?

Interstellar dust consists of macroscopic particles larger than atoms and molecules but smaller than meteorites; the typical range of sizes is from 10 Åto 1 μm. The size distribution roughly follows a power law, scaling as $N(a) \sim a^{-\gamma}$, where a is the dust grain size and $\gamma \sim 3.5$. Dust grains are made mostly of silicates, carbonaceous material, and iron.

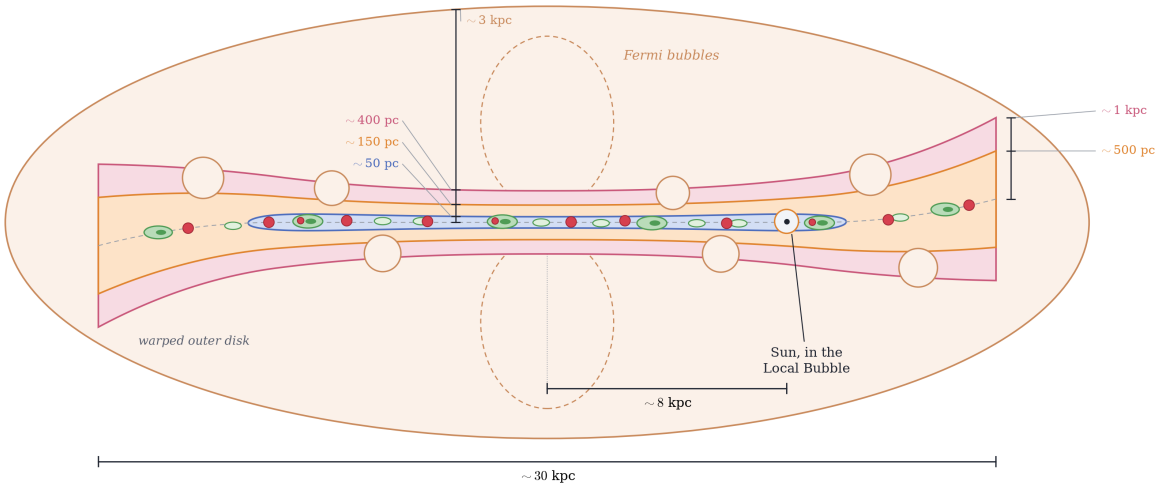
Dust is observed through the extinction and reddening of background starlight, polarization measurements, and through its thermal emission spectrum in the infrared.

Dust is important because it scatters and extincts light along the line of sight, reprocessing it into infrared modified blackbody emission. Dust also exhibits spectral features, such as line emission associated with polycyclic aromatic hydrocarbons (PAHs) and diffuse interstellar bands (DIBs). We do not know the "carrier" grains for many of these lines. Dust also polarizes background starlight: grains are asymmetric, often elongated along a certain axis, and they can align with each other on macroscopic scales due to magnetic fields, so when starlight passes through these aligned grains, the dust preferentially absorbs and scatters light oscillating along the grain's long axis, imprinting a linear polarization. In molecular clouds and star-forming regions, dust grains act as catalysts for surface chemistry, enabling, for example, the formation of molecular hydrogen, and shield dense gas clouds from ambient UV radiation, allowing for stars and planets to form. Grains also provide electrons for photoelectric heating.

24 Describe the major phases of the ISM in the Milky Way. What are their size and mass scales?

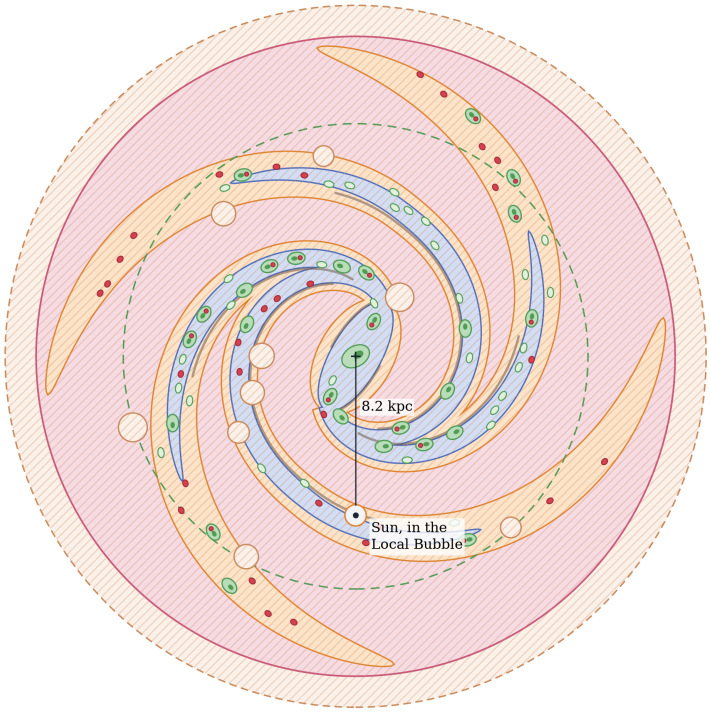
The properties of the major ISM phases in the Milky Way are shown in Fig. 3.1. A sketch of the planar and vertical views of the Milky Way are presented in Fig. 3.1, and likewise a sketch of molecular clouds and H II regions is presented in Fig. 3.2. Some additional notes:

- Molecular clouds consist of molecular hydrogen which collapses to form stars.
- We do not see 21 cm line emission in the CNM because it is not at a high enough temperature for the line to emit.
- The WIM consists of discrete regions, such as H II regions which are star-forming, planetary nebulae around intermediate mass stars which have shed their envelopes, and SNR, and diffuse regions.



	T (K)	n (cm^{-3})		T (K)	n (cm^{-3})
Hot ionized medium	10^6	10^{-3}	Giant molecular clouds cores: $T\sim 10$ K, $n\sim 10^4$	50	10^3
Warm ionized medium	10^4	0.2	Diffuse H_2 clouds	100	100
Warm neutral medium (HI)	5000	0.5	HII regions some embedded in GMCs	10^4	10^4
Cold neutral medium (HI)	100	30	Hot supernova bubbles	10^6	—

Schematic only: vertical scale heavily exaggerated; hydrogen phases only; values approximate.



Diffuse gas

- Hot ionized medium — pervades the whole disk and halo (hatched)
- Warm ionized medium — pervades most of the disk
- Warm neutral medium (HI) — traces the spiral arms; extends further
- Cold neutral medium (HI) — traces the spiral arms
- Dust lanes — mostly along the spiral arms
- Extent of most diffuse H_2

Clouds and bubbles

- Giant molecular cloud — with a dense core
- GMC with an embedded HII region
- Diffuse H_2 cloud
- HII region
- Hot (coronal) gas bubble — lower-density ISM inside

Objects drawn much larger than their true size.

Figure 3.1: Views of the ISM.

Phase	H Form	$n\,(\mathrm{cm}^{-3})$	$T\,(\mathrm{K})$	$f_{\mathrm{vol}}\,(\%)$	$f_{\mathrm{mass}}\,(\%)$	Tracer
Mol. Clouds	H_2	$\gtrsim 100$	~ 10	< 1	20	CO
CNM	H I	~ 100	~ 100	$2-4$	40	H I absorption (UV, optical)
WNM	H I	~ 1	$5000-8000$	30	30	21 cm line emission
WIM	H II	~ 0.1	$\sim 10^4$	20	10	Hα emission (6563 Å)
HIM	H II	$< 10^{-3}$	$\sim 10^6$	50	1	X-ray, UV absorption

Table 3.1: Phases of the ISM.

25 How is the gas in the different phases observed?

See the rightmost column of Table 3.1.

- Molecular clouds are traced by CO, the second most abundant molecule. H_2 does not have a permanent dipole, so its rotational transitions are extremely weak and nearly impossible to excite at cold temperatures. See Question 30.
- The CNM is traced by UV absorption lines in neutral hydrogen (e.g., Lyman series) and optical metal absorption lines (forbidden transitions). Can also be traced by 21 cm line absorption.
- The WNM is traced by 21 cm line emission.
- The WIM is traced by Hα emission
- The HIM is traced by X-rays thermal emission (bremsstrahlung) and high-ionization UV absorption lines.

26 What are Strömgren spheres and how do we estimate their sizes?

Consider a central massive star, with $T \geq 20,000\,\mathrm{K}$, surrounded by gas. Note, for example, an O8V star has $T_{\mathrm{eff}} \approx 35,000\,\mathrm{K}$. About 30% of the energy from this star comes out in photons with $h\nu > 13.6\,\mathrm{eV}$, i.e., that can ionize hydrogen. We are interested in how the star will affect the surrounding gas, which will be highly ionized (H II region) with strong emission lines (Balmer series, [O III], etc.). A **Strömgren sphere** is the ionized sphere of gas around such an ionizing star, with radius out to the radius at which photoionization and recombination balance.

The photoionization rate per atom is

$$\zeta_{\mathrm{pi}} = \int_{\nu_0}^{\infty} \frac{4\pi J_\nu}{h\nu} \sigma_\nu d\nu, \tag{3.1}$$

where ζ_{pi} has units of inverse time, ν_0 is the threshold frequency (for hydrogen, corresponding to a photon with energy 13.6 eV), σ_ν is the photoionization cross section, and $J_\nu = (4\pi)^{-1} \int I_\nu d\Omega$ is the mean intensity. The rate per unit volume of radiative recombination is $n_e n_{\mathrm{II}} \alpha_B(T)$, where we consider case B recombination because direct recombination to the ground state releases a photon with energy 13.6 eV that can immediately be reabsorbed to ionize another atom. To balance photoionization and recombination, we require

$$n_{\mathrm{I}} \zeta_{\mathrm{pi}} = n_e n_{\mathrm{II}} \alpha_B(T). \tag{3.2}$$

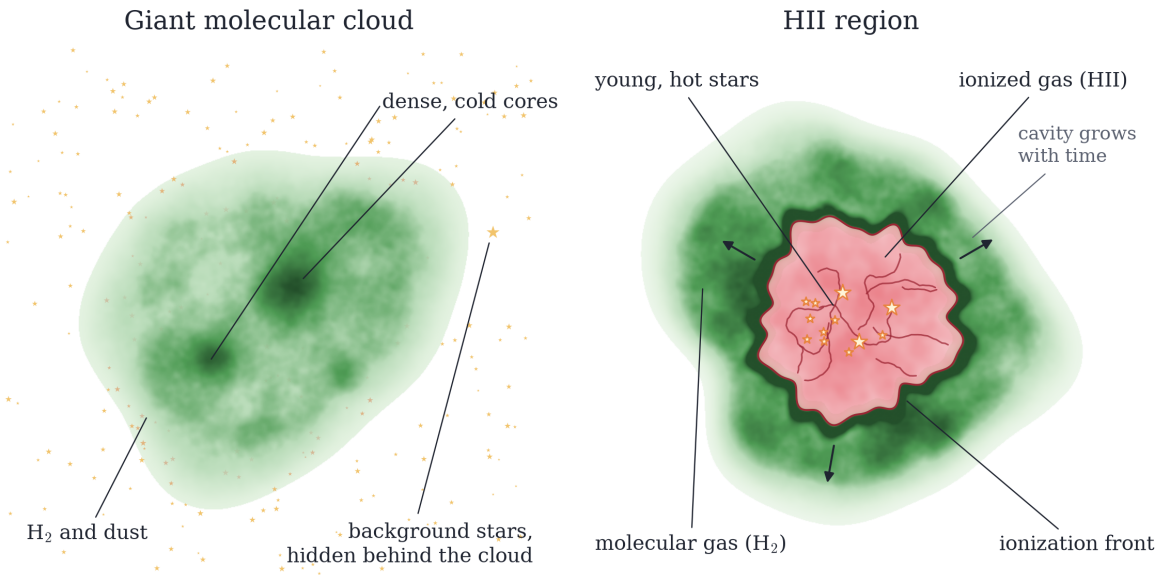


Figure 3.2: Molecular clouds and H II regions. The cores of molecular clouds consist of the densest, coldest gas in the ISM. They are opaque to starlight at high enough densities of H_2 and dust, and perturbations can lead to collapse and star formation. Such star formation triggers ionizing radiation that carves out an ionized cavity growing over time. These H II regions exhibit filamentary structure because of inhomogeneous starting material, radiative and hydrodynamic instabilities, and compression and radiation pressure from stellar winds.

We integrate over the sphere. If every ionizing photon is absorbed inside the sphere, then the left side integrates to Q_0, the number of ionizing photons per second from the source, so we recover

$$Q_0 = \frac{4}{3}\pi R_{\rm S}^3 n_e n_{\rm II} \alpha_B; \tag{3.3}$$

thus the total rate of ionizing photons from the source equals the total recombination rate within the sphere. We can solve for the **Strömgren radius**, giving

$$R_{\rm S} = \left(\frac{3Q_0}{4\pi n_e n_{\rm II}\alpha_B}\right)^{1/3} = 2.9\,{\rm pc}\left(\frac{Q_0}{10^{49}\,{\rm s}^{-1}}\right)^{1/3}\left(\frac{n_{\rm H}}{100\,{\rm cm}^{-3}}\right)^{-2/3}\left(\frac{T}{8000\,{\rm K}}\right)^{0.28}, \tag{3.4}$$

where we have taken helium to be singly ionized, adding 9.6% to the electron density (otherwise $n_e = n_{\rm II} = n_{\rm H}$), and we have used that, for hydrogen,

$$\alpha_B(T) \approx 2.54\times10^{-13}\left(\frac{T}{10^4\,{\rm K}}\right)^{-0.8}\,{\rm cm}^3\,{\rm s}^{-1}. \tag{3.5}$$

How long does it take for the Strömgren sphere to develop (i.e., if we just turn on the star in such a gas cloud)?

$$\frac{1}{\alpha_B n_{\rm H}} \sim 1.2\times10^3\left(\frac{n_{\rm H}}{100\,{\rm cm}^{-3}}\right)^{-1}\,{\rm yr} \tag{3.6}$$

$$t_{\rm O^*} \sim 3-10\,{\rm Myr} \tag{3.7}$$

$$t_{\rm cross} \sim 10^5\,{\rm yr}. \tag{3.8}$$

So, this ionized bubble is formed quite quickly and lasts until the lifetime of the O star. Note also that the ionization boundary is very sharp.

27 Describe the different stages of star formation and approximate timescales.

There are many stages of star formation.

- Giant molecular clouds of cold gas and dust gradually, over millions of years, become dense and unstable, triggered by external shocks or gravity.
- Dense cores within a cloud exceed their critical mass (Jeans mass, see Question 28) and collapse inward, fragmenting into individual clumps over 100,000 to 1 million years. If we assume that existing pressure gradients are negligible and gas is isothermal throughout collapse, we are essentially in free fall. From Question 3 (Eq. 1.11), the free-fall time depends only on density and is independent of the initial radius, so we have homologous collapse: all parts of the cloud take the same amount of time to collapse and density increases by the same rate everywhere. If the cloud is already centrally condensed when collapse begins, we have inside-out collapse where density increases more rapidly near the center, because the free-fall time is shorter for material in the center. Note that during fragmentation, only about 1% of the total mass of the cloud actually forms stars per free-fall time, and over the lifetime of the cloud, about 10% of the mass goes into stars.
- While the gas remains isothermal, the Jeans mass ($\propto T^{3/2}\rho^{-1/2}$, per Eq. 3.14) falls as density rises, so the cloud continues to fragment. At this point, the cloud is optically thin, so thermal energy generated by gravitational compression escapes as radiation, hence the isothermal collapse. But once the gas is dense enough (around $10^{-13}\,\mathrm{g\ cm^{-3}}$), it becomes optically thick, so photons are trapped and we are in an adiabatic compression stage: the gravitational work done on the system is trapped inside. The temperature (and pressure) rises, so the Jeans mass increases, ending fragmentation. This is called **hierarchical opacity-limited fragmentation**.
- As the pressure also rises, the rate of collapse is slowed, and eventually the central region will be nearly in hydrostatic equilibrium. This state is called the **first hydrostatic core** of the **protostar**. This protostellar core is still embedded in a massive envelope of material still in free-fall, and when infalling material meets the nearly hydrostatic core, a shock wave develops. At this shock front, infalling material loses a significant fraction of its kinetic energy, which produces much of the protostar's luminosity.
- Once the temperature reaches $\sim 1500\,\mathrm{K}$ through adiabatic heating, dust begins to vaporize and opacity drops. Photons can thus escape more easily from the core, and the photosphere (boundary at which $\tau \sim 1$) recedes inward. Luminosity, $L \propto R^2 T_{\mathrm{eff}}^4$, is still held high by accretion, so this must mean the effective temperature further increases. Once the temperature reaches $\sim 2000\,\mathrm{K}$, H_2 molecules dissociate into atomic hydrogen, which absorbs energy that would otherwise go into pressure maintaining hydrostatic equilibrium. This drop in pressure triggers a second, final collapse into the **second hydrostatic core**; temperatures rise in the core until deuterium can be fused.
- With the formation of a quasi-static protostar, and the surrounding envelope begins to clear out, the rate of evolution is controlled by the rate at which the star can thermally adjust to the collapse, which is the **Kelvin-Helmholtz timescale** (see, e.g., Question 47). This is the time it takes for a star to radiate its total thermal energy content at its current luminosity. $\tau_{\mathrm{KH}} \gg \tau_{\mathrm{ff}}$, so protostellar evolution is slower than free-fall collapse, which is of order $10^4 - 10^5\,\mathrm{yr}$. It can take almost $40\,\mathrm{Myr}$ for a $1\,M_\odot$ star to contract to its main sequence structure.
- Pre-main sequence stars with $M \lesssim 2\,M_\odot$ travel first along the **Hayashi track** of the HR diagram. This is a nearly vertically downward track, where luminosity decreases while effective temperature increases only slightly. The envelope of the contracting protostar is convective.

- Eventually, at the minimum luminosity and as temperature rises, increasing ionization decreases opacity in the center of the protostar and a radiative core develops. This allows energy to escape into the convective envelope more efficiently, causing luminosity to increase. The effective temperature continues to increase, since the star is still shrinking. These are the nearly horizontally leftward (or tilted slightly upwards) **Henyey tracks** as the star approaches its main sequence position. Nuclear reactions can begin, but not at their equilibrium rates, including the first steps of the proton-proton I (PP-I) chain ($p + p \rightarrow {}^2\mathrm{H} + e^+\nu_e$, ${}^2\mathrm{H} + p \rightarrow {}^3\mathrm{He} + \gamma$). Once the temperature is high enough for the PP-I chain to become dominant, and the establishment of a stable fusion energy source means that gravitational energy terms become insignificant and the star arrives at the **zero age main sequence (ZAMS)**. See Fig. 3.3.

- For stars with $M \lesssim 0.5\,M_\odot$, the Henyey track is missing because the central temperature is never high enough to burn ${}^{12}\mathrm{C}$. Also, since temperatures remain cool and opacity is sufficiently high, a radiative core never develops. If $M \lesssim 0.072\,M_\odot$, the core is never hot enough to start nuclear reactions, so the main sequence is never obtained; these objects can still burn lithium above $0.06\,M_\odot$ and deuterium above $0.013\,M_\odot$, and are called **brown dwarfs**.

- For massive stars, the temperature at the center from gravitational collapse quickly becomes high enough to burn ${}^{12}\mathrm{C}$ and convert ${}^1\mathrm{H}$ to ${}^3\mathrm{He}$. There is no convective stage, so they bypass the Hayashi phase and have nearly horizontal Henyey tracks. The CNO cycle becomes the dominant mechanism for hydrogen fusion in these main sequence stars. Because the CNO cycle is so strongly temperature dependent, the core remains convective even after the main sequence is reached, but their envelopes are radiative, the opposite of low mass stars.

- The visible Hayashi and Henyey phase objects are **T Tauri stars** at low mass and **Herbig Ae/Be** stars at intermediate mass. Massive stars reach the main sequence while still accreting and deeply embedded, in $10^4 - 10^5\,\mathrm{yr}$ (compared to $\sim 40\,\mathrm{Myr}$ for the Sun, see Table 3.2), so there is no visible pre-main sequence object we can observe.

Initial Mass ($M_\odot$)	Contraction Time (Myr)
60	0.0282
25	0.0708
15	0.117
9	0.288
5	1.15
3	7.24
2	23.4
1.5	35.4
1	38.9
0.8	68.4

Table 3.2: Pre-main sequence contraction times, copied from Table 2 of Bernasconi & Maeder (1996), which are for $Z = 0.020$. The contraction time is defined as from the point when the protostar has formed its hydrostatic core and emerges onto the Hayashi track to the point where it settles at the ZAMS.

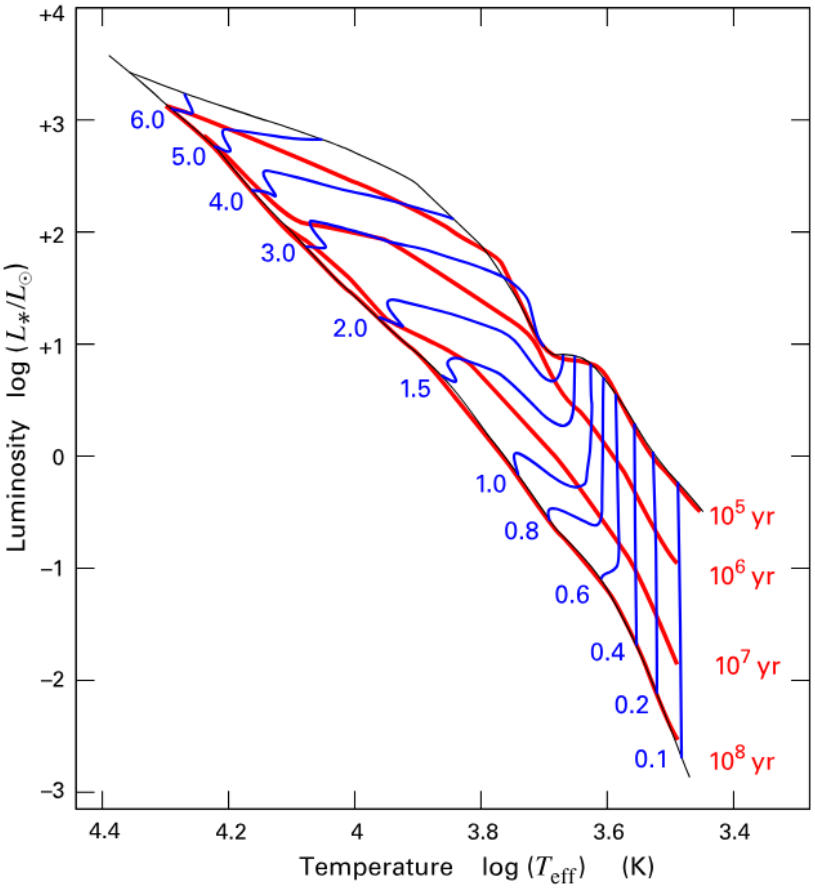


Figure 3.3: Pre-main sequence Hayashi and Henyey tracks.

28 Describe the Jeans instability and sketch the derivation of the Jeans mass.

Under which conditions can small perturbations of a gas cloud grow exponentially, leading to the collapse of the cloud and SF? The **Jeans instability** describes the gravitational instability of a self-gravitating gas cloud.

The simplest order of magnitude derivation is from force balancing: collapse occurs if the inward gravitational force is bigger than the outward pressure force. The mean forces per unit volume are

$$\frac{F_p}{V} \sim \frac{F_P}{Ar} \sim \frac{p}{r}, \qquad \frac{F_G}{V} \sim \frac{GM^2}{r^2 V} \sim \frac{GM\rho}{r^2} \sim G\rho^2 r, \tag{3.9}$$

so we get

$$\frac{p}{r} = G\rho^2 r \Longrightarrow r = \sqrt{\frac{\rho}{G\rho^2}}. \tag{3.10}$$

Note that $r \propto \sqrt{p}$, so that higher pressure means more stability, and $r \propto 1/\rho$ so higher density means less stability. With $p = k\rho T/\mu m_{\mathrm{H}} = c_s^2 \rho$ (where c_s is the isothermal sound speed), we get

$$r = \frac{c}{\sqrt{G\rho}}. \tag{3.11}$$

We can also just consider the relevant timescales. The sound crossing time is $\tau_{\mathrm{sound}} = r/c_s$. The dynamical timescale is $\tau_{\mathrm{dyn}} \approx 1/\sqrt{G\rho}$, which is the gravitational free fall timescale in the absence of pressure support:

$$\tau_{\mathrm{dyn}} = \frac{r}{v_{\mathrm{ff}}} \text{ and } v_{\mathrm{ff}} \approx \sqrt{\frac{2GM}{r}} \Longrightarrow \tau_{\mathrm{dyn}} \approx \frac{r}{\sqrt{2GM/r}} = \sqrt{\frac{3}{8\pi G\rho}} \sim \frac{1}{\sqrt{G\rho}}. \tag{3.12}$$

The instability criterion is thus also $\tau_{\mathrm{dyn}} \lesssim \tau_{\mathrm{sound}}$, which again gives $r > \frac{c}{\sqrt{G\rho}}$. Intuitively, this timescale criterion says that instability occurs if sounds waves, which propagate "information," do not have time to traverse the cloud over a dynamical timescale. This means that pressure cannot increase in time to counterbalance gravity, as, for example, the center of the cloud does not yet know what is going on at the surface.

In reality, if we do the whole computation, we find that $\tau_{\mathrm{dyn}} = \sqrt{\frac{3\pi}{32G\rho}}$, which yields the actual **Jeans length**

$$R_{\mathrm{J}} = c_s \sqrt{\frac{3\pi}{32G\rho}}. \tag{3.13}$$

The **Jeans mass** is the volume of a sphere with radius equal to the Jeans length and average density ρ,

$$M_{\mathrm{J}} = \frac{4\pi R_{\mathrm{J}}^3 \rho}{3} = \frac{\sqrt{3}\,\pi^{5/2} c_s^3}{32\sqrt{2} G^{3/2} \rho^{1/2}} = \frac{\sqrt{3}\pi^{5/2} k^{3/2}}{32\sqrt{2}(G\mu m_{\mathrm{H}})^{3/2}} \frac{T^{3/2}}{\rho^{1/2}} \propto T^{3/2} \rho^{-1/2}. \tag{3.14}$$

We can also use the virial theorem (see Question 4), which gives roughly the same answer, although with some different constants. First, we derive the gravitational potential energy of a sphere with constant density $\rho = \frac{M}{4\pi R^3/3}$. Mass contained a sphere of radius $0 \le r \le R$ is $m(r) = M\left(\frac{r}{R}\right)^3$, and $dm = \frac{3M}{R^3} r^2 dr$. Then,

$$dU = -\frac{Gm(r)dm}{r} = -\frac{3GM^2}{R^6} r^4 dr, \quad U = -\frac{3GM^2}{R^6} \int_0^R r^4 dr = -\frac{3GM^2}{5R}. \tag{3.15}$$

The total thermal kinetic energy in the cloud is $K = \frac{3}{2} NkT$, where $N = M/\mu m_{\mathrm{H}}$ is the total number of particles. The virial theorem gives $\langle K \rangle = -\frac{1}{2} \langle U \rangle$. For collapse to occur, we must have that $\langle T \rangle < -\frac{1}{2} \langle U \rangle$; the gravitational potential energy overtakes the thermal kinetic energy. So, we need

$$3NkT < \frac{3GM^2}{5R} \Longrightarrow \frac{kT}{\mu m_{\mathrm{H}}} < \frac{GM}{5R} = \frac{4\pi R^3 G\rho}{15R}. \tag{3.16}$$

So, again using $c_s^2 = kT/\mu m_{\rm H}$, we get

$$R_{\rm J} = c_s \sqrt{\frac{15}{4\pi G\rho}}. \tag{3.17}$$

29 Describe at least three heating and cooling processes in the ISM and their importance in various domains.

Process	Regime	Volume rate $\propto$
Heating		
Photoionization	H II regions	$n_e n(\mathrm{H}^+)$
Photoelectric (dust)	CNM, WNM, PDRs	$n_{\rm H} G_0 \epsilon (G_0 T^{1/2}/n_e)$
Cosmic rays	molecular clouds	$n_{\rm H}\zeta_{\rm CR}$
X-ray photoionization	near X-ray sources	$n_{\rm H} F_X$
H_2 formation, UV pumping	dense PDRs	$n_{\rm H} n(\mathrm{H}^0)$, $G_0 n(\mathrm{H}_2)$
Turbulence, MHD waves	molecular clouds	uncertain
Cooling		
[C II] 158 μm	CNM	$n_{\rm H}^2 e^{-91\,{\rm K}/T}$
[O I] 63 μm	WNM, CNM	$n_{\rm H}^2 e^{-228\,{\rm K}/T}$
Lyα	WNM near 10^4 K	$n_e n(\mathrm{H}^0) e^{-1.18\times10^5\,{\rm K}/T}$
Grain recombination	WNM	$n_e n_{\rm H}$
Forbidden lines	H II regions	$n_e n_i$
Radiative recombination	H II regions	$n_e n(\mathrm{H}^+)$
Free-free	H II regions, hot gas	$n_e n_i T^{1/2}$
CO rotational lines	molecular clouds	$n_{\rm H}^2$ to $n_{\rm H}$
Gas-grain collisions	dense cores	$n_{\rm H}^2 T^{1/2}(T - T_{\rm d})$
Metal lines in CIE	hot gas, $10^5 - 10^7$ K	$n_e n_{\rm H} \Lambda(T)$
Compressive Work		
Compression and expansion	shocks, SNR interiors	$P\,dV$

Table 3.3: Heating and cooling processes in the ISM.

The processes are summarized in Table 3.3. Heating processes include:

- **Photoionization**: a photon of energy $h\nu$ will hit an atom with an ionization potential I and will release an electron with energy $h\nu - I$. The electrons share their energy with other particles on collisional timescales. We can write

$$\Gamma_{\rm pi} \simeq \alpha_\beta n_e n_{\rm H} \langle E_{\rm pe} \rangle . \tag{3.18}$$

- **Photoelectric heating** from dust: photons can also release electrons from dust grains. Graphite has

a bulk work function $W \approx 4.5\,\mathrm{eV}$, so photoelectrons have energy $h\nu - W$. We get

$$\Gamma_{\mathrm{pe}} \simeq 10^{-24} \epsilon G_0 n_{\mathrm{H}}, \tag{3.19}$$

with $\epsilon \sim 0.02$ and $G_0 \equiv \frac{u(6-13.6\,\mathrm{eV})}{5.29\times10^{-14}}\,\mathrm{erg\ cm^{-3}}$. In this case, the electron does not "re-attach" to the dust grain, unlike in photoionization, in which electrons can (and do) recombine. Photoelectric heating is the dominant heating source in the (cold and warm) neutral medium.

- **Cosmic rays**: atomic nuclei that have been accelerated to energies of 1 GeV or higher will also release electrons with high energy from atoms. These high energy electrons again collide with other atoms, releasing more electrons. The final heating energy is less than the initial energy because some goes into ionization. The expression is

$$\Gamma_{\mathrm{cr}} = \zeta_{\mathrm{cr}} E_{\mathrm{H}} n_{\mathrm{H}}, \tag{3.20}$$

  where $\zeta_{\mathrm{cr}} = 4\pi \int j(E)\sigma(E)dE$ is the primary ionization rate, with σ the cross-section and j the cosmic ray flux. Cosmic rays are the dominant heating source in molecular clouds and dense cores, which are shielded from ionizing photons. Note that FUV photons are absorbed by the same dust that produces the photoelectrons in photoelectric heating, so that photoelectric heating falls as $\sim e^{-1.9A_V}$, so cosmic rays take over beyond $A_V \sim 2$.

- **H_2 formation and UV pumping**: H_2 molecules form on the surfaces of dust grains, and each formation releases $\sim 4.5\,\mathrm{eV}$ of binding energy, which is enough to eject it from the grain and give it some kinetic energy. H_2 also absorbs FUV photons in its Lyman and Werner bands (between 11.2 and 13.6 eV) and often undergoes collisional de-excitation with surrounding gas particles, heating the gas. This is a dominant heating process in photon-dominated regions (PDRs), the dense gas bordering H II regions.

- **Mechanical heating**: processes such as bulk motion dissipation, the decay of turbulence, MHD wave damping, and friction can all heat gas.

Cooling processes include:

- **Radiative cooling**: collisional de-excitation and radiative de-excitation gives $\Lambda = n_u A_{u\ell} E_{u\ell}$; in the case where $n_c \ll n_{\mathrm{crit}}$, this becomes $\Lambda = n_\ell n_c g_{\ell u} E_{u\ell} \propto n^2 e^{-E_{u\ell}/kT}$. In the CNM, [C II] at 158 μm is the primary cooling transition, with $E/k \sim 91\,\mathrm{K}$ and $I = 11.3\,\mathrm{eV}$. [O I] at 63 μm in the CNM and WNM is also dominant, and near $10^4\,\mathrm{K}$, Lyα is very strong, preventing neutral gas from reaching temperatures above this limit. In molecular clouds, CO is the dominant coolant.

- **Gas-grain collisions**: this scales as $\Lambda_{\mathrm{gd}} = \alpha_{\mathrm{gd}} n_{\mathrm{H}}^2 T^{1/2}(T_{\mathrm{g}} - T_{\mathrm{d}})$. Note that this can also be a heating term in some cases, depending on the gas and dust temperatures. This is a dominant cooling term in dense cores.

- **Radiative recombination**: a free electron that recombines with an ion removes its kinetic energy from the thermal gas; the photon carrying the ionization energy is not a loss of thermal energy, because it was supplied by an ionizing photon rather than drawn from the gas. Electrons also recombine with dust grains and PAHs. Recombination is negligible in cold gas, but it offsets a substantial fraction of the photoelectric heating in warm gas, and near $T \sim 10^4\,\mathrm{K}$, it becomes the dominant exchange of energy between gas and grains.

- **Free-free (bremsstrahlung) emission**: electrons lose kinetic energy when deflected in the electric field of ions, converting thermal energy into radiation. Because free-free cooling increases with temperature while line cooling declines once metals are highly ionized, free-free emission dominates the cooling of the hottest gas and of metal-poor gas at lower temperatures.

30 Most of the ISM is hydrogen, but we instead often observe CO to characterize the ISM. Why is this the case? Under what conditions can we directly measure hydrogen in the ISM?

Molecular hydrogen has no permanent dipole and thus no dipole rotational transitions. The lowest rotational lines (weak electric quadrupole lines) have an upper-level energy of $E/k \sim 510\,\mathrm{K}$ and $A \sim 3 \times 10^{-11}\,\mathrm{s}^{-1}$, so that level is essentially unpopulated in molecular clouds. On the other hand, CO, the second most abundant molecule (with $N(\mathrm{CO})/N(\mathrm{H_2}) \sim 10^{-4}$ on average), has a small permanent dipole. The CO $J = 1 \to 0$ at 115 GHz has $E/k \sim 5.5\,\mathrm{K}$. The small dipole also gives CO a critical density $n_\mathrm{c} \sim 10^3\,\mathrm{cm}^{-3}$, so it is collisionally excited in typical molecular gas. That line falls in the atmospheric millimeter window (see Question 10), so ground-based radio telescopes can observe it. Note that ^{12}CO is typically optically thick, so its intensity is more a measure of cloud temperature and velocity dispersion than column density; we can use the isotopologues ^{13}CO or C^{18}O to measure column density.

It is sometimes possible to measure $\mathrm{H_2}$ directly, either through UV electronic absorption (Lyman and Werner bands) or the infrared rotational quadrupole rotational lines in warm or shocked gas where those upper levels are populated. We can measure atomic hydrogen through UV absorption lines or the 21 cm line.

Chapter 4

Planets and the Solar System

31 Describe three techniques for detecting exoplanets.

There are many exoplanet detection techniques. See Fig. 4.1 for a summary.

1. **Radial velocities (RV)**: credited with finding the first "real" exoplanet in 1995 (51 Peg b). A planet and its star will orbit around their common center of mass, so that the star will undergo a small orbit or "wobble" that is observable via shifting of the star's spectral lines from the Doppler effect. When the star is moving away from Earth, its light is redshifted, and when it is moving towards Earth, it is blueshifted. This measures the planet's minimum mass, specifically $M_{\rm p} \sin i$ for inclination angle i, and the orbital period. This is most sensitive to large planets, though modern spectrographs can detect super-Earths, and short-to moderate orbital periods.

2. **Transits**: the current golden method, with a boom from Kepler and TESS. With the proper geometry/inclination with respect to our line of sight, a planet will pass in front of its host star, causing a small dip in its observed brightness. This measures the planet's radius relative to the star's radius, derived from the transit dip depth, and the orbital period, from the time between dips. This is most sensitive to short periods and close orbits, because the geometric probability of alignment is higher and transits occur more frequently, and large planets, though Kepler and TESS have made it good at detecting super-Earths and sub-Neptunes. In combination with RV data, we can estimate the planet's density and thus its bulk composition (i.e., rocky versus gaseous).

3. **Direct imaging**: we can directly observe a planet's light, usually by blocking out the host star's bright light with a coronagraph. This is the only method where we can directly detect the planet, as all others are indirect methods. This measures photometric light and possibly spectra of the planet itself, allowing direct atmospheric composition and thermal emission analysis. This is most sensitive to wide separations, where the planet is physically far enough from the star to be resolved separately, and for young, massive planets, particularly self-luminous gas giants that still glow brightly from the thermal energy of their recent formation.

4. **Microlensing**: when a star and its planet pass in front of a background source, light from the background source can be bent. As this occurs, the brightness of the source can be magnified. The star, the more massive object, will create a stronger magnification, but small peaks or deviations can occur from a companion planet. This is expected to be a big method with Roman. This measures the planet-to-star mass ratio and projected orbital separation, normalized to the Einstein radius (which is determined by the mass of the lens star, the distance to the lens, and the distance to the background source). This is sensitive to distant planets and is independent of host brightness, so that it can also be used to detect free-floating planets not bound to stars.

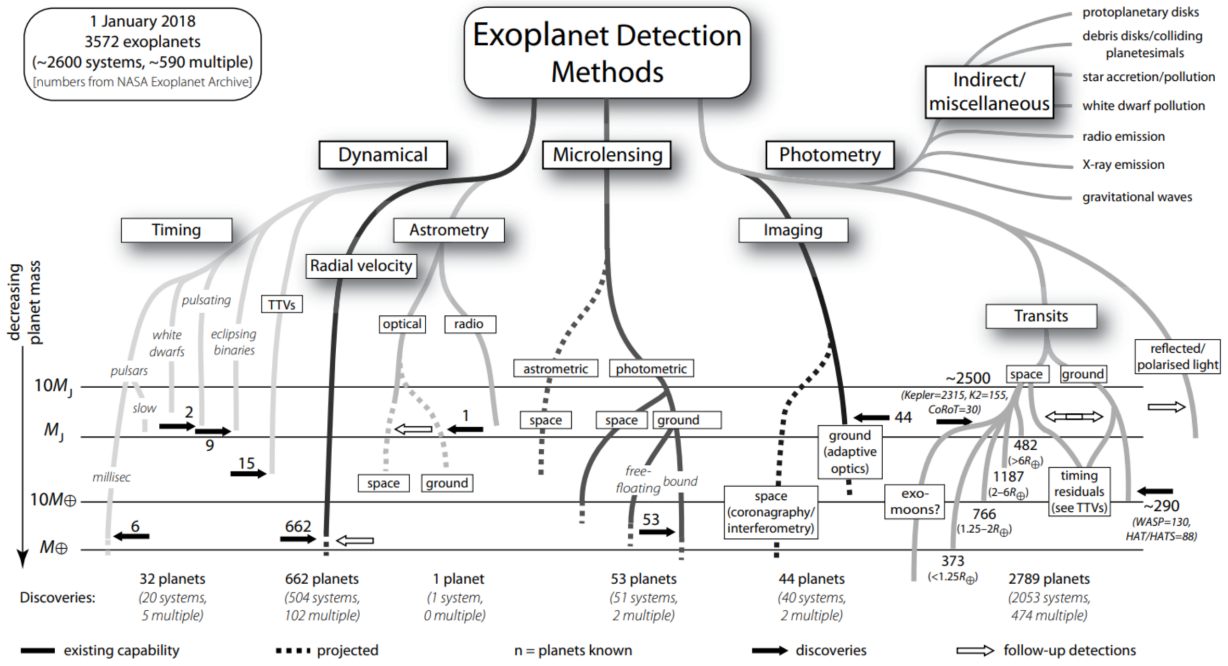


Figure 4.1: Exoplanet detection methods, with approximate planet mass detection sensitivities. Note this data is from 2018, so discovery numbers and projected capabilities are not updated. Taken from Kaushik et al. (2024).

5. **Astrometry**: We can track a star's precise position and proper motion over time to infer an orbiting companion. This is similar to the RV method in that we track small, periodic movements, but through spatial data instead of spectrographic data. There is only a small number of confirmed planets via astrometry, but we anticipate Gaia to find many. This measures the planet's true mass (not the minimum mass $M_\mathrm{p} \sin i$), the orbital inclination, and semi-major axis. This is most sensitive to wide-separation orbits and massive planets orbiting nearby, low-mass stars (with Gaia getting sensitive to Jupiter or Saturn analogs).

32 What physical properties of a planet can you determine from RV and transit observations, respectively? What else do you need to know about the system in order to determine physical properties?

See Question 31. From RV observations, we can determine the minimum masses of planets. The sinusoidal variations (in velocity versus time) due to the small orbit induced by the planet on the star will have a semi-amplitude proportional to the planet mass, as

$$K_1 = \left(\frac{2\pi G}{P}\right)^{1/3} \frac{m_\mathrm{p} \sin i}{m_\star^{2/3}} \frac{1}{\sqrt{1-e^2}}, \tag{4.1}$$

where m_p is the planet mass, $m_\star$ is the stellar mass, i is the inclination angle, P is the orbital period, and e is the orbit eccentricity. An edge-on orbit (to the observer) will maximize $m_2 \sin i$; if the inclination is not constrained, we measure the minimum mass. We can estimate eccentricity from the shape of the curve; K_1 and P come directly from the time-series data. The stellar mass is generally uncertain; we can use spectroscopy to get T_eff, $\log g$, and metallicity, for example, and use stellar evolution models to estimate mass.

From transits, we can determine the radius of the planet. A transiting planet will block an amount of flux proportional to its cross-sectional area, so the transit depth is defined by

$$\frac{\Delta F}{F} = \frac{\pi R_{\rm p}^2}{\pi R_\star^2} = \left(\frac{R_{\rm p}}{R_\star}\right)^2, \tag{4.2}$$

where $R_{\rm p}$ is the planet radius and $R_\star$ is the stellar radius. The detection of a transit also tells us that the inclination must be (extremely close to) 90 degrees, as transits can only occur with the proper observing geometry.

From both methods, we can determine the orbital period (and thus the semi-major axis via Kepler's third law; see Question 1. Combining RV and transit observations, we can determine the density of a planet, and hence its bulk composition. This information is limited by how well the stellar properties, particularly mass and radius, are constrained.

33 What information about a planetary system is encoded in the ingress and egress duration of a transit light curve? What information is encoded in the transit duration (full width at half maximum)?

Consider a planet of radius $R_{\rm p}$ on a circular orbit of semi-major axis a and period P around a star of radius $R_\star$. Three quantities can be read off a transit light curve: the depth δ, the duration T (measured between the midpoints of ingress and egress, which is approximately the full width at half maximum), and the ingress/egress duration τ. To first order, the depth gives the radius ratio, $\delta \approx k^2$ with $k \equiv R_{\rm p}/R_\star$. Note that limb darkening modifies this slightly. The two timescales encode the orbital geometry.

The **ingress** and **egress** phases are the intervals during which the planet's disk overlaps the stellar limb. Their duration depends on how the planet's path meets the limb, which is set by the **impact parameter**,

$$b = \frac{a\cos i}{R_\star}, \tag{4.3}$$

the sky-projected distance of the planet's path from the stellar centre, in units of $R_\star$. For a central transit ($b = 0$) the planet crosses the limb head-on and ingress is brief. For a high impact parameter ($b \to 1$) the path meets the limb at an oblique angle, so the planet takes longer to move fully onto the disk. Grazing transits therefore have long ingress and egress and appear V-shaped, while central transits have sharp edges and a flat bottom, i.e., are U-shaped.

The transit duration is the length of the chord the planet traverses across the stellar disk divided by its orbital speed. In the limit $k \ll 1$ and $a \gg R_\star$, the two timescales are

$$T \approx T_0\sqrt{1-b^2}, \qquad \tau \approx \frac{T_0\, k}{\sqrt{1-b^2}}, \qquad T_0 \equiv \frac{P}{\pi}\frac{R_\star}{a}, \tag{4.4}$$

where T_0 is the duration of a central transit. The duration on its own is degenerate: a short transit may come either from a wide orbit (large $a/R_\star$) or from a grazing geometry (large b). The ingress/egress duration breaks this degeneracy, because τ and T respond to b in opposite directions. Combining the two gives

$$b^2 = 1 - k\,\frac{T}{\tau}, \qquad \frac{a}{R_\star} = \frac{P}{\pi}\sqrt{\frac{k}{T\tau}}. \tag{4.5}$$

Once $a/R_\star$ is known, Kepler's third law (see Question 1) converts it into the mean stellar density. Assuming $M_{\rm p} \ll M_\star$ and a circular orbit,

$$\rho_\star = \frac{3\pi}{GP^2}\left(\frac{a}{R_\star}\right)^3. \tag{4.6}$$

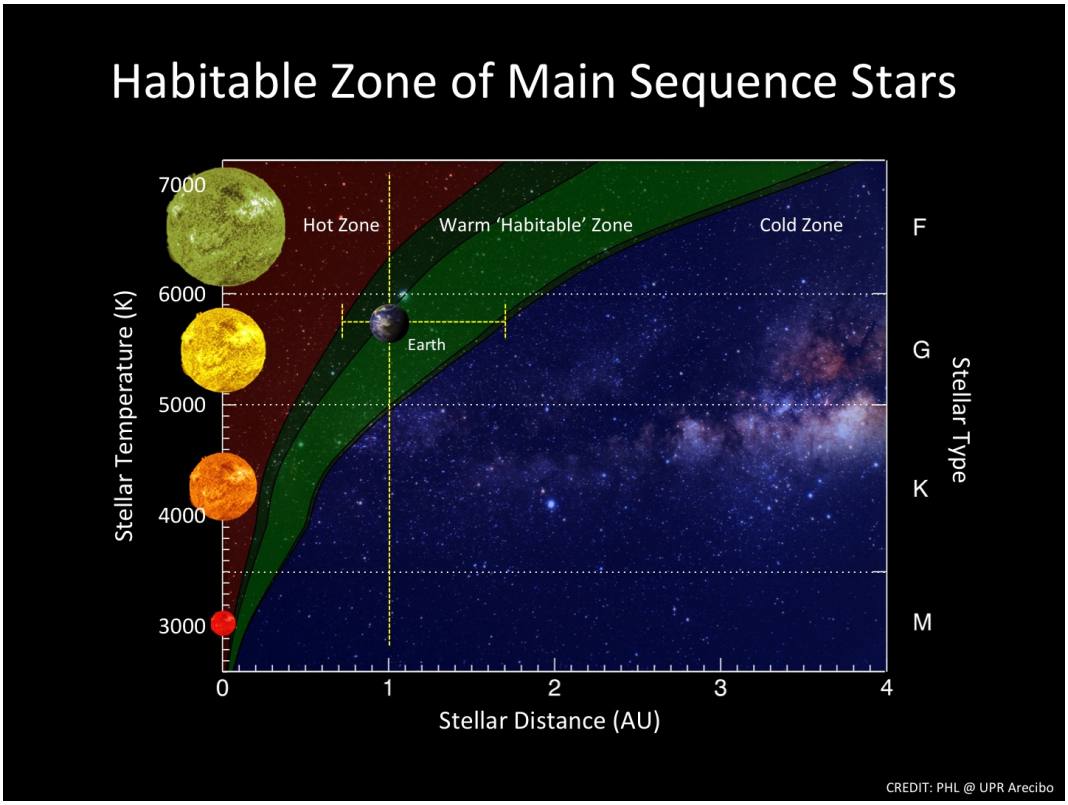


Figure 4.2: The habitable zone.

Seager & Mallén-Ornelas (2003) showed that the depth, the total and flat-bottom durations, and the period can be inverted analytically for b, $a/R_\star$, and $R_\mathrm{p}/R_\star$, and hence for $\rho_\star$. The stellar density obtained this way requires neither the distance to the star nor a stellar evolution model. The light curve alone constrains only ratios, however, so $R_\star$ and a are not determined individually.

In summary, the depth gives $R_\mathrm{p}/R_\star$. The duration and the ingress/egress duration jointly determine b and $a/R_\star$; neither suffices alone. Together with the period, these yield the inclination and the mean stellar density directly from the light curve.

34 What is the "habitable zone," and how does it vary with the mass of the host star?

The **habitable zone (HZ)** of a star is the region, or range of orbits, around a star in which the conditions allow liquid water to exist on the surface of a planet (see Fig. 4.2). Earth is, naturally, in the Sun's HZ. Too close in to the star and water evaporates; too far and water freezes. Usually, HZs move further out with more massive host stars, since more massive stars output more energy. The distance to the HZ scales as $d \propto \sqrt{L}$, since flux drops off with the square of the distance; for main sequence stars, we often estimate $L \propto M^{3.5}$, so that $d \propto M^{1.75}$ to M^2.

We can also account for a star's changing luminosity over time (the **optimistic habitable zone**). Note that our definition of the HZ simply scales with the luminosity of the host star, but this is further complicated by stellar activity (for example, in M dwarfs) or the presence of planetary atmospheres (runaway greenhouse effects, clouds, etc.).

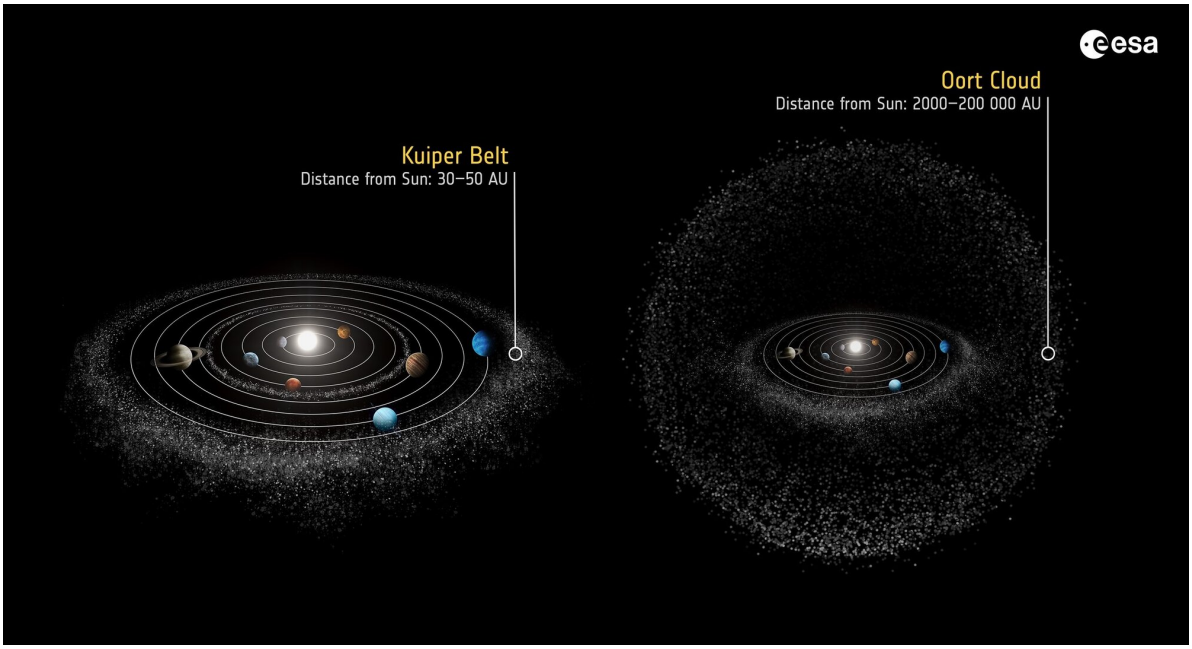


Figure 4.3: The Kuiper Belt and the Oort Cloud.

35 What is the Kuiper Belt? What is its approximate size? What are the properties of Kuiper Belt objects? What is the Oort cloud and how do we think it formed?

See Fig. 4.3. The **Kuiper Belt** consists of small, icy bodies organized in a disk (torus) outside of Neptune's orbit ($\sim 30\,\text{AU}$) to approximately 50 AU. Similar to how the asteroid belt is largely perturbed by Jupiter, the Kuiper belt is strongly influenced by Neptune. The belt can be roughly divided into classes based on orbital resonances:

- the 2:3 resonance group, at $\sim 39.4\,\text{AU}$ (the "plutinos," named after Pluto),
- the 1:2 resonance group, at $\sim 47.7\,\text{AU}$ (the "twotinos"),
- and the "Classical Kuiper Belt" between these two resonances, ~ 42 to 48 AU (the "cubewanos"). These tend to have lower orbital inclinations ($< 30°$), while there are also a family of "scattered" KBOs that have higher orbital inclinations.

The Kuiper Belt is the source of short-term comets in the Solar System, with periods less than $\sim 200\,\text{yr}$. The belt can be considered to be leftover icy planetesimals from the protoplanetary disk that were unable to grow into larger bodies due to gravitational disruptions from the outward migration of the ice giants to create the group of small bodies in this region.

The **Oort Cloud** is a collection of small bodies (10^{12} to 10^{13} members, with a combined mass of around $100\,M_\oplus$) much further out in the Solar System, hypothesized to have an inner disk ($\sim 1000\,\text{AU}$) and an outer shell ($\sim 10,000$ to $100,000\,\text{AU}$). These bodies are thought to have been perturbed by the migrating gas giants during the planetesimal phase and flung far outward to large distances, but not far enough to escape the Sun's gravitational force. At these distances, they can easily be perturbed by passing stars or gas clouds, which can randomize their orbits, causing the outer sphere. Some of these perturbations can cause long-term comets, with periods greater than $\sim 200\,\text{yr}$.

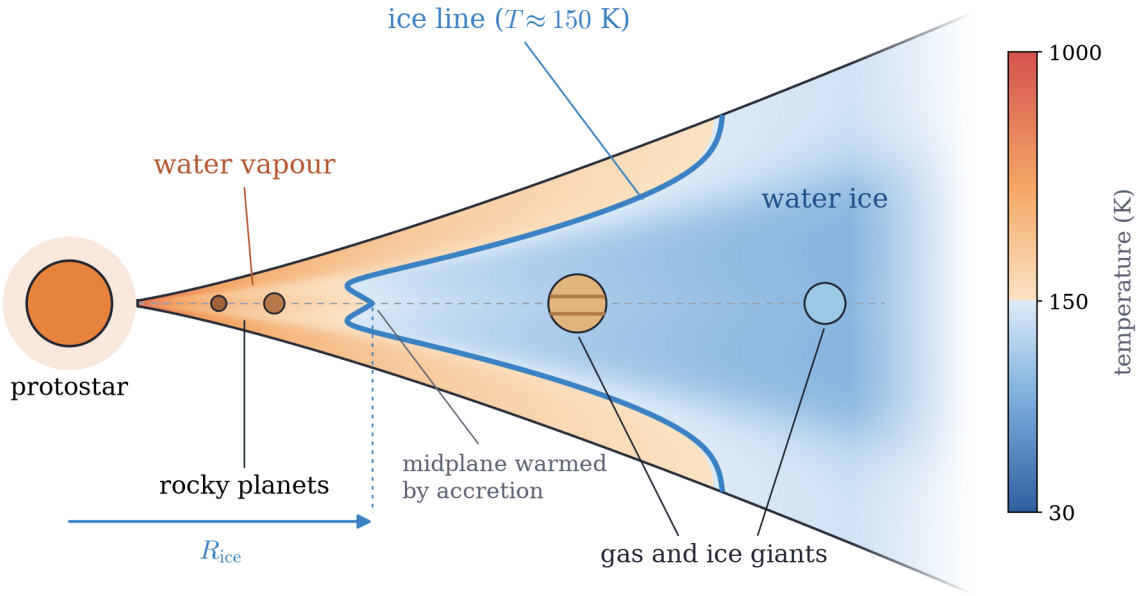


Figure 4.4: Ice line of a protoplanetary disk. The (water) ice line is where the temperature falls to about 150 K, inside which water ice sublimates into vapor. In the midplane, heat released by accretion keeps the gas warm out to $R_{\rm ice}$. Just above and below the midplane, that heating has faded, so the gas is cooler and ice survives closer to the star, which makes the kink. Higher still, starlight heats the disk's surface layer, so the ice line sweeps outward. Rocky planets form inside the ice line, and gas and ice giants form beyond it, where solid ice adds to the building material. Vertical scale exaggerated.

36 Describe the main elements of planet formation as we currently understand it.

Planet formation is thought to be a natural consequence of star formation. As a new star forms through the gravitational collapse of a gas cloud (see Question 27), conservation of angular momentum causes the cloud material to settle into a flat, rotating accretion disk around the star, called a **protoplanetary disk**. Protoplanetary disks are relatively short-lived, lasting ≤ 10 Myr, so planet formation must be a fast process.

Globally, there is a temperature and composition gradient such that the inner protoplanetary disk material is hot, and it cools with distance from the star. This forces relatively refractory materials (high condensation temperatures, e.g., metals) closer inward and volatile materials (low condensation temperatures, e.g., ices) further outward. These regions can be separated by the **ice/frost line** which is the distance at which a material condenses (see Fig. 4.4).

In the standard **core-accretion model**, planets are thought to have been built from the "bottom-up," starting from small (μm-sized) dust grains that begin to coagulate and collide, sticking together and building up into larger pebbles or planetesimals through electrostatic/Van der Waals interactions. At a certain point, a planetesimal can become massive enough to start gravitationally accreting solid material in a process called **runaway growth**, eventually growing into planet-sized objects. For objects closer to the (proto)star (i.e., within the ice line), the planets are expected to be mostly rocky, like the terrestrial planets. For objects further out (i.e., beyond the ice line), ices can accrete onto planetesimals, allowing the formation of larger planetary cores that are massive enough to accrete gas, forming the larger gas giants.

While the core-accretion model can explain most planets, it does not explain the existence of extremely large planets, so-called "Super-Jupiters," on wide orbits, because the disk would not have enough material

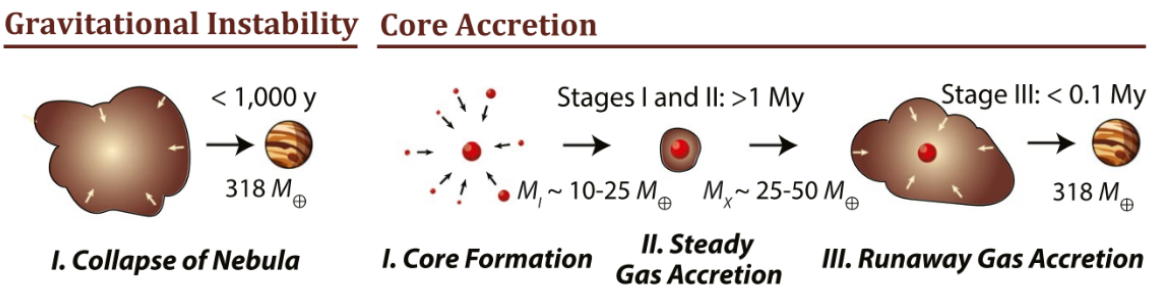


Figure 4.5: Core-accretion and gravitational instability ("bottom-up" and "top-down," respectively) models of planet formation.

and time to build these giant planets at these distances. The alternate **gravitational instability method** forms larger planets "top-down," with the disk fragmenting and directly self-collapsing, similar to star formation. See Fig. 4.5.

37 What is the 'meter-size barrier' and why does it matter for planet formation?

In the core-accretion model (see Question 36), once dust grains grow large enough, they will experience drag from the gas in the disk, which is strongest for meter-sized objects. Due to drag, these bodies will begin drifting toward the central star on very fast timescales, of order $\sim 1000\,\mathrm{yr}$. Additionally, by this size, particles move too fast for them to grow larger through collisions, as they will either bounce or fragment other particles at their speeds. In conjunction, these effects make planet formation at this point theoretically difficult, meaning planetesimal formation into larger, kilometer-sized objects at this stage must be very rapid to survive through the drifting timescales. Several mechanisms are proposed as solutions (pressure traps or streaming instability, for example).

The physical details: Gas in a protoplanetary disk has slightly sub-Keplerian velocities as it is partially supported against gravity by an outward pressure gradient. The solid (dust) particles are not pressure-supported and move with Keplerian velocities. As a consequence, they will experience a "headwind" from the gas, creating a drag effect. Very small particles are strongly coupled to the gas and will move with roughly the same velocity, so the drift is not very strong; for very large particles, the gas drag does not make a large enough effect to cause rapid inward migration. The radial drift velocity peaks characteristically at the meter scale.

38 Describe the architecture of the Solar System, i.e., basic characteristics, distance and size/mass scales for the Sun, planets, asteroids, comets, etc.

The **Sun** is a G2V star, with $R_\odot \approx 6.96 \times 10^8\,\mathrm{m}$ and $M_\odot \approx 1.989 \times 10^{30}\,\mathrm{kg}$. The Sun contains 99% of the mass of the Solar System, but only $\sim 1\,\%$ of the angular momentum.

The Solar System planets can be divided into the inner terrestrial and outer gas planets, which are separated by the asteroid belt. The terrestrial planets are:

- **Mercury**, which has $M \approx 0.05\,M_\oplus$, $R \approx 0.38\,R_\oplus$, an orbit with $a \approx 0.38\,\mathrm{AU}$ and eccentricity $e \approx 0.2$ (the highest of any planet), and 3:2 spin-orbit resonance;
- **Venus**, which has $M \approx 0.82\,M_\oplus$, $R \approx 0.95\,R_\oplus$, an orbit with $a \approx 0.72\,\mathrm{AU}$, extremely high temperatures due to a thick CO_2 atmosphere, and retrograde rotation;

- **Earth**, which has $M_\oplus \approx 6 \times 10^{24}\,\mathrm{kg}$ and $R_\oplus \approx 6.4 \times 10^6\,\mathrm{m}$, both of which are the largest of the terrestrial planets;
- and **Mars**, which has $M \approx 0.1\,M_\oplus$, $R \approx 0.53\,R_\oplus$, and an orbit with $a \approx 1.52\,\mathrm{AU}$.

Between the terrestrial and gas planets ($\sim 2 - 3.5\,\mathrm{AU}$) is the asteroid belt, containing approximately tens of millions of rocky small ($\sim 1 - 10^5\,\mathrm{m}$) bodies. These are mainly influenced by perturbations from Jupiter, which some asteroids are in orbital resonance with. The gas and ice giants, which have rock and ice cores with either substantial hydrogen and helium (Jupiter and Saturn) or ice (Uranus and Neptune) interiors, are:

- **Jupiter**, which has $M \approx 318\,M_\oplus \approx 0.001\,M_\odot$, $R \approx 11\,R_\oplus$, and an orbit with $a \approx 5.2\,\mathrm{AU}$;
- **Saturn**, which has $M \approx 95\,M_\oplus$, $R \approx 9.4\,R_\oplus$, and an orbit with $a \approx 9.6\,\mathrm{AU}$;
- **Uranus**, which has $M \approx 14\,M_\oplus$, $R \approx 4\,R_\oplus$, and an orbit with $a \approx 19.2\,\mathrm{AU}$;
- and **Neptune**, which has $M \approx 17\,M_\oplus$, $R \approx 3.8\,R_\oplus$, and an orbit with $a \approx 30\,\mathrm{AU}$.

Past Neptune is the Kuiper belt and Oort cloud, which are the primary reservoirs for comets (see Question 35).

39 What planet-plus-star configuration is most favorable for detecting a radial velocity signal? What about direct imaging and microlensing?

See Questions 31 and 32.

Radial velocities favor:

1. massive planets, for which the orbit about the barycenter induced by the planet on the host star is larger;
2. edge-on orbits, because the radial velocity signal scales as $\sin i$ (Eq. 4.1) so that inclination angles $i \approx 90°$ produce stronger signals while those near $i \approx 0°$ (i.e., face-on orbits) may be missed entirely;
3. and shorter periods, both because the signal strength increases with shorter periods (Eq. 4.1) and because it is observationally favorable to search for planets on shorter periods.

Since the signal scales as $m_\star^{-2/3}$ (Eq. 4.1), low-mass host stars give a stronger signal. Precision of the velocity measurements also depends on the stellar properties; most favored are bright, cool (F through M), slowly rotating, chromospherically quiet stars with many sharp absorption lines. Hot (A or hotter), rapidly rotating stars have few, (rotationally) broadened lines. Stellar activity is a strong noise floor at current capabilities.

Direct imaging favors:

1. large orbits, because we can better separate and resolve the planet from the light of the host star when the orbit is on the order of tens to thousands of AU;
2. large planets, especially Jupiter-sized or larger, because they retain more formation heat and cool more slowly, and thus emit significantly more thermal light;
3. and young systems, because (at current capabilities) we find direct imaging candidates emitting their own thermal light rather than reflected starlight, and young, newly-formed planets are still hot from formation and thus brighter in thermal emission.

Naturally, direct imaging favors nearby systems, because what matters is angular separation, and faint, low-mass hosts such that the planet-to-star contrast is less extreme.

Microlensing favors planets located near the Einstein ring radius of the host star, which is typically on the order of a few AU, because this is where their induced gravitational perturbation is the strongest. This method is uniquely powerful for detecting small planets at large orbits, which other detection methods are not sensitive to, and is the primary way to find free-floating planets.

40 Why is the surface of Venus so much hotter than Earth?

The surface temperature on Venus ($\sim 737\,\mathrm{K}$) far exceeds that on Earth ($\sim 288\,\mathrm{K}$) despite Venus absorbing less sunlight due to its albedo of ~ 0.75 from its global sulfuric-acid cloud cover. The difference is therefore entirely due to the greenhouse effect, which warms Venus by $\sim 500\,\mathrm{K}$ compared with $\sim 33\,\mathrm{K}$ on Earth (see Question 41).

Venus's atmosphere is $\sim 96\,\%$ CO_2 with a surface pressure of $\sim 92\,\mathrm{bar}$. At these pressures, pressure broadening and collision-induced absorption close the infrared windows through which Earth's surface cools, and SO_2, trace H_2O, and the clouds block the remaining gaps. The atmosphere is so optically thick at thermal infrared wavelengths that the planet radiates to space only from high, cold layers near T_{eq}.

Interestingly, Earth and Venus have comparable total carbon inventories, but the majority on Earth is locked in carbonate rocks formed through liquid water and plate tectonics. Venus, receiving $\sim 1.9\times$ Earth's incoming solar radiation, likely underwent a runaway greenhouse effect: water vapor reached the upper atmosphere, was photodissociated, and the hydrogen escaped to space, as evidenced by Venus's D/H ratio, ~ 100–$150\times$ Earth's. Without oceans to sequester it, the CO_2 remained in the atmosphere.

41 What is the equilibrium temperature of the Earth assuming it had no atmosphere? Sketch out how you would calculate it. Given your answer, why is Earth's atmosphere important for habitability?

Assume both the Sun and Earth radiate like a blackbody. The Sun's luminosity is given as:

$$L_\odot = 4\pi R_\odot^2 \sigma T_\odot^4. \tag{4.7}$$

At Earth, the flux received drops with orbital distance ($a_\oplus = 1\,\mathrm{AU}$), and flux absorbed is only across its cross-sectional area. Thus, the total energy received at Earth is

$$W_\oplus = \overbrace{\left(\frac{L_\odot}{4\pi a_\oplus^2}\right)}^{\text{Flux at } a_\oplus} \underbrace{\left(\pi R_\oplus^2\right)}_{\text{Area}} \overbrace{(1-A)}^{\text{abs. frac.}}. \tag{4.8}$$

Assuming that Earth will re-radiate this energy thermally across its entire surface area, it will have a luminosity of

$$L_\oplus = 4\pi R_\oplus^2 \sigma T_\oplus^4, \tag{4.9}$$

where $T_\oplus$ is the uniform surface temperature of Earth, assuming no atmosphere. Setting the powers received and emitted to be equal, we get the **equilibrium temperature**:

$$T_{\oplus,\,\mathrm{eq}} = \left(\frac{R_\odot}{a_\oplus}\right)^{1/2} \left(\frac{1-A}{4}\right)^{1/4} T_\odot. \tag{4.10}$$

Plugging in Solar values and using an average albedo $A \approx 0.3$, we get that $T_{\oplus,\,\mathrm{eq}} \approx 255\,\mathrm{K} = -17\,^\circ\mathrm{C}$. This is too cold for liquid water to exist on the surface. That Earth does have liquid water on its surface is of

clear importance for habitability, and it is the greenhouse effect from the atmosphere (see Question 40) that drives the temperature high enough for liquid surface water.

Note that $T_{\rm eq}$ is independent of planetary radius, so it can be applied to exoplanets even without known radii (assuming orbital and stellar properties are constrained).

42 Describe two difficulties with direct imaging of exoplanets.

One difficulty is the extreme contrast between star and planet. In reflected light (i.e., typically visible), the planet-to-star flux ratio is

$$C \approx A_g \, \phi(\alpha) \, (R_{\rm p}/a)^2, \tag{4.11}$$

where A_g is the geometric albedo, α is the phase angle, $\phi(\alpha)$ is the phase function (the fraction of full-phase brightness seen at α), $R_{\rm p}$ is the planet's radius, and a is the semi-major axis of the planet's orbit. For Jupiter at 5.2 AU, this is a few $\times 10^{-9}$ (and only $\sim 10^{-10}$ for an Earth analog). Thermal emission is more favorable only if the planet is hot: young, self-luminous giant planets that retain heat from formation reach $C \sim 10^{-4} - 10^{-6}$ in the near-infrared, whereas a mature Jupiter is far fainter. Reaching these contrasts requires adaptive optics, coronagraphs to suppress the stellar light, and post-processing (e.g., angular and spectral differential imaging). Consequently, imaging surveys are strongly biased toward young, massive planets observed in the infrared (see Question 39).

A second difficulty is angular resolution. A planet at separation a from a star at distance d appears at $\theta \approx a/d$, which must be compared to the diffraction- or seeing-limited angular resolution. Coronagraphs additionally block an inner working angle of a few λ/D.

These two difficulties are in tension: longer wavelengths improve the contrast for thermally emitting planets but degrade the resolution, since $\theta \propto \lambda$ (see Question 16).

43 Stars rotate and often have spots about the size of a Jupiter-mass planet. How is that spot-rotation and planet transits can be distinguished in a light curve?

There are several ways to distinguish the two in a light curve.

- **Morphology and duration**: A planetary transit is a sharp, brief event typically lasting a few hours, characterized by rapid ingress and egress phases and a relatively flat bottom. In contrast, stellar rotation produces a smooth, continuous, quasi-sinusoidal brightness modulation over days to weeks, on the order of the star's rotation period. The shape is smooth due to projection effects (the projected area when the spot is near the edge is small) and limb darkening.
- **Periodicity and stability**: Transits are strictly periodic, repeating with the planet's orbital period. Spots, however, are transient features. They evolve, shear due to stellar differential rotation, and decay over timescales of weeks to months, so that the spot-rotation modulation pattern changes shape and phase over time.
- **Wavelength dependence**: Transits are nearly achromatic, since they are caused by physical blocking of stellar light. Starspots are cooler than the surrounding photosphere, so their photometric signature is wavelength dependent (deeper in optical than infrared wavelengths), so light curves in multiple bands can help distinguish the two. Kaylee: I would be careful with this one since transits can also be wavelength dependent (e.g., how we find atmospheres, though I agree that it is not as exaggerated.)

Additionally, while spot-rotation can mimic transit-like variations, confirmation of a planet typically relies on ruling out stellar variability via multi-wavelength photometry, spectroscopy, or confirming the mass of the companion via radial velocities. we also note that when a transiting planet crosses a starspot, it produces a temporary bump in the transit light curve dip.

Chapter 5

Stars

44 How do we measure the temperature, luminosity, surface gravity, and metallicities of stars?

All four parameters are coupled and usually fit together with model atmospheres, but there are ways to disentangle the degeneracies.

44.1 Temperature

Generally, the temperature we can measure is the **effective temperature**, which is the temperature that would explain the star's luminosity via the Stefan-Boltzmann law:

$$T_{\text{eff}} = \left(\frac{L_\star}{4\pi R_\star^2 \sigma}\right)^{1/4}. \tag{5.1}$$

Practically, this is often measured by fitting a blackbody Planck curve, which approximates T_{eff}, or a custom spectral model, to the stellar spectrum. For stars hotter than the Sun, the broad profiles of the Balmer lines serve as excellent, pure thermal indicators because their excitation states are heavily temperature-dependent while remaining virtually unaffected by the star's chemical composition (though for A stars and hotter, the profiles are pressure broadened and become a surface gravity indicator). Conversely, for cool K- and M-type stars, the presence of molecular bands (such as titanium oxide, TiO) tracks temperature changes via molecular dissociation thresholds.

Color indices, such as $B - V$, are a simple photometric method to estimate temperature, since hotter stars peak at shorter wavelength (Question 5) and thus put more flux in B than V such that F_B/F_V is higher and

$$B - V = m_B - m_V = -2.5 \log_{10}\left(\frac{F_B}{F_V}\right) + C \tag{5.2}$$

is smaller.

44.2 Luminosity

If we have the stellar radius, we can use the effective temperature to solve for the luminosity as

$$L_\star = 4\pi R_\star^2 \sigma T_{\text{eff}}^4. \tag{5.3}$$

However, the radius is typically difficult or impossible to determine because stars are far away; it is possible with interferometric angular diameters or if the star is in an eclipsing binary system. If we instead measure the bolometric flux (taking into account bolometric and extinction corrections) and we know the star's

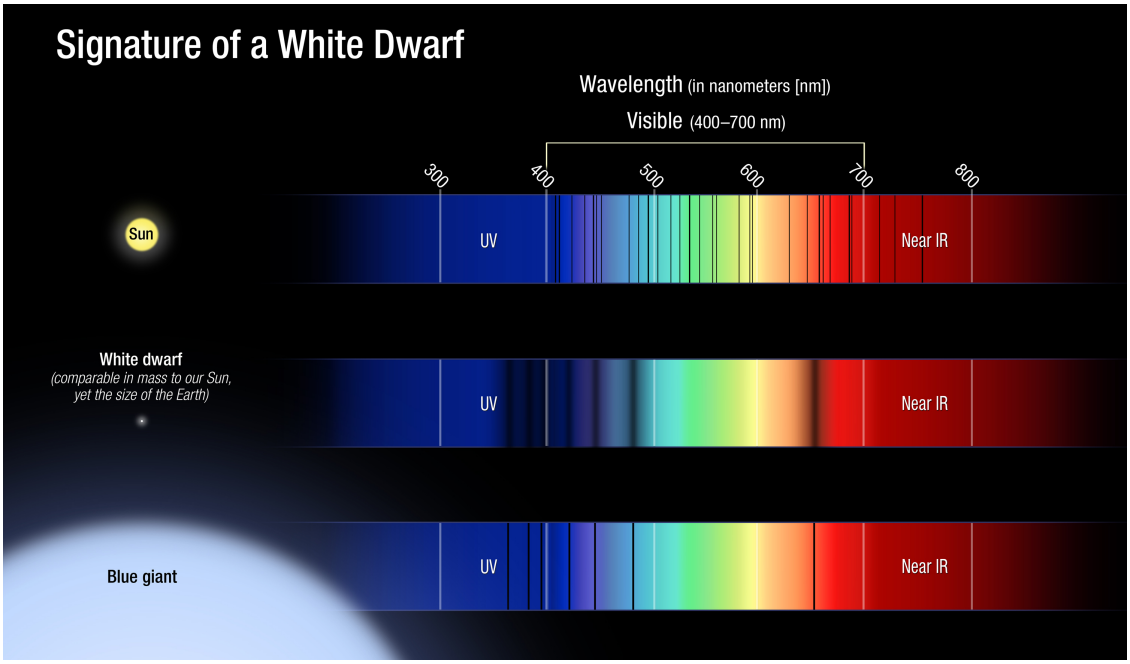


Figure 5.1: The spectrum of a white dwarf exhibits strongly pressure broadened lines.

distance (e.g., from Gaia parallaxes, Question 11), we can get the luminosity as (Question 12)

$$L_\star = 4\pi d^2 F_\star. \tag{5.4}$$

44.3 Surface gravity

Recall that one way lines are broadened is via pressure (Question 7). Stars with higher surface gravity have a greater pressure in the photosphere and the surrounding regions, leading to stronger broadening. High-pressure dwarf atmospheres display prominent, broad damping wings on strong lines, whereas low-density giant stars feature extremely narrow profiles for the same transitions. This effect is maximized in white dwarfs (see Fig. 5.1).

Surface gravity can also be constrained by measuring ionization balances of the same element (e.g., the ratio of Fe II/Fe I), as the ion fraction in equilibrium given by the Saha equation is highly sensitive to electron density, and thus the electron pressure and surface gravity. Lower gravity means lower electron pressure and thus more ionization, so Fe II lines are relatively stronger in giants. Note that if we know the stellar mass and radius, we can calculate the surface gravity directly as

$$\log_{10} g = \log_{10} g_\odot + \log_{10}\left(\frac{M}{M_\odot}\right) - 2\log_{10}\left(\frac{R}{R_\odot}\right), \tag{5.5}$$

where this comes from

$$g = \frac{GM}{R^2}. \tag{5.6}$$

Studying stellar pulsations, i.e., asteroseismology, can also constrain surface gravity by tracking the propagation of acoustic and gravity waves through the stellar interior.

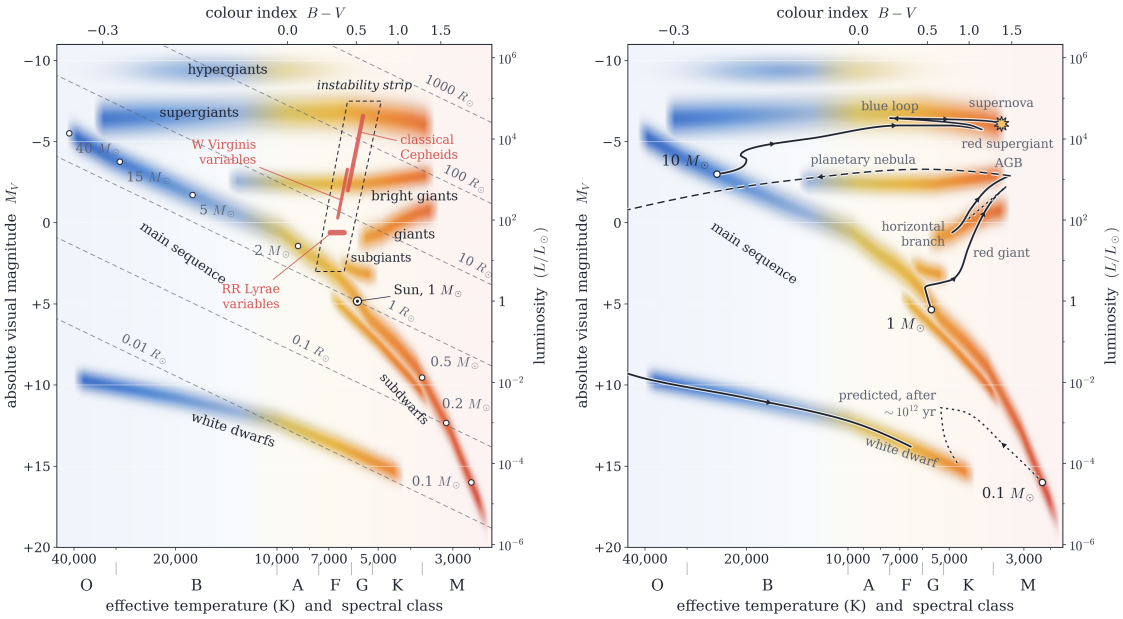


Figure 5.2: HR diagram and evolutionary tracks for 0.1, 1.0, and 10 solar mass stars.

44.4 Metallicity

Again, we look at spectra to determine stellar metallicity. The more of a metal there is, the stronger its absorption lines will generally be. However, analyzing metallicity introduces a fundamental degeneracy with temperature: an increased abundance of metal lines in the blue/UV region causes line blanketing, which absorbs high-energy light and re-radiates it at redder wavelengths, making a metal-rich star appear artificially cool. This is analogous to the degeneracy between age and metallicity in galaxy stellar populations.

To break this degeneracy in single stellar spectra, we can use high-resolution spectroscopy to construct a curve of growth from tens to hundreds of isolated, unblended atomic transitions, allowing us to solve for T_{eff}, $\log g$, and metallicity simultaneously given synthetic stellar atmosphere models. Molecular lines also can separate temperature and metallicity, because, for example, cool stars will exhibit strong molecular bands, but a hot star with high metallicity will only have strong atomic metal lines but no deep molecular bands since the molecules will have been dissociated.

Often, we look at the metallicity relative to the Sun, so that a star with ten times the solar iron Fe/H ratio would have

$$[\text{Fe/H}] = \log_{10}\left(\frac{N_{\text{Fe}}}{N_{\text{H}}}\right)_\star - \log_{10}\left(\frac{N_{\text{Fe}}}{N_{\text{H}}}\right)_\odot = 1.0. \tag{5.7}$$

The overall metallicity is denoted by M, or Z if mass-weighted, giving $[M/\text{H}]$ and $[Z/\text{H}]$.

45 Sketch the HR diagram and label major stellar phases. Draw evolutionary sequences for 0.1, 1.0, and 10 solar mass stars. Label the axes of the diagram.

See Fig. 5.2. The right panel shows evolutionary sequences for 0.1, 1.0 and 10 $M_\odot$ stars. A 0.1 $M_\odot$ star moves from a M-type MS star diagonally left and up, then curves down into the white-dwarf region. However, because such stars have lifetimes longer than the age of the universe, this track is theorized. A 1.0 $M_\odot$ star moves from a G-type MS star through the subgiant phase into the red giant phase. It then undergoes a helium flash to the horizontal branch, climbs the asymptotic giant branch (AGB), and then crosses left as a post-AGB planetary nebular path, reappearing on the white-dwarf cooling track. A 10 $M_\odot$ star moves from

a B-type MS star up and right into the supergiant phase, with a small hook at core-hydrogen exhaustion. It moves left and right as luminosity slightly increases (a "blue loop," during core helium burning), and there are very small fluctuations near the end of its life before going supernova due to carbon, neon, oxygen, and silicon burning (which take $\sim 10^3$ years, ~ 1 year, and ~ 1 day, respectively, compared to a few tens of Myr on the MS and a few Myr of helium burning).

46 What is the stellar main sequence? Why are stellar parameters on the main sequence not simple power-law functions of mass (hint: describe how the internal structure and primary energy generation mechanisms vary with mass along the main sequence)?

The **stellar main sequence** is the relationship between temperature and luminosity that all stars follow when they are fusing hydrogen into helium in their cores while in hydrostatic equilibrium. It forms a continuous band on the HR diagram (Question 45). The stellar parameters on the main sequence are not simple power-law functions of mass because of changes in the stellar internal physics across different mass regimes. The internal structure changes from fully convective at low mass ($\lesssim 0.35\,M_\odot$) to a radiative core and convective envelope and intermediate mass to a convective core and radiative envelope at high mass ($\gtrsim 1.3\,M_\odot$). These changes alter how efficiently energy is transported to the surface. Furthermore, with increasing mass and core temperature, the dominant fusion mechanism transitions from the **p-p chain**, which scales roughly as T^4, to the highly temperature-sensitive CNO cycle, which scales as T^{16} or steeper.

47 Describe at least three timescales relevant for stars and how they affect stellar evolution.

1. **Dynamical timescale**: how long does it take a perturbation to reach hydrostatic equilibrium? This is the characteristic time it takes for a sound wave to travel from the center to the surface of the star. If this it too long, the star can have a substantial non-HDE shape (see Betelgeuse, for example). $t_{\rm dyn} \sim R/c_s \sim 1/\sqrt{G\rho}$. Note the similarity to the free-fall time (Question 3); the time it takes for a star to feel a pressure change ($t_{\rm dyn}$) is the same order of magnitude as the time it would take to collapse if that pressure vanished ($t_{\rm ff}$). This timescale is typically short, on the order of hours to days for the Sun, so any deviation from HDE is rapidly fixed; however, when HDE fails catastrophically, such as in core-collapse SNe or stellar pulsations, this is the timescale on which it happens.

2. **Radiation (photon diffusion) timescale**: this is the time it takes for a photon to randomly walk its way out of the dense interior of a star, given by $t_{\rm rad} \sim R^2/\ell c$, where ℓ is the mean free path. This governs how efficiently radiative zones transport energy outward, influencing the overall temperature gradients and structure of the star.

3. **H-burning (nuclear) timescale**: how long does the hydrogen fuel in a star last? This is equal to the available fusion energy divided by the stellar luminosity, $t_{\rm nuc} \sim \eta Mc^2/L$, where η is the mass-to-energy efficiency factor, and it is the main driver of evolution. This is by far the longest timescale, dictating how long a star spends on the MS.

4. **Kelvin-Helmholtz (thermal) timescale**: once the star runs out of fuel, how long does it take to collapse; what is the time it takes for a star to radiate away its total gravitational binding energy at its current luminosity? This is equal to the gravitational potential energy divided by the stellar luminosity, $t_{\rm KH} \sim GM^2/RL$. This is the timescale of the pre-MS phase (see Question 27), when a protostar contracts and heats up before core hydrogen ignition, and it governs how a star adjusts thermally when nuclear fuel zones shift.

48 Why do more massive stars spend less time on the main sequence compared to lower mass stars? Derive the main-sequence lifetime of a 100 solar mass star given that the Sun has a main-sequence lifetime of 10 Gyr. Derive key scaling relations that define the main sequence.

More massive stars spend less time on the main sequence because, although they possess a larger nuclear fuel supply, scaling with mass, their luminosity scales cubically ($L \propto M^3$), so they burn through their fuel budget faster.

From Question 47, we have that the nuclear timescale is

$$t_{\rm nuc} \propto \frac{M}{L} \approx 10^{10}\,{\rm yr}\left(\frac{M}{M_\odot}\right)\left(\frac{L_\odot}{L}\right). \tag{5.8}$$

We can fit the main sequence mass-luminosity relation approximately as $M^3 \propto L$ such that

$$t_{\rm nuc} \sim 10^{10}\,{\rm yr}\left(\frac{M}{M_\odot}\right)^{-2}. \tag{5.9}$$

So a $100\,M_\odot$ star will have a MS lifetime of order

$$t_{\rm nuc}(100\,M_\odot) \sim 10^6\,{\rm yr}. \tag{5.10}$$

We can derive the mass-luminosity relation as follows.

Proof. From HDE (Eq. 1.24), we get

$$\frac{dP}{dr} \approx -\frac{GM(r)\rho(r)}{r^2} \Rightarrow P \sim \frac{GM\rho}{R} \sim \frac{GM^2}{R^4}. \tag{5.11}$$

Assuming that the stellar interior behaves like an ideal gas, we have that $P \propto \rho T$, so that

$$\frac{MT}{R^3} \sim \frac{GM^2}{R^4} \Rightarrow T \propto \frac{M}{R}. \tag{5.12}$$

The Virial Theorem (Eq. 1.15) also gives this relation:

$$2K + |U| = 0 \Rightarrow \frac{M}{m_{\rm H}}kT \sim GM^2/2R \Rightarrow T \sim \frac{M}{R}. \tag{5.13}$$

This tells us that more massive stars or more compact stars must be much hotter inside to prevent gravitational collapse.

The stored radiation energy density is aT^4, so that the total stored radiation energy inside a star is $E_{\rm rad} \sim aT^4R^3$. The photon diffusion time (Question 47) is set by the opacity κ, with

$$t_{\rm rad} \sim R^2\kappa\rho/c, \text{ with } \kappa\rho = \ell^{-1}. \tag{5.14}$$

The luminosity is essentially the total stored energy divided by the time it takes for that energy to radiate out,

$$L \sim \frac{E_{\rm rad}}{t_{\rm rad}} \sim \frac{T^4R^3}{R^2\kappa\rho} \sim \frac{(M/R)^4R^3}{\kappa MR^{-1}} \sim \frac{M^3}{\kappa} \Rightarrow L \propto \frac{M^3}{\kappa}. \tag{5.15}$$

□

49 Why do stars die? Describe the different stellar endpoints as a function of mass.

Stars die when they exhaust the nuclear fuel available for fusion in their cores, ending the outward thermal pressure that supports them against gravity. Note that when the core exhausts one fuel, it contracts and heats until the next fuel ignites; the star dies when this cycle can longer continue, either because electron degeneracy pressure halts the contraction before the next ignition temperature is reached, or the core is made of iron-group elements, for which fusion becomes endothermic. Which of these happens first, and what remnant is left behind, depends primarily on the initial mass.

- $M < 0.08\,M_\odot$: the core never reaches the temperature needed for sustained hydrogen fusion before electron degeneracy halts contraction. These are **brown dwarfs**, which simply cool and dim indefinitely.
- $0.08 < M < 0.35\,M_\odot$: these stars are fully convective, so the entire star is mixed and enarly all of its hydrogen is burned rather than just the core. Over $\sim 10^{12}$ yr, they heat up and become more luminous ("blue dwarfs"), moving up and left in the HR diagram, and directly become **(helium) white dwarfs**. There has not been enough time in the universe for this to happen yet.
- $0.35 < M < 8 - 10\,M_\odot$: after core hydrogen exhaustion, hydrogen burning moves to a shell around the inert helium core and the envelope expands into a red giant. The core contracts and heats until helium ignites at $\sim 10^8$ K; for $M \lesssim 2\,M_\odot$, the core is degenerate, so ignition is runaway (the **helium flash**). The star then burns helium in its core on the horizontal branch. Once core helium is exhausted, the carbon-oxygen core contracts, and the star ascends the asymptotic giant branch with alternating hydrogen and helium shell burning (**thermal pulsations**). Eventually, the strong winds strip the envelope, which is ionized by the hot exposed core as a **planetary nebula**, leaving behind a **(carbon-oxygen) white dwarf**. Between about 8 and $10\,M_\odot$, the core becomes hot enough to ignite carbon but becomes degenerate before neon, leaving behind an **oxygen-neon white dwarf**.
- $8 - 10 < M \lesssim 30\,M_\odot$: an iron core eventually develops, after successive burning stages, each shorter than the last. The star collapses once electron degeneracy fails, producing a Type II supernova and leaving behind a **neutron star**.
- $M \gtrsim 30\,M_\odot$: the same picture as before, but now neutron degeneracy also fails, and a **black hole** forms.

Note that for supernovae, whether a neutron star or black hole forms depends on factors like the metallicity of the star and whether the core exceeds the TOV limit of $\sim 3\,M_\odot$, the maximum mass beyond which neutron degeneracy pressure fails. High metallicity can prevent a star from forming a black hole, since the increased opacity causes greater mass loss before the SN event.

50 What are Type II and Type Ia SNe? Approximately how bright are they, what are the typical energy scales, etc.? Why is the latter important for cosmology?

Type II SNe are core-collapse SNe of massive stars. Their luminosity varies but in general $M \sim -17.5$ and a total of $\sim 10^{54}$ erg of energy are released.

Type Ia SNe are thermonuclear detonations of white dwarfs that have reached the Chandrasekhar limit. In the single degenerate case, the white dwarf accretes material from a companion star until it reaches the limit. In the double degenerate case, two white dwarf stars in a binary system merge. In the former case, the peak luminosity is theoretically always the same, since the SN event occurs once the mass has reached

the Chandrasekhar limit, and so these events are standard candles that can be used to calculate distances. The peak magnitude is $M \sim -19$ and 10^{51} erg of energy are released.

Even though Type II SNe release more energy, they have lower luminosities than Type Ia SNe; this is because 99% of the energy of a Type II SN is released in neutrinos escaping the collapsing core, via either electron capture ($p + e^- \to n + \nu_e$) or gamma-ray collisions mediated by electron-positron annihilation ($e^+ + e^- \to \nu_e + \bar{\nu}_e$, and the remaining $< 1\%$ becomes kinetic energy and light. Also, the light has to travel through a thick hydrogen envelope, since the core collapse happens deep inside a supergiant star. The envelope absorbs and diffuses the light, stretching the explosion out over months. For Type Ia SNe, there is no core collapse and thus no neutrino burst, so most of its total energy goes into kinetic energy and radiation; also, white dwarfs are compact, so the SN is unobstructed.

51 Stars can pulsate for a variety of reasons. Describe at least two types of variable stars and the underlying physical mechanisms driving this variation.

- **Cepheid variables**: these are high-mass ($4-20\,M_\odot$), evolved supergiant stars on the instability strip, useful for distance measurement through the period-luminosity relationship. Their radial pulsations are driven by the **κ-mechanism** in the partial ionization zone of helium ($\mathrm{He}^+ \rightleftharpoons \mathrm{He}^{2+}$). As the star contracts and heats up, doubly ionized helium accumulates, such that the rise in temperature and density increases opacity (κ) rather than decreasing it. The higher opacity traps radiation from the core, which heats the gas, increasing the internal pressure and forcing expansion. As the layer expands, it cools and recombines, and the opacity drops, so the trapped energy escapes and gravity pulls the layer back inward to restart the cycle, which has a period of order weeks to months. The lower-mass relative are **RR Lyrae variables**.

- **Mira variables**: these are highly evolved, low- to intermediate-mass ($0.8-3\,M_\odot$) stars on the asymptotic giant branch (AGB) undergoing large-amplitude radial pulsations, with periods of months to years. They are dynamical envelope oscillations driven by the κ- and γ-mechanisms in the outer and first helium partial-ionization zones. These pulsations cause brightness variations over up to three orders of magnitude; these variations are enhanced by chemistry, because their low surface temperatures in the extended, cool envelope allow simple molecules to form, obscuring optical light and shifting energy into the infrared. These pulsations are not the same as the thermal pulses that AGB stars undergo over $10-100\,\mathrm{kyr}$: on the asymptotic giant branch for solar mass star, once the helium shell detonates, the star temporarily expands but the hydrogen-burning shell deactivates, causing contraction (dimming). The star brightens as hydrogen fusion resumes, and over time, helium re-accumulates, repeating the cycle.

52 What is the stellar Initial Mass Function (IMF)? Plot the canonical Milky Way IMF. How do we measure the IMF in the Galaxy? Why is it difficult to measure?

Given a population of stars, the **Initial Mass Function (IMF)** is the probability density function of stellar masses during star formation. It is observed to be approximately (and strikingly) universal in the Milky Way. It differs from the **present-day mass function (PDMF)**, which describes the current distribution of stellar masses after evolution and mass loss.

The original formulation was by Salpeter (1955), which prescribed a single power law,

$$\xi(m) = \frac{dN}{dm} \propto m^{-2.35}. \tag{5.16}$$

However, this overpredicts the number of low-mass stars, so it only applies to $m > 0.5\,M_\odot$. The standard

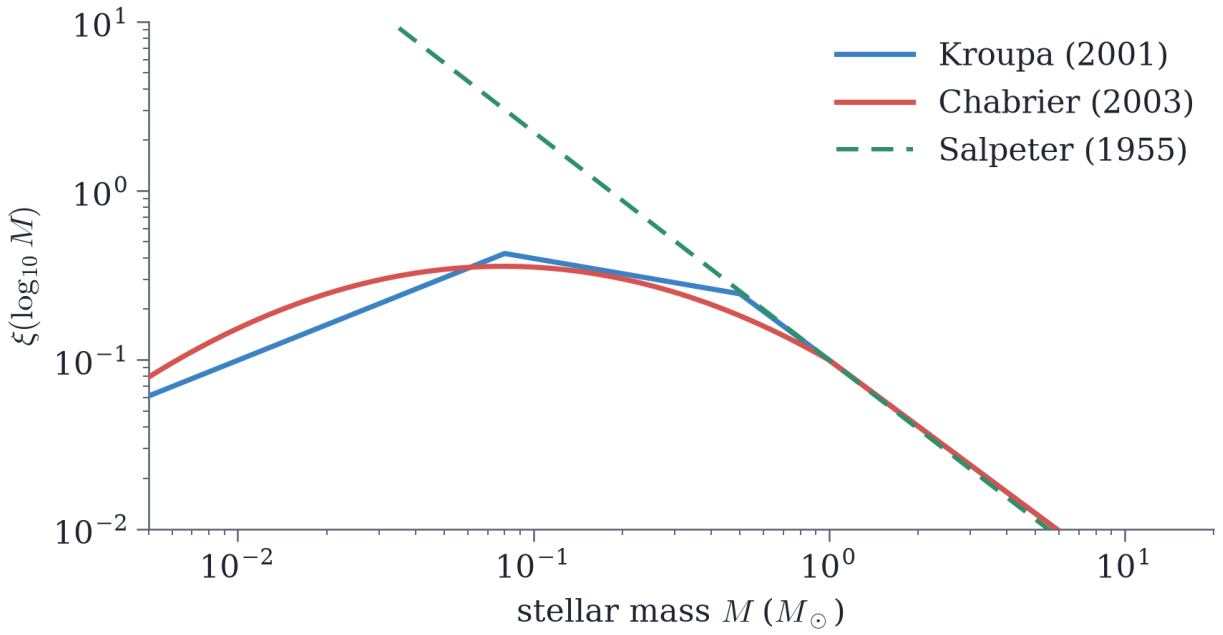


Figure 5.3: Comparison between the Salpeter, Kroupa, and Chabrier IMFs (not properly normalized).

reference for the Milky Way is the multi-part power law formulated by Kroupa (2001) as

$$\xi(m) \propto m^{-\alpha_i}, \tag{5.17}$$

where

$$\alpha_0 = 0.3 \pm 0.7, \qquad 0.01 \leq m/M_\odot < 0.08, \tag{5.18}$$
$$\alpha_1 = 1.3 \pm 0.5, \qquad 0.08 \leq m/M_\odot < 0.50, \tag{5.19}$$
$$\alpha_2 = 2.3 \pm 0.3, \qquad 0.50 \leq m/M_\odot < 1.00, \tag{5.20}$$
$$\alpha_3 = 2.3 \pm 0.7, \qquad 1.00 \leq m/M_\odot. \tag{5.21}$$

Another formulation is that by Chabrier (2003), which is a smooth distribution connecting the Salpeter IMF for $m \gtrsim 1\,M_\odot$ to a log-normal distribution centered around $0.08\,M_\odot$ for $m \lesssim 1\,M_\odot$. See Fig. 5.3 for a comparison of these three IMFs.

To measure the IMF, we must first measure the PDMF, particularly in young star clusters or in the solar neighborhood, where all stars formed from the same molecular cloud at roughly the same time. Stellar lifetime models (to account for high-mass stars that have already evolved and vanished), completeness corrections (to account for dim, low-mass stars and that we deal with small-number statistics at the high-mass end), and dynamical evolution considerations (to account for gravitational ejection of stars and mass segregation) must be used to convert the PDMF into the IMF. This is difficult because computing stellar masses (from photometry, maybe spectra) in the first place is hard, and there are many uncertainties in the modeling conversion. An important difficulty is also that many "stars" are actually unresolved multiple star systems; counting them as single objects flattens the low-mass end of the IMF. Thus, Chabrier, for example, gives separate single-star and system IMFs. And, there is no guarantee that the IMF is universal, so the solar neighborhood is not necessarily representative of the Milky Way IMF.

53 Name the primary stellar spectral types and their defining characteristics.

The approximate properties of the spectral types are listed in Table 5.1 and characteristics are listed in Table 5.2. Some characteristics of the spectral types are summarized here:

- O stars are the hottest ($\sim 40,000\,\mathrm{K}$) and emit most of their energy in the UV. They are so hot that they can exhibit ionized helium lines. They are extremely rare.
- B stars generally do not have ionized helium lines, but they do have weaker hydrogen lines than A stars.
- A stars exhibit the strongest hydrogen lines, because hotter stars start ionizing the hydrogen (lowering abundance) and cooler stars often do not populate the $n = 2$ level enough (the optical hydrogen lines are the Balmer series, describing transitions down to $n = 2$).
- In FG stars, metal lines become important (e.g., Ca II, Na I). Neutral metals are important for later stars.
- In KM stars, molecular bands start to appear as the star becomes cool enough for molecules (such as TiO and VO) to exist.

Class	T_{eff} **(K)**	**MS** M $(M_\odot)$	**MS** R $(R_\odot)$	**MS** L **(bol.,** $L_\odot$**)**	**Hydrogen lines**	**% of all MS stars**
O	$\geq$ 33,000	≥ 16	≥ 6.6	$\geq$ 30,000	Weak	0.00003%
B	10,000–33,000	2.1–16	1.8–6.6	25–30,000	Medium	0.12%
A	7,300–10,000	1.4–2.1	1.4–1.8	5–25	Strong	0.61%
F	6,000–7,300	1.04–1.4	1.15–1.4	1.5–5	Medium	3.0%
G	5,300–6,000	0.8–1.04	0.96–1.15	0.6–1.5	Weak	7.6%
K	3,900–5,300	0.45–0.8	0.7–0.96	0.08–0.6	Very weak	12%
M	2,300–3,900	0.08–0.45	≤ 0.7	≤ 0.08	Very weak	76%

Table 5.1: Harvard spectral classification of main-sequence stars.

54 Describe how $B - V$ color is related to a star's temperature.

$B - V$ color is the comparison of a star's magnitude measured through a blue ($B \approx 440\,\mathrm{nm}$) and visual ($V \approx 550\,\mathrm{nm}$) filter, $B - V = m_B - m_V$. Because stellar emission approximates a blackbody, Wien's law dictates that hotter stars peak at shorter wavelengths, so that $F_B > F_V$ and $m_B < m_V$ (see Questions 5 and 44). The more negative $B - V$ is, the bluer and hotter the star is. See Fig. 5.2.

55 What are the main energy production and transport mechanisms in stars?

55.1 Energy Production Mechanisms

In young, pre-MS stars prior to fusion ignition, the main energy production mechanism is gravitational collapse via the Kelvin-Helmholtz mechanism, i.e., the conversion of gravitational potential energy into thermal energy and radiation via the virial theorem (see Questions 27 and 47). In all other stars, it is fusion:

Spectral Type	Characteristics
O	Hottest blue-white stars with few lines Strong He II absorption (sometimes emission) lines. He I absorption lines becoming stronger.
B	Hot blue-white He I absorption lines strongest at B2. H I (Balmer) absorption lines becoming stronger.
A	White Balmer absorption lines strongest at A0, becoming weaker later. Ca II absorption lines becoming stronger.
F	Yellow-white Ca II lines continue to strengthen as Balmer lines continue to weaken. Neutral metal absorption lines (Fe I, Cr I).
G	Yellow Solar-type spectra. Ca II lines continue becoming stronger. Fe I, other neutral metal lines becoming stronger.
K	Cool orange Ca II H and K lines strongest at K0, becoming weaker later. Spectra dominated by metal absorption lines.
M	Cool red Spectra dominated by molecular absorption bands, especially TiO and VO. Neutral metal absorption lines remain strong.
L	Very cool, dark red Stronger in infrared than visible. Strong molecular absorption bands of metal hydrides (CrH, FeH), H_2O, CO, and alkali metals (Na, K, Rb, Cs). TiO and VO are weakening.
T	Extremely cool, Infrared Strong methane (CH_4) bands but weakening CO bands.
Y	Coolest ($\lesssim$ 500 K), Mid-Infrared Strong methane (CH_4) and ammonia (NH_3) bands; water ice clouds.

Table 5.2: Defining characteristics of the different spectral types.

- On the **main sequence**, low- to intermediate-mass stars ($M \lesssim 1.5\,M_\odot$) generate energy predominantly via the proton-proton chain. High-mass stars ($M \gtrsim 1.5\,M_\odot$) are dominated by the carbon-nitrogen-oxygen (CNO) cycle due to its steeper temperature sensitivity ($T^{16} - T^{20}$).
- **Post-main sequence** stars burn heavier elements in shells and cores (triple-α helium fusion, carbon fusion, etc.).
- **Stellar remnants**, i.e., white dwarfs and neutron stars, produce no significant nuclear energy; they simply cool over cosmic time, with neutrino emission dominating early thermal losses.

55.2 Energy Transport Mechanisms

The primary energy transport mechanisms are radiation and convection; there is also conduction, which is generally negligible in non-degenerate gas but dominates in degenerate cores (e.g., white dwarfs and degenerate helium cores of low-mass red giants). On the main sequence, the configuration of radiative and convective zones depends on stellar mass and local opacity:

- Low-mass stars ($M \lesssim 0.35\,M_\odot$) are fully convective, driven by high atmospheric opacities and low temperatures.
- Intermediate-mass stars ($0.35\,M_\odot \lesssim M \lesssim 1.5\,M_\odot$) have a radiative inner layer and a convective envelope, where lower surface temperatures increase opacity.
- High-mass stars ($M \gtrsim 1.5\,M_\odot$) have a convective core, driven by the intense central energy generation rate and steep temperature gradient of the CNO cycle, and a radiative envelope where high temperatures keep opacity low.

The criterion for convective instability is the **Schwarzshcild Criterion**, which dictates that a fluid layer in HDE becomes convectively unstable if the temperature gradient is steeper than the adiabatic temperature gradient,

$$\frac{dT}{dz} > \left(1 - \frac{1}{\gamma}\right)\frac{T}{P}\frac{dP}{dz}. \tag{5.22}$$

Intuitively, if a parcel of gas is pushed upward, it moves into a higher, cooler layer. Because this happens quickly, it cools at the adiabatic rate; if its surroundings have temperature drop off very steeply (faster than the adiabatic rate), the parcel will find itself becomes hotter and more buoyant than its new surroundings, it will keep rising, driving stellar convection.

56 What is the equation of state and what is required to compute it?

The **equation of state** relates pressure to other state variables, usually density and temperature. Formally, to compute this equation means to integrate the momentum flux over the appropriate momentum distribution function.

Often, stellar exteriors follow the ideal gas law, $P \propto \rho T$; for very massive stars, the radiation pressure is so strong that there is a $P \propto T^4$ term.

For white dwarfs and compact cores, the situation is more complicated. Fermions (e.g., electrons) cannot share the same state, so once their density becomes high enough they experience **degeneracy pressure**, which notably has no temperature dependence. Initially, $P \propto \rho^{5/3}$, and adding mass in this state decreases the radius: from HDE (Eqs. 1.24 and 5.11), $P \sim GM^2/R^4$, so

$$M^2/R^4 \sim P \sim \rho^{5/3} \sim M^{5/3}R^{-5} \Rightarrow R \propto M^{-1/3}. \tag{5.23}$$

As density increases, electrons become ultra-relativistic and $P \propto \rho^{4/3}$. With this weaker degeneracy pressure scaling, the internal energy density is

$$U \propto PV \propto \rho^{4/3}V \propto M^{4/3}R^{-4}R^3 = M^{4/3}R^{-1}, \tag{5.24}$$

the same scaling as the gravitational potential energy, $\Omega \propto -M^2/R$. The total energy is $E = U + \Omega = (k_1 M^{4/3} - k_2 M^2)/R \propto C/R$. As long as $k_1 M^{4/3} > k_2 M^2$, then $E > 0$ and the star expands until the electrons become non-relativistic again, restoring equilibrium. If M exceeds the **Chandrasekhar limit** ($1.4\,M_\odot$) at which the coefficient vanishes, the star loses dynamical stability against contraction because $E \to -\infty$ as $R \to 0$. What happens next depends on the core composition:

- In massive stars, as surrounding shells dump more iron onto the core, it approaches the Chandrasekhar limit. Electron capture ($e^- + p \to n + \nu_e$, reducing electron density) and endothermic photodisintegration of iron remove energy from the core, driving rapid gravitational collapse into a neutron star or black hole, and the outer layers rebound off the collapsing core to cause a Type II, Ib, or Ic SN (though the exact mechanism for overcoming the stalled shock is not fully settled).

- In accreting CO white dwarfs, compressional heating ignites carbon fusion just before the Chandrasekhar limit. Because the degenerate electrons dominate pressure independently of temperature, the core cannot thermally expand to cool itself. The thermonuclear runaway unbinds the entire star in a Type Ia SN, leaving no remnant behind. This thermal runaway mechanism is analogous to the helium core flash in low-mass red giants, except that in red giants the energy released during the runaway heats the degenerate core, eventually lifting the degeneracy and allowing the core to expand and settle into stable helium burning, with the overlying envelope absorbing and redistributing much of the energy, preventing a significant, immediate change in the observable surface luminosity.

In reality, many stars (such as the Sun as it ages) have "partial degeneracy" in their cores, as the ideal-gas regime starts transitioning into the degeneracy regime due to quantum effects starting to matter. You need a good model of stellar structure to calculate this accurately.

Neutron stars are kept stable by **neutron degeneracy pressure**, and they are much smaller since neutrons have a higher mass and as such have much smaller wavelengths (need to be packed tighter before quantum effects matter).

57 How do pre-main sequence stars evolve along the HR diagram?

See Question 27. Pre-MS stars start above and to the right of the ZAMS. As they contract and arrive at the ZAMS once fusion begins, they get less luminous (move downward) and hotter (move left). Their luminosity is entirely via gravitational contraction (the Kelvin-Helmholtz mechanism) rather than nuclear fusion. Sun-like pre-MS stars ($\lesssim 2\,M_\odot$) are observed as **T Tauri stars**, while larger ones ($2-8\,M_\odot$) appear as **Herbig Ae/Be stars**. The most massive stars do not have observable objects, because they contract so rapidly that they arrive at the ZAMS before finishing accreting and so are embedded in their natal accretion envelopes..

First is the Hayashi track, nearly vertically downward on the HR diagram. Energy transport at this stage is convective throughout the entire interior, maintained by high opacity, so the star contracts at a nearly constant effective surface temperature (though the core temperature continues to rise). As the radius decreases, the luminosity also drops. As the core heats up, the opacity drops, $\kappa \propto \rho T^{-7/2}$ (**Kramer's opacity law**, where opacity is approximated as produced by bound-free transitions, i.e., photoionization, and free-free transitions, i.e., thermal bremsstrahlung) allowing radiative transport to take over in the interior while the outer convective envelope remains. The protostar at this point moves to the Henyey track. This is characterized by contraction at nearly constant luminosity. Since $T_{\text{eff}} \propto (L/R^2)^{1/4}$, the protostar moves to the left, eventually arriving at the ZAMS once it is hot enough to fuse hydrogen.

Note that very low-mass stars ($M \lesssim 0.35 - 0.5\,M_\odot$) remain fully convective all the way to the ZAMS and thus never enter a Henyey track. Conversely, intermediate-to-high mass stars ($M \gtrsim 3-4\,M_\odot$) are born mostly radiative and lack an observable Hayashi phase.

58 What is the red giant branch (RGB)? What is the significance of the tip of the RGB?

The **red giant branch (RGB)** is the evolutionary track on the HR diagram populated by low- to intermediate-mass stars ($M \lesssim 2\,M_\odot$) after core hydrogen exhaustion. Structurally, an RGB star consists of an inert, electron-degenerate helium core surrounded by a hydrogen-burning shell and an expansive, highly convective envelope ("puffing out" to tens or hundreds of solar radii as surface temperatures drop). The tip of the RGB marks the end of this phase, when helium ignites in the core via the triple-α process at $T \sim 10^8$ K. Because the core is electron degenerate, its structure depends almost entirely on mass, so ignition only occurs once the core contracts and accumulates helium "ash" from the shell fusion reaching a nearly universal critical mass $\sim 0.45\,M_\odot$. The overlying hydrogen-burning shell luminosity is governed almost entirely by

core mass, so stars across a wide range of initial masses achieve virtually the same bolometric luminosity at the moment of ignition. And because degeneracy pressure is independent of temperature, the core cannot expand and cool in response to the energy release, so burning runs away in the **helium flash**. Thus, the tip of the RGB is a sharp, nearly constant-luminosity feature and is thus an accurate extragalactic standard candle, particularly in the I-band with $M_I \approx -4.0$, for measuring distances out to $\sim 50\,\mathrm{Mpc}$ (with JWST). The ultimate ceiling is blending and crowding; at these distances, stars are difficult to separate. Stars above $\sim 2\,M_\odot$ ignite helium before their cores become degenerate, so there is no helium flash, the core mass at which helium ignites depends on the star's total mass, and they do not share this constant luminosity. Thus, unlike Cepheids, which are a young Population I star found in active star-forming galactic disks, this method relies on stars in old halos and elliptical or dwarf galaxies, where all RGB stars are of low mass.

59 What do we mean by a star's metallicity and how does it affect its appearance? How does metallicity affect the internal stellar structure?

A star's **metallicity** (Z) is the mass fraction composed of all elements heavier than helium It is conventionally quantified relative to the Sun, for example as

$$[\mathrm{Fe/H}] = \log_{10}\left(\frac{N_{\mathrm{Fe}}}{N_{\mathrm{H}}}\right)_\star - \log_{10}\left(\frac{N_{\mathrm{Fe}}}{N_{\mathrm{H}}}\right)_\odot . \tag{5.25}$$

For a given mass and evolutionary stage, a star with higher metallicity appears cooler, redder, and slightly fainter for primarily two reasons. The first is line blanketing: abundant metal atoms and ions provide many absorption lines primarily at blue and UV wavelengths (see Question 44). This absorbed energy is redistributed and re-emitted as continuum flux at longer optical and IR wavelengths, shifting the observed color redward. The second is that greater atmospheric opacity due to metals traps radiation, causing the outer layers to expand, which lowers the photospheric effective temperature.

Metallicity also affects internal stellar structure. Metallicity also contribute to the opacity of stars, and higher interior opacity impedes radiative energy transport out of the core. Because heat is trapped more effectively, the tar expands to radiate its luminosity, resulting in a larger radius and lower mean density for a given mass on the main sequence. The onset of convection is governed by the Schwawrzschild criterion (see Question 55); note that the radiative flux passing through a layer is given by

$$F_{\mathrm{rad}} = -\frac{16\sigma T^3}{3\kappa\rho}\frac{dT}{dz}, \tag{5.26}$$

such that $dT/dz \propto \kappa$, so that the LHS of Eq. 5.22 increases with opacity. In other words, higher metallicity steepens the radiative temperature gradient, promoting convection. Consequently, higher-metallicity stars feature deeper outer convective envelopes, or they develop convective envelopes at higher stellar masses than their low-metallicity counterparts.

60 What are globular clusters, and how do they differ from open clusters? Consider ages, metallicities, spatial distribution in the Milky Way, internal density structure, etc.

Globular clusters are massive, roughly spherical, gravitationally tightly bound systems containing $10^4 - 10^6$ stars. They consist primarily of old, metal-poor Population II stars (see Question 61), with ages $\gtrsim 10-13\,\mathrm{Gyr}$ and [Fe/H] ~ -2.5 to -0.5. In the Milky Way, globular clusters are spherically distributed throughout the Galactic halo and bulge, on eccentric, highly inclined orbits that pass through the disk. They exhibit high central densities ($10^3 - 10^5\,\mathrm{stars/pc}^3$. Most globular clusters follow a King profile with a flat, constant-density central core, but some have undergone gravothermal core collapse to form a steep power-law central

cusp. Two-body relaxation timescales (the characteristic time it takes for random close encounters to deflect a star's velocity vector by roughly 90°, which is typically $\sim 10^{14}\,\mathrm{yr}$ in the MW but $\sim 10^8 - 10^9\,\mathrm{yr}$ in the compact globular cluster environment) are shorter than their age, so they have had time to undergo many relaxation cycles, driving pronounced mass segregation (i.e., the cluster acts like a thermal gas reaching equipartition of kinetic energy, $\langle E_{\mathrm{kin}} \rangle = \frac{1}{2} m \langle v^2 \rangle \approx$ constant, such that heavier stars sink deeper into the gravitational potential well and lighter stars migrate to wider orbits or get ejected).

Open clusters are less massive, more loosely bound, irregular collections of $\sim 10^2 - 10^4$ stars). They consist of young, metal-rich Population I stars, with ages $\sim 1 - 1000\,\mathrm{Myr}$ and [Fe/H] ~ -0.2 to $+0.2$. They are generally found in the thin disk of the Milky Way, where stars are actively forming. They have moderate-to-low central densities ($1 - 100\,\mathrm{stars/pc^3}$) with smoothly declining, flat-cored profiles. Because their binding energy is low, they do not undergo core collapse; instead, they dissolve over $10^8 - 10^9\,\mathrm{yr}$ due to stellar mass loss, disk shocking, and tidal encounters with giant molecular clouds.

61 What are the differences between Population I, II, and III stars? Where do we find these stars in the Milky Way? Which stars formed first?

These stars are ordered by metallicity, or equivalently age.

61.1 Population III stars

These were the first to form in the Universe, of order $\sim 100 - 400\,\mathrm{Myr}$ after the Big Bang ($z \sim 10 - 30$); they were the primary triggers for the Epoch of Reionization (Question 82).

- Metallicity: $Z = 0$, and [Fe/H] < -6. Pop III stars are composed strictly of primordial gas from BBN (hydrogen, helium, and trace lithium, see Questions 73 and 75), because they formed before repeated star formation, stellar nucleosynthesis, and stellar evolution released metals into the universe.

- Properties: without metal cooling channels, primordial molecular clouds were warmer, yielding a much larger Jeans mass (see Question 28); Pop III were thus predominantly massive $\gtrsim 10 - 100\,M_\odot$ and short-lived.

- Location: Pop III stars have not been observed in the present day and are largely theoretical. Their legacy exists primarily as stellar remnants (black holes) and the initial enrichment of the ISM.

61.2 Population II stars

These formed $\sim 10 - 13\,\mathrm{Gyr}$ ago during the early assembly of proto-galaxies.

- Metallicity: low metallicity, with $-3 \lesssim$ [Fe/H] $\lesssim -0.5$. Because they formed rapidly in environments enriched primarily by Type II core-collapse SNe, they are **α-enriched**, i.e., enriched in chemical elements with even atomic numbers built by sequential capture of helium nuclei inside stars, such as O, Mg, Si, Ca, and Ti.

- Properties: surviving Pop II stars are very old, with ages $\geq 10\,\mathrm{Gyr}$, and have low-mass $M \leq 0.8\,M_\odot$; any higher-mass Pop II stars would have already evolved and died.

- Location: extreme Pop II stars are found in the stellar halo, concentrated in globular clusters, with highly eccentric, random orbits (see Questions 60 and 64). There are also so-called intermediate Pop II stars in the Galactic bulge and thick disk, with properties intermediate between Pop II and Pop I stars.

61.3 Population I stars

These formed continuously over the last $\sim 8\,\mathrm{Gyr}$ and are still being formed today.

- Metallicity: high metallicity, with $-0.3 \lesssim [\mathrm{Fe/H}] \lesssim +0.3$, or $Z \approx 0.01 - 0.03$.
- Properties: Pop I stars have a broad age spread and contain stars across all spectral types. They also have high planet-formation fractions due to abundant metals.
- Location: they are concentrated in the thin disk of the Milk Way, particular in spiral arms and open clusters (see Questions 60 and 64). They move on nearly circular planar orbits around the Galactic center, where cold molecular gas continued to trigger active star formation.

62 What are pulsars? Explain one or more examples of their historical importance in astronomy.

Pulsars, or pulsating radio sources, are highly magnetized, rapidly rotating neutron stars (formed in the core-collapse SNe of massive stars with $M \sim 8 - 25\,M_\odot$). Pulsars possess intense magnetic dipole fields, and relativistic charged particles accelerate along magnetic field lines above the magnetic poles, emitting tightly collimated beams of radiation, primarily in radio but extending to optical, X-ray, or gamma ray wavelengths. Because the magnetic axis is tilted relative to the star's rotation axis, these beams sweep through space as the star spins. An observer detects a periodic pulse each time a beam crosses the line of sight.

Standard pulsars have spin periods of $P \sim 0.01 - 10\,\mathrm{s}$, but **millisecond pulsars (MSPs)** spin at hundreds of Hz ($P < 10\,\mathrm{ms}$). MSPs are "recycled" old neutron stars spun up through angular momentum transfer via mass accretion from a binary companion, and they have relatively weak magnetic fields and very low spin-down rates. This last characteristic makes them extremely regular, rivaling the precision of human-made atomic clocks.

There are several examples of their historical importance:

- **Neutron Star Confirmation**: The confirmed discovery of the first pulsar (PSR B1919+21) by Jocelyn Bell Burnell and Antony Hewish in 1967 provided the first observational proof that degenerate neutron stars, theorized in the 1930s, exist in nature (1974 Nobel Prize to Hewish).
- **First Evidence for GWs**: Russell Hulse and Joseph Taylor discovered the first binary pulsar (PSR B1913+16), which provided the first indirect evidence of gravitational waves. By tracking the orbital decay (inward spiral) of the two neutron stars over years of observations, they showed that the orbital energy loss matched GR predictions for quadrupole gravitational radiation (1993 Nobel Prize).
- **First Exoplanets**: Aleksander Wolszczan and Dale Frail detected periodic variations in the MSP PSR B1257+12, revealing the first confirmed exoplanetary system, years before the first planet around a MS star was found. one of these exoplanets is still the least massive exoplanet known.
- **Nanohertz GW Detectiom**: Pulsar timing arrays, such as NANOGrav, monitor dozens of ultra-stable MSPs across the Milky Way. Correlated timing residuals that follow the Hellings-Downs curve would provide evidence for a stochastic, nanohertz-frequency gravitational wave background, likely produced by inspiring SMBH binaries (see Questions 18 and 99).

63 Name two ways of measuring the mass of a star. Describe how these methods work.

Two methods for measuring the mass of a star are:

1. **Binary systems**: if we know the orbital period of the stellar system and the total semi-major axis, from Kepler's third law (Question 1), we can calculate the total mass of the system. The center-of-mass relation also gives

$$\frac{M_1}{M_2} = \frac{a_2}{a_1} = \frac{v_2}{v_1}. \tag{5.27}$$

   In practice, true physical separations (i.e., a) are projected onto the sky by an unknown orbital inclination i. The ideal scenario is with double-lined eclipsing binaries, because radial velocity curves yield the orbital period P and velocity amplitudes K_1, K_2, providing the mass ratio and minimum masses, and eclipse light curves constrain the inclination angle ($i \approx 90°$) to break the $\sin i$ degeneracy. Alternatively, for visual binaries, measuring the apparent orbital ellipse over time can yield i and the angular semi-major axes, which convert to physical distances a if the parallax or distance is known.

2. **Asteroseismology**: a star's mass affects any pulsations or acoustic oscillations traveling through it, driven by convective turbulence in their outer layers. High-precision photometric time series extract two characteristic frequency parameters which can be combined (with a spectroscopically determined effective temperature) to yield an estimate of the stellar mass.

3. **Astrometric microlensing**: like in exoplanet detection (Question 31), when an isolated foreground star passes almost directly in front of a distant background source, its gravitational field bends the light, creating two unresolved images whose centroid shifts across the sky. By measuring both the photometric brightening curve (which gives the Einstein crossing time t_E) and the sub-milliarcsecond astrometric deflection of the source's centroid, one measures the angular Einstein radius θ_E. Combining θ_E with the distances to the lens and source breaks all degeneracies to yield the lens star's mass directly as

$$M = \frac{c^2}{4G}\,\theta_\mathrm{E}^2\,\frac{D_\mathrm{L} D_\mathrm{S}}{D_\mathrm{S} - D_\mathrm{L}}. \tag{5.28}$$

   This is difficult; Gaia can potentially do this for a handful of events (see Klüter et al. (2020)). Or see McGill et al. (2023) for using this technique (with Gaia + Hubble) to get the first mass measurement of a single, isolated white dwarf.

Chapter 6

Galaxies

64 Describe the major structural components of the Milky Way, including their sizes and masses. How do we think these major components formed?

The three major structural components of the MW are the **disk**, **bulge**, and **halo**.

64.1 Descriptions

The disk is the most luminous component. Its size is difficult to define; stars can be seen out to $R \sim 15\,\mathrm{kpc}$ from the Galactic center, but gas (as observed in the 21 cm line) stretches out to $R \sim 25\,\mathrm{kpc}$. Relative to the radius of the disk, its thickness is small: most disk stars are less than 0.5 kpc from the disk midplane. We can in fact divide the disk into two components, the **thin disk** and the **thick disk**. The thin disk contains stars of all ages, including active star formation, and the thick disk contains stars older than $t \sim 5\,\mathrm{Gyr}$. Near the Sun, the distribution of stars falls off exponentially with distance z from the disk plane. For the thin disk, $n(z) = n_{\mathrm{thin}} \exp(-|z|/h_{\mathrm{thin}})$, with scale height $h_{\mathrm{thin}} \sim 0.3\,\mathrm{kpc}$; for the thick disk, $n(z) = n_{\mathrm{thick}} \exp(-|z|/h_{\mathrm{thick}})$ with $h_{\mathrm{thick}} \sim 1\,\mathrm{kpc}$. Typically $n_{\mathrm{thin}} \approx 10 n_{\mathrm{thick}}$.

The bulge is spheroidal, extending out $\sim 1\,\mathrm{kpc}$ radially (exponential scale length $\sim 700\,\mathrm{pc}$) and vertical scale height $\sim 350\,\mathrm{pc}$. We see it extend "above" and "below" the thin disk, most clearly in, say, IR wavelengths to minimize the effects of dust. At the center of the bulge is a small nucleus that is bright in radio wavelengths.

The halo is a roughly spherical distribution of stars about 100 kpc in radius, extending far beyond the bulge and disk. The halo has a similar luminosity as the bulge (at least in B band), but its stars are spread out over a volume $\sim 10^6$ times larger.

Some rough estimates of properties are given in the following table.

Component	$L_B\,(L_\odot)$	$M_\star\,(M_\odot)$	Height scale (kpc)	Radial scale (kpc)
Disk	19×10^9	5×10^{10}	$h_{\mathrm{thin}} \sim 0.3$, $h_{\mathrm{thick}} \sim 1$	$R_\star \sim 15$, $R_{\mathrm{gas}} \sim 25$
Bulge	2×10^9	10^{10}	~ 0.35	$\sim 0.7 - 1$
Halo	2×10^9	10^9	~ 100	~ 100

Table 6.1: Caption

The stellar populations in each component also differ. The thin disk contains relatively young stars, and is where most active star formation is. These stars tend to be rich in metals, with $Z \geq 0.01$ (Population I stars). The halo contains stars that are relatively old and metal poor, with $Z \leq 0.001$ (Population II stars). The thick disk contains stars with intermediate properties between Pop. I and II stars ("intermediate

populations" of stars). The bulge contains a mixture of young and old stars, i.e., a mixture of Pop. I and II stars.

Why are the halo and bulge B band luminosities so similar if the bulge has ~ 10 times more stellar mass? The halo is metal-poor, so even though the stars are old, the lack of metals makes them more transparent to blue light compared to some of the more metal-rich stars in the bulge (metal content can absorb blue light and re-radiate at longer wavelengths). Of course, the disk is so much more luminous because the stars are young and hot, which outweighs the effects of line blanketing by metals.

64.2 Formations

The halo is the oldest component of the galaxy, built primarily through major galactic mergers. The proto-MW began forming stars $> 13\,\mathrm{Gyr}$ ago; the last major collision was a merger with the Gaia-Enceladus dwarf galaxy. Its stars were sent into highly eccentric orbits, creating the loosely bound stellar halo. It is estimated that $\sim 30 - 40\,\%$ of the stellar halo formed in situ, from gas streams from infalling galaxies or galactic disk outflows. Naturally, the DM halo formed first via gravitational collapse of CDM density perturbations.

The disk formed through conservation of angular momentum as gas collapsed, spun faster, and flattened into the disk we see today. The original disk already had begun forming stars in the proto-MW; when mergers occurred, particularly the Gaia-Enceladus merger, the gravitational impact dynamically heated the existing disk, expanding it out vertically into the thick disk. For the past 8 to 9 Gyr, as the mergers subsided, the gas settled back into the quiet organized rotating plane, forming the thin disk. Gas continues to feed into the disk from the inside out, so the innermost regions formed stars first, while the outer regions still grow today.

The bulge (and bar) formed primarily through internal dynamical processes. The high density of gas and stars in the inner disk became gravitationally unstable, causing circular orbits to warp into an elongated bar (maybe $\sim$ 5 to 7 Gyr ago). As the bar rotated, stellar orbits became vertically unstable, so the disk "buckled" out of the galactic plane, creating the bulge.

65 How massive, and what sizes are the most massive and least massive galaxies?

Note here: the half-light radius is the distance from the center of a galaxy that contains half of its total emitted light.

The most massive known galaxy is ESO 146-5, with an estimated mass of $\sim 2.7 \times 10^{13}\,M_\odot$ and diameter of $\sim 200\,\mathrm{kpc}$. This is an elliptical galaxy in Abell 3827 cluster, with its mass measured by its gravitational lensing of two other galaxies (which means this dominated by dark matter). Its stellar mass is of order $10^{12}\,M_\odot$. Most massive galaxies have stellar mass of order $5 \times 10^1 1\,M_\odot$, with diameters $\sim 100 - 150\,\mathrm{kpc}$. Their DM halos can reach up to $10^{13}\,M_\odot$. This is in comparison to $6 \times 10^{10}\,M_\odot$ of stellar mass, $10^{11}\,M_\odot$ of baryonic mass, $10^{12}\,M_\odot$ total halo mass, and diameter $\sim 50\,\mathrm{kpc}$ for the MW.

The least massive galaxies have dynamical masses of order $5 \times 10^5\,M_\odot$. An example is Segue 2, an extremely small and faint MW satellite, with a half-light radius of about 34 pc and dynamical mass within the half-light radius of $\sim 1.5 \times 10^5\,M_\odot$. Its integrated luminosity is only $\sim 800\,L_\odot$ (lower than most globular clusters!), but this means it has a fairly high mass-to-light ratio of 650 (compared to about 10-50 for MW, this is solar masses/solar luminosities). If Ursa Major III/Unions 1 is confirmed as a dwarf MW satellite galaxy (as it is debated whether or not it is just a stellar cluster), it will be far lower, with only about 60 star systems, a half-light radius of 5 pc, and a mass-to-light ratio of 6500. Typical ultra-faint dwarf galaxies have a half-light radius of 20-500 pc, stellar masses of 10^3 to $10^5\,M_\odot$, and total DM halo mass of 10^8 to $10^9\,M_\odot$.

Some luminosities for scale: The most luminous elliptical galaxies have $M_V \sim -23\,\mathrm{mag}$, or $L_V \sim 10^{11} L_{V,\odot}$. MW and Andromeda have $M_V \sim -21$ or $L_V \sim 2 * 10^{10} L_{V,\odot}$. Dwarf ellipticals generally have $M_V > -18$ or $L_V < 10^9 L_{V,\odot}$, and dwarf spheroidals, the dimmest dwarf ellipticals, have $M_V > -14$ or

$L_V < 3 \times 10^7 L_{V,\odot}$. Just as M dwarfs are the most common type of star, dwarf galaxies are the most common type of galaxy!

66 What differentiates elliptical and spiral galaxies (give at least two examples)?

66.1 Morphology & dynamics

Of obvious importance is the different shapes of elliptical and spiral galaxies. Spiral galaxies consist of a flat rotating disk, a bright central core (the bulge), and, as their name suggests, curved spiral arms winding outward. Elliptical galaxies, as their name suggests, are more spheroidal/ellipsoidal and look relatively smooth. Most stars and much of the gas in spiral galaxies tend to orbit in one direction and in the flat plane of the disk, whereas in elliptical galaxies, stars can orbit in all different directions.

66.2 Star formation & population

Spiral galaxies contain lots of cold gas and dust, which act as stellar nurseries, or sites of active star formation. These are particularly common in the spiral arms, leading to, in the optical, bright blue clumps of intense star formation. The centers of spiral galaxies are redder and less clumpy than the outer parts, indicating the presence of older stars and less active star formation. On the other hand, elliptical galaxies lack reservoirs of cold gas and dust, so they do not experience much active star formation. They also tend to be redder than spiral galaxies, because their stellar populations are old and metal-rich (particularly for massive ellipticals). Though note that dwarf elliptical galaxies are typically metal-poor.

66.3 Formation

Elliptical galaxies are thought to form mostly through mergers, which can scramble disk orbits. Spiral galaxies grow through quiescent gas accretion.

66.4 Environment

Elliptical galaxies tend to dominate the cores of galaxy clusters, at the bottom of the clusters' gravitational potential wells. These massive ellipticals grow through mergers with smaller neighbor galaxies, featuring large, diffuse stellar halos and extended envelopes, and typically containing virtually no active star formation. Spiral galaxies are primarily found in the field, i.e., regions with low-density environments, or in smaller, gravitationally looser galaxy groups. These isolated, "gentle" conditions tend to have more gas available for active star formation, whereas the "harsher" conditions (stronger gravitational forces, crowded, high-velocity) of massive clusters can destroy spiral nature through ram-pressure stripping (which is when hot ICM blows out cold gas, shutting off star formation) and other processes.

66.5 Light profiles

The surface brightness of bright elliptical galaxies, $I(r)$, tend to follow $\log I$ decreasing as $-r^{1/4}$. Spiral galaxies fall off exponentially, with $\log I$ decreasing as $-r$. Note that the bulges of spiral galaxies can frequently be fit with the same 1/4 law that applies to elliptical galaxies.

67 What determines a galaxy's color and star formation rate? What are the main differences between the optical spectra of blue and red galaxies?

Visible light emitted by a galaxy is typically produced by stars, with an absorption line spectrum, and by hot gas, with an emission line spectrum. Most starlight comes from a small number of very luminous stars, not

from the massive number of M dwarfs (note a single MS O star with $M_V \ -5$ produces as much visible light as 100 million MS M stars each with $M_V \ +15$). In spiral and irregular galaxies with active star formation, the brightest stars are young, hot O and B MS stars. In elliptical galaxies, where star formation has usually stopped long ago, the brightest stars are red giants. Thus, elliptical galaxies are relatively cool and redder in color than spiral and irregular galaxies, whose bright stars are hot. Indeed, a galaxy will only appear blue if it is actively and continuously forming new stars, because O and B type stars only live for up to tens of millions of years.

The most important factor in determining the SFR, which in turn determines the color as above, is the availability of cold, dense molecular gas. If there is a large gas reservoir, this triggers gravitational collapse, leading to high SFRs. Without gas, SF is quenched. Other factors include

- AGN feedback, which can heat up or eject cold gas, suppressing star formation.
- Galaxy collisions and mergers, since when two gas-rich galaxies collide, gas is compressed, causing a temporary spike in SFR (i.e., a starburst). Note that this consumes the gas quickly and accelerates the transition to red elliptical galaxies.
- The cosmic environment, since galaxies in dense galaxy clusters can lose gas to ram-pressure stripping (see Sect. 66.4) or strangulation (the cluster blocks the inflow of new gas).

Red galaxies typically have strong absorption lines and no emission lines, because they lack the hot stars needed to ionize gas. They can have emission lines if an AGN is present. The integrated light of all the stars in an elliptical galaxy produces a spectrum similar to a star of spectral type K. Blue galaxies have both strong absorption and moderately strong emission lines, like Hα and ionized oxygen, since O and B stars can ionize surrounding gas clouds. The metal absorption lines (sodium, mag, calcium) are weaker than in red stars, because they are highly ionized and these transitions have lines outside the optical spectrum, but they have, for example, strong Balmer absorption. Integrated light of stars in blue galaxies tend to have similar absorption lines to a star of spectral type F or G.

A primary feature in the continuum component is the 4000 Åbreak that differs in strength between types of galaxies. This break is the transition from higher flux at wavelengths above 4000 Åand lower flux at shorter wavelengths, and it caused by blanket metal absorption of blue and UV light, and also near tail-end of Balmer series. Older stellar populations (like in red galaxies) feature a much stronger 4000 Åbreak because metals in atmospheres of the old stars absorb the bluer wavelengths. A weak break indicates active, ongoing SF (i.e., blue galaxies).

This is all ignoring dust, which can complicate the picture. Some galaxies look red because they are trapped behind dust, which scatters blue light. This can be confirmed by checking the Balmer decrement, which is the Hα/Hβ emission ratio. If a galaxy has red continuum but still has strong Hα emission lines, it is likely actually a blue galaxy obscured by dust.

68 What physical processes are believed to influence galaxy growth over cosmic time?

There are many such processes.

- Gravitational collapse is the scaffolding for structure and galaxy formation. Fluctuations in the density of the early universe create slightly overdense regions where dark matter gathers by gravity. These gravitational potential wells (DM halos) pull in gas to grow galaxies.
- Galaxies continuously grow through gas accretion, either through cold-mode or hot-mode accretion of gas from the surrounding web, ultimately leading to star formation. Cold-mode accretion dominates in halos below $\sim 10^{12}\,M_\odot$; gas flows in along dense streams without shock-heating to the virial temperature, which efficiently feeds star forming disks. For halos above $\sim 10^{12}\,M_\odot$, a stable virial shock

forms, causing hot-mode accretion and forming a hot CGM atmosphere; gas must cool out of a hot halo, and these cooling times can be long, particularly at low z and low densities.

- Star formation converts gas accreted onto galaxies into stars, but it is broadly slow and inefficient.
- Smaller galaxies and clumps of gas collide and merge over billions of years under mutual gravity; this is hierarchical merging. Mergers can trigger starbursts and AGN episodes, and they can transform disks into spheroids (elliptical galaxies). Minor mergers dominate the size growth of massive ellipticals after $z \sim 2$. Over most of cosmic time, mergers contribute less to the total stellar mass growth than in situ SF, except in the most massive systems.
- Stellar feedback, through SNe, stellar winds, radiation pressure, and photoionization, injects energy into galaxies and can blow out gas through galactic winds, reducing SF. These dominate in low-mass galaxies. Feedback also produces the galactic fountain and recycled accretion, and it enriches the CGM/IGM with metals.
- AGN feedback and AGN outflows quench SF in massive galaxies.
- Environmental stripping in crowded galaxy clusters can strip away gas and alter galaxy shapes through processes like ram pressure stripping, strangulation, and tidal interactions (see Questions 66.4 and 67). This often leads to satellite quenching, where satellites of a larger host galaxy stop forming new stars.
- There is also secular evolution, which are slow, internal changes to a galaxy, often driven by non-axisymmetric structures such as bars and spiral arms. These exert torques that can that can drive gas inward, which can fuel central starbursts and, potentially, black hole growth, and over time build up disky pseudobulges. The resulting central mass concentration can in turn stabilize the gas disk against fragmentation, suppressing further star formation (morphological quenching). Unlike other processes, morphological quenching does not require gas stripping; it can cause galaxies to be filled with older, redder stars even though gas is still actively flowing inward.

69 What influences the rotation curve in a galaxy? What does it mean for a rotation curve (or velocity dispersion profile) to be "isothermal"?

A galaxy's rotation curve is primarily influenced by the distribution of mass, both luminous matter and dark matter. If we assume matter is on circular orbits in a roughly spherical mass distribution, which we typically do for rotation curves, balancing the centripetal and gravitational accelerations gives

$$\frac{v(R)^2}{R} = \frac{GM(R)}{R^2} \quad \Longrightarrow \quad v(R) = \sqrt{\frac{GM(R)}{R}}, \tag{6.1}$$

where $v(R)$ is the orbital speed of a star at radius R and $M(R)$ is the mass contained inside a sphere of radius R centered at the galaxy center. For a flattened disk this is only approximate, and in practice the measured curve is also affected by inclination, non-circular motions (e.g., from bars), and pressure support in the gas or stars (asymmetric drift).

A system is **isothermal** if its velocity dispersion σ is the same at all radii. This is the analogue of an ideal gas at uniform temperature, with σ^2 playing the role of kT/m: the stars have a Maxwellian velocity distribution with the same σ everywhere. A self-gravitating system in this state is an isothermal sphere. The general solution has a roughly constant-density core and falls off as $\rho \propto r^{-2}$ at large radii. The simplest case is the **singular** isothermal sphere, where

$$\rho(r) = \frac{\sigma^2}{2\pi G r^2}. \tag{6.2}$$

This gives

$$M(R) = \int_0^R 4\pi r^2 \cdot \frac{\sigma^2}{2\pi G r^2} dr = \frac{2\sigma^2 R}{G} \quad \Longrightarrow \quad v(R) = \sqrt{2}\sigma, \tag{6.3}$$

so that $v(R)$ is constant. Thus an isothermal halo produces a flat rotation curve. A flat rotation curve therefore means the enclosed mass keeps growing linearly with radius, since $M(R) = v(R)^2 R/G$. The mass has not converged at the last measured point, so we obtain only a lower limit on the total mass. That the curves stay flat well beyond the visible disk is a key piece of evidence for extended dark matter halos.

Elliptical galaxies are different. Massive ellipticals rotate slowly compared with spirals of similar size (though many lower-mass ellipticals are fast rotators). Their absorption lines are, however, much broader than thermal broadening in stellar atmospheres would produce. The broadening comes from the stellar velocity dispersion σ. In elliptical galaxies, stars are not on near-circular orbits like we assume to make a rotation curve. Instead, they are on eccentric, randomly oriented orbits. Along any LOS through the galaxy, we see stars with a range of radial velocities, hence a broad line. Ellipticals are therefore dispersion-supported rather than rotation-supported, and the rotation-curve method does not apply.

Instead, we can estimate their masses with the virial theorem. For a system in equilibrium, the total mass is

$$M \approx \frac{2.5\langle v^2 \rangle r_h}{G}, \tag{6.4}$$

where r_h is the three-dimensional half-mass radius and the factor 2.5 comes from $r_h \approx 0.4\, r_g$, with r_g the gravitational radius. If the velocities are isotropic, the mean-square speed is related to the observed line-of-sight dispersion by

$$\langle v^2 \rangle = 3\sigma^2. \tag{6.5}$$

Alternatively, if the measured dispersion profile $\sigma(r)$ is approximately flat, i.e. independent of radius, we can model the galaxy as an isothermal sphere. This is a stronger assumption than the virial estimate, which needs only the galaxy-averaged $\langle v^2 \rangle$, but it gives the enclosed mass at any radius, $M(R) = 2\sigma^2 R/G$.

These estimates are harder to make reliable than rotation-curve masses. We observe the projected half-light radius, not the three-dimensional half-mass radius. Dark matter makes the mass-to-light ratio vary with radius, so half-mass and half-light radii are different.

70 At what redshift did the cosmic star formation rate peak? Why, and how do we know this?

Cosmic SFR peaked around $z \sim 2 - 3$, which is about 3 to 3.5 Gyr after the BB or around 10 to 11 Gyr ago. This was when the universe had an abundance of cold molecular gas. Following this peak, SFR density declined by roughly one OOM to present day levels as gas reservoirs depleted and feedback quenched SF. Half of the stellar mass observed today was formed before $z \approx 1.3$.

Before $z \sim 3$, most accretion was cold-mode accretion (see Sect. 68), because typical DM halos were smaller and cosmic gas densities were higher so that cooling times were shorter. Thus, cooling was not required to begin SF. As the universe continued to expand, density dropped and accretion transitioned to primarily hot-mode accretion. This made accreted gas harder to cool once inside DM halos as it gets shock-heated to millions of degrees. Also, gravitational mergers became less frequent as cosmic expansion separates clusters (and within clusters, galaxies move too fast to merge), which cut off these gas-rich disruptions that trigger starbursts. Additionally, the huge amount of SNe from young stars being formed blew large amounts of gas out of galaxies, and AGNs grew rapidly, which further heated gas and suppressed hot-mode accreted gas from cooling down.

UV light from distant galaxies is a direct tracer of the current SFR. The period of peak SFR was very dusty, and dust absorbs UV and re-radiates it in the IR, so IR telescopes also trace SFR. If we measure galaxy redshifts and SFR through those tracers, we can plot SFR density ($M_\odot$ of SF per year per cubic

Mpc) against redshift in a so-called Madau-Dickinson plot. Such plots, as in Fig. 6.1, inform us of the timing of the peak. To confirm why it peaked and dropped, telescopes like ALMA look at molecular gas (primarily CO) and find that galaxies at $z \sim 2-3$ had much more cold molecular gas than galaxies today.

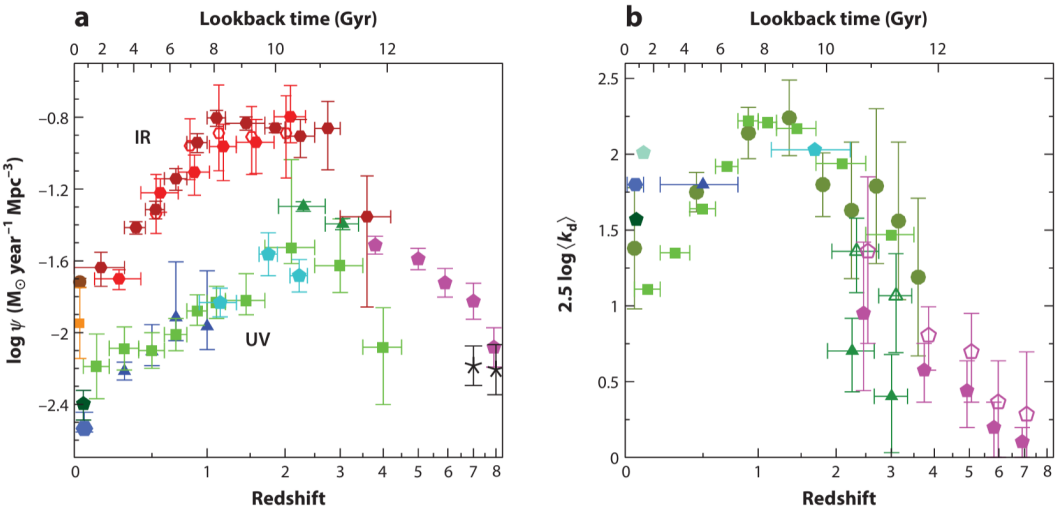


Figure 8

(*a*) SFR densities in the FUV (uncorrected for dust attenuation) and in the FIR. The data points with symbols are given in **Table 1**. All UV and IR luminosities have been converted to instantaneous SFR densities using the factors $\mathcal{K}_{\rm FUV} = 1.15 \times 10^{-28}$ and $\mathcal{K}_{\rm IR} = 4.5 \times 10^{-44}$ (cgs units) valid for a Salpeter IMF. (*b*) Mean dust attenuation in magnitudes as a function of redshift. Most of the data points shown are based on UV spectral slopes or stellar population model fitting. The symbol shapes and colors correspond to the data sets cited in **Table 1**, with the addition of Salim et al. (2007) (*cyan pentagon*). Two versions of the attenuation factors are shown for UV-selected galaxies at $2 < z < 7$ (Reddy & Steidel 2009, Bouwens et al. 2012a) (offset slightly in the redshift axis for clarity): one integrated over the observed population (*open symbols*), the other extrapolated down to $L_{\rm FUV} = 0$ (*filled symbols*). Data points from Burgarella et al. (2013) (*olive green dots*) are calculated by comparing the integrated FIR and FUV luminosity densities in redshift bins, rather than from the UV slopes or UV-optical spectral energy distributions. Abbreviations: FIR, far-infrared; FUV, far-UV; IMF, initial mass function; IR, infrared; SFR, star-formation rate.

Figure 6.1: From Madau & Dickinson (2014).

71 What gives rise to the spiral arms and bars in spiral galaxies?

Spiral arms are driven by the gravitational field of spiral galaxies. Originally, it was thought that because spiral galaxy disks are in differential rotation, with angular velocity decreasing with radius, galactic rotation shears compressed regions into spiral patterns. However, this gives way to the winding problem: if arms were material features carried along by gas and stars, then differential rotation would wind them up within a few rotations, or a few hundred Myr, but observed arms are not that tightly wound. Instead, Lin & Shu (1964) proposed that they are stable, rigid density waves that sweep through the disk, packing gas and stars tighter together. They are a quasi-stationary pattern that rotates rigidly at a single pattern speed Ω_p, and as gas clouds collide with these high-density waves, they compress, triggering star formation in the arms.

Bars form when stars in a thin galactic disk experience non-axisymmetric gravitational instabilities, so that circular orbits destabilize and stretch into ellipses. This creates a coherent bar structure.

72 How does surface brightness (flux per solid angle) of a galaxy change with distance (ignore cosmological effects)?

Ignoring cosmological effects, surface brightness Σ does not change with distance. Surface brightness is apparent brightness/flux density per unit angular area of an extended object (in this case, a galaxy), i.e., F/Ω. As the distance to a galaxy increases, the flux density decreases as d^{-2}, but the angular area of the galaxy also decreases by d^{-2}. Specifically, the total observed flux for a galaxy of intrinsic luminosity L and physical surface area A at distance d goes as

$$F = \frac{L}{4\pi d^2} \propto d^{-2}, \tag{6.6}$$

and the solid angle subtended by the galaxy goes as

$$\Omega \approx \frac{A}{d^2} \propto d^{-2} \tag{6.7}$$

Fundamentally, bolometric surface brightness corresponds to the integrated specific intensity $I = \int I_\nu d\nu$. Recall from Lecture 1 of Radiative Processes that specific intensity is conserved along rays regardless of distance.

Note that this invariance only holds when the object is resolved. Once the angular size is below the instrumental PSF or the atmospheric seeing, the light is spread over a fixed solid angle, causing the apparent surface brightness to drop.

Not ignoring cosmology, surface brightness dims proportional to $(1+z)^{-4}$. One factor is from energy loss of photons as they are redshifted. Another factor is from the reduced arrival rate of individual photons due to cosmological time dilation. The last two factors are from the fact that the galaxy subtends a larger angle on the sky than we would expect from its comoving distance alone, i.e.,

$$\Omega \propto \theta^2 \propto \left(\frac{D}{d_\mathrm{A}}\right)^2 = \frac{D^2(1+z)^2}{r_c^2} \propto (1+z)^2, \tag{6.8}$$

where D is the physical size, d_A is the angular diameter distance, and r_c is the comoving distance. This is why low surface brightness features at high redshift are so hard to detect.

73 What are the main sources of "heavy elements" ($Z > 3$) in the Universe? Describe the processes that can change their abundances over time within a galaxy. Consider both elements locked in stars and elements in the gas. What do we learn from measuring the relative abundances of the heavy elements?

There are several processes that produce the heavy elements in the Universe. Note in the below that the **r-process** is rapid neutron capture and the **s-process** is slow neutron capture. In the r-process, a massive wave of neutrons bombards seed nuclei (such as iron-56), which absorb dozens of neutrons before they have the chance to β-decay (neutron in the nucleus becomes a proton, emitting an electron and an electron antineutrino, increasing atomic number by 1 and keeping the mass number the same),, pushing them into unstable, neutron-rich territory. Once the neutron flood subsides, these overloaded nuclei cascade backwards via β-decays into stable heavy elements, such as gold, platinum, uranium, and thorium. In the s-process, the neutron flux is low, so seed nuclei absorb a neutron and undergo β-decay before another neutron comes along. This means the atom moves steadily along the stable band of the periodic table, producing elements such as barium, strontium, and lead.

- **Big Bang nucleosynthesis (BBN)**: produced lithium ($Z = 3$) and trace amounts of beryllium

($Z = 4$). Intense heat and pressure in the first few minutes of the Universe allowed proton and neutron fusion, but expansion and cooling was too fast to form significant amounts of $Z > 3$ elements.

- **Core collapse (Type II) SNe:** produce oxygen through iron ($Z = 8$ to 26). See Questions 50 and 56. Stars with $M \gtrsim 8\,M_\odot$ fuse progressively heavier elements in their cores until iron, when fusion becomes endothermic, and once the core collapses and explodes as a Type II SNe, these elements (primarily α-elements (see Question 61) are ejected into space. Even heavier elements may be formed from the burst of neutrons (r-process), though this is less certain.

- **Thermonuclear (Type Ia) SNe:** produce iron group elements. See Questions 50 and 56. CO white dwarfs reaching runaway thermonuclear ignition release heavy elements that have been converted from carbon and oxygen during the explosion itself. Intermediate-mass elements like silicon and sulfur are produced in the intermediate layers, and the outermost layer is blown away so fast that the carbon and oxygen escape unburnt. These produce more than half of all the iron in the universe.

- **Low-mass stars**: produce up to carbon and oxygen through triple-α fusion of helium, but are never hot enough to fuse heavier elements. The s-process is important in the evolved stages of their lives. In the AGB phase, thermal pulses release free neutrons, which can create elements up to lead and bismuth ($Z \leq 83$). Stellar winds and planetary nebulae can blow these elements into space.

- **Neutron star mergers (kilonovae):** produce the heaviest elements, including gold, platinum, uranium, and iodine. When two neutron stars collide, a large number of free neutrons are released, creating these heavy elements through the r-process.

- **Cosmic ray spallation**: high-energy particles strike matter and knock out protons and neutrons to create lighter elements that stellar fusion struggles to make, such as lithium, beryllium, and boron.

The metallicity of galaxies constantly changes through chemical evolution, which is fundamentally driven by the birth and death of stars. Type II SNe from short-lived massive stars occur within just millions of years after a starburst event, flooding the galaxy with elements like oxygen, magnesium, neon, and silicon. Massive stellar winds, such as in Wolf-Rayet stars, also contribute elements such as carbon. AGB stars and Type Ia SNe take hundreds of millions to billions of years to evolve, so their release of carbon, nitrogen, and elements around iron are delayed. An old galaxy will thus have a higher ratio of iron to oxygen.

Gas inflows from the IGM tend to be low-metallicity hydrogen, which dilutes the abundance of heavy elements, although their total mass remains the same. SNe are a primary source of both heavy elements and thermal energy, so galactic winds from SNe often blow out gas that is highly enriched with metals, which removes heavy elements from the galaxy into the IGM/CGM.

Turbulence in the ISM can mix heavily enriched regions, such as dense star-forming spiral arms, with less enriched regions, which smooths out local spikes in abundance. This also forms a metallicity gradient, where denser inner regions of galaxies are more metal-rich over time than outer edges (inside-out formation, see Question 64). The stars in the inner region of the MW formed first, so there have been more stellar lifecycles and thus more metals in that region, whereas the outer regions are fed by infalling, low-metallicity gas.

The speed at which a galaxy converts gas to stars also affects the speed of enrichment. Stars lock up gas and elements within them for millions of years (or more!); low-mass stars and compact remnants can lock them permanently. A galaxy undergoing starburst rapidly consumes gas, which leads to spikes in metal production; on the other hand, if star formation is quenched, the chemical evolution of a galaxy can be frozen.

Measuring abundances of heavy elements can teach us about many things.

- **Timeline and rate of SF**: we often look at the [α/Fe] ratio, where α denotes O, Mg, Si, etc. (see Question 61; these are elements produced by sequential capture of helium nuclei in massive stars and

released by Type II SNe). Since iron comes primarily from Type Ia SNe, which take $\sim 10^9$ yr from starburst, if the ratio of α elements to iron is high, a star must have formed early in the Universe from gas enriched by Type II SNe before Type Ia SNe began exploding. A galaxy with a high ratio across its stellar population likely had a short burst of SNe and then was quenched.

- **NS mergers**: r-process elements compared to s-process elements can help us map out when the first neutron star mergers occurred, how common they were, etc.
- **MW archaeology**: stars born in small dwarf galaxies have distinct element ratios (often lower [α/Fe] ratios at a given metallicity) than stars native to the Milky Way. This can help us trace back stars to the galaxies they belonged to, such as for the famous Gaia-Enceladus merger (see Question 61).

Chapter 7

Cosmology

74 Describe Olber's Paradox and its solution.

Olber's paradox points out the apparent contradiction of a dark night sky and the classical cosmological model of a Universe that is infinitely large, infinitely old, static, and homogeneously populated with stars. Under these assumptions of the Universe, there should be a star along every line of sight, and thus the night sky should be completely illuminated and bright. Divide space into concentric spherical shells of radius r and thickness dr. The volume of a shell scales as $dV = 4\pi r^2 dr$, so the number of stars in the shell is $dN = n \cdot 4\pi r^2 dr \propto r^2$ (for constant number density n). The flux received from an individual star of luminosity L falls off via the inverse-square law: $F = \frac{L}{4\pi r^2} \propto r^{-2}$. Thus, the total flux contribution from each shell is independent of distance:

$$dF_{\text{shell}} = dN \times F = (n \cdot 4\pi r^2 dr)\left(\frac{L}{4\pi r^2}\right) = nL\,dr \tag{7.1}$$

Integrating over all shells from $r = 0$ to $r = \infty$ yields a divergent integral:

$$F_{\text{total}} = \int_0^\infty nL\,dr \to \infty \tag{7.2}$$

Accounting for stars obscuring those behind them, every line of sight must eventually terminate on the surface of a star. Consequently, the entire night sky should be uniformly as bright as the surface of an average star.

The paradox is resolved because modern cosmology invalidates the classical assumptions:

1. **Finite Age of the Universe**: The universe is not infinitely old; it originated in the Big Bang approximately 13.8 Gyr ago. Because light travels at a finite speed c, an observer can only receive light from within the **observable universe** bounded by the particle horizon (current comoving radius $\sim$ 46.5 billion lyr). There has not been nearly enough cosmic time for light from arbitrarily distant stars to reach Earth, leaving the vast majority of sightlines empty of visible stars.

2. **Finite Stellar Lifetimes and Energy Budget**: Stars do not shine forever. Even if the universe were spatially infinite and static, the total mass-energy density available for thermonuclear fusion in baryons is far too small to illuminate the entire sky to stellar temperatures (the stars would burn through all available nuclear fuel within $\sim 10^{10}$ yr, roughly 10^{13} times faster than the time required to establish thermal equilibrium with the cosmic volume).

3. **Cosmological Expansion and Redshift**: Because the universe is actively expanding, radiation from distant sources undergoes cosmological redshift ($E_{\text{obs}} = E_{\text{emit}}/(1+z)$), diminishing energy density. In

fact, an analog to Olbers' bright sky does exist: the Cosmic Microwave Background (CMB) fills every line of sight from the surface of last scattering ($T \approx 3000\,\mathrm{K}$), but expansion has redshifted it by a factor of $(1+z) \approx 1100$ into faint, invisible microwave radiation at $T_0 \approx 2.73\,\mathrm{K}$.

75 What observations support the Big Bang theory?

The **Big Bang Theory** is the proposal that the Universe expanded from a hot, dense early state. There are several lines of evidence supporting it.

1. **Hubble Expansion**: We observe that objects that are farther away are moving away from us faster. The relationship that describes this is $v = H_0 d$, with H_0 constrained to be between 67 and 73 km/s/Mpc (see Question 77). This observation is consistent with an expanding Universe model. The value of Hubble's constant H_0 has changed throughout the history of the Universe depending on its matter content and energy.

2. **The Cosmic Microwave Background**: The CMB is background radiation with an observed temperature of $T = 2.7255\,\mathrm{K}$. The blackbody radiation from the CMB can be explained by a hot and dense early universe when matter was fully ionized, and the free electrons frequently scattered photons, causing the Universe to be opaque. With these frequent interactions, the early universe reached thermal equilibrium, resulting in a near-perfect blackbody spectrum we observe today. As the universe expanded and cooled down according to the Big Bang theory, ions and electrons formed neutral atoms during recombination (z $\approx$ 1100), allowing the universe to become transparent and for photons to freely travel through space. Through time, these photons have been stretched into the microwave wavelengths, dropping the CMB's temperature to today's value of $T = 2.7255\,\mathrm{K}$. A "steady state" universe cannot explain the Universe's beginning in a hot, dense state or the transition from an opaque to a transparent universe.

3. **Abundance of Light Elements**: In the first few minutes after the Big Bang, the universe was hot and dense enough to fuse light elements through **Big Bang Nucleosynthesis (BBN)** (75% ^{1}H, 25% ^{4}He, trace amounts of ^{2}H, ^{3}He, and ^{7}L). As the universe expanded and cooled after the first $\sim$ 20 minutes, it became too cold for fusion to continue and essentially locked these elemental ratios. The ratios themselves can be derived from the ratio of photons to baryons (which can be independently calculated from the CMB). Observations have shown that around 25% of the universe's ordinary matter is helium, which matches what is predicted by BBN; Stars cannot have produced this much helium over the universe's lifetime. Deuterium is also a great marker since ^{2}H is only destroyed in stars, so any observed deuterium in the universe must have originated from BBN, and observed abundances match excellently to what is predicted.

4. **Evolution of Large Scale Structure**: The first quasars and galaxies formed around a billion years after the Big Bang, with larger structures (e.g., galaxy clusters) following. These structures can be thought of as small initial density perturbations growing into larger-scale structure by gravity through time, proving that the universe must be evolving.

76 What problems with the standard Big Bang theory does the theory of inflation alleviate?

The **theory of inflation** posits a short period of rapid accelerated expansion of the Universe, $10^{-36} - 10^{-32}\,\mathrm{s}$ after the Big Bang. Inflation resolves three main problems with the standard Big Bang theory:

76.1 The Horizon Problem

The CMB exhibits remarkable isotropy, with temperature fluctuations of order $\Delta T/T \sim 10^{-5}$. In the standard hot Big Bang model without inflation, regions separated by more than roughly 1° on the CMB sky were outside each other's particle horizon at the time of recombination and could never have been in causal contact or thermal equilibrium. Inflation solves this problem by postulating that the entire observable Universe originated from a sub-horizon patch that was in causal contact and thermal equilibrium prior to the inflationary epoch. The rapid expansion stretched this thermalized region far beyond the comoving Hubble radius, explaining the observed large-scale uniformity.

76.2 The Flatness Problem

The Friedmann equation can be expressed in terms of the density parameter $\Omega(t)$ as:

$$|1 - \Omega(t)| = \frac{|k|}{a^2 H^2}, \tag{7.3}$$

where k is the discrete curvature constant in the FLRW metric, with $k \in \{+1, 0, -1\}$ in normalized coordinates (for closed, flat, and open geometries, respectively; see Question 83). In a standard Universe dominated by matter or radiation without dark energy, the comoving Hubble radius $(aH)^{-1}$ grows with time, meaning that any deviation from flat geometry ($|1 - \Omega| = 0$) grows rapidly. To observe $\Omega_0 \approx 1$ today, the early Universe must have been fine-tuned to extraordinary precision (e.g., $|1 - \Omega| \lesssim 10^{-16}$ at Big Bang nucleosynthesis, and $\lesssim 10^{-60}$ at the Planck scale). Positive curvature would lead to expansion slowed by gravity until the Universe contracts, and we get the Big Crunch. Negative curvature would lead to expansion that cannot be slowed by gravity, and we get thermal death.

Inflation solves this because during accelerated expansion ($\ddot{a} > 0$), the comoving Hubble radius $(aH)^{-1}$ decreases exponentially. As a result, $|1 - \Omega(t)|$ is driven exponentially toward zero, driving the spatial geometry of the observable patch to near-perfect flatness ($k \approx 0$).

76.3 The Monopole Problem

Grand unified theories (GUTs) that have strong, weak, and EM forces be the same force at high energies predict that magnetic monopoles should have formed in the early Universe and that we could detect them today. They would have been formed in phase transitions right after the Big Bang and would dominate the energy density of the Universe today. The problem is that we have never detected magnetic monopoles. Inflation alleviates this because the exponential expansion dilutes the monopole number density by a factor of $\gtrsim 10^{78}$; provided that the temperature immediately after inflation is below the GUT energy scale, monopoles were not thermally regenerated, and the expected number of monopoles within our current observable Universe is fewer than one.

77 What is Hubble's Law, and how was it discovered?

Hubble's Law states that galaxies recede from us at speeds proportional to their distance,

$$v \approx cz = H_0 d, \tag{7.4}$$

where H_0 is the Hubble constant, currently constrained to $H_0 \approx 67 - 73\,\mathrm{km/s/Mpc}$. This holds in the low-redshift ($z \ll 1$) limit. This linear relationship is the natural consequence of a homogeneous, isotropic, expanding Universe described by the FLRW metric, where $H(t) \equiv \frac{\dot{a}}{a}$. It implies that observers anywhere in the Universe observe the same apparent expansion.

Note that this does not represent motion through space, but rather the metric expansion of space itself. There is currently a **Hubble Tension** because early-Universe probes (e.g., *Planck* CMB observations

yielding $\approx 67.4\,\mathrm{km/s/Mpc}$) and late-Universe distance-ladder measurements (e.g., Cepheid-calibrated Type Ia supernovae yielding $\approx 73\,\mathrm{km/s/Mpc}$) yield statistically significant discrepancies in the Hubble constant.

The history of Hubble's Law is long.

- In 1908, Henrietta Swan Leavitt found the Cepheid variable period-luminosity relation from variable stars in the Small Magellanic Cloud, which gave astronomers a standard candle for measuring distances (see Questions 51 and 78).
- In 1912, Vesto Slipher measured the first radial velocities of spiral nebulae via spectroscopic Doppler shifts, discovering that the vast majority were redshifted and moving away from Earth at hundreds of kilometers per second.
- In 1922, Alexander Friedmann found dynamic solutions to Einstein's equations. He showed that the Universe might expand (or contract) with a time-dependent scale factor $a(t)$.
- In 1924, Hubble found Cepheids in M31 (Andromeda) and M33, finding their distance to be well outisde of the Milky Way.
- In 1927, Lemaître derived $v \propto d$ from expanding GR models, and using Slipher's velocities and Hubble's distances, he estimated the constant to be $H_0 \approx 600\,\mathrm{km/s/Mpc}$.
- In 1929, Hubble used 24 galaxies with distances from Cepheids and found the constant to be $H_0 \approx 500\,\mathrm{km/s/Mpc}$.
- Subsequent corrections found that Hubble's initial value was an order of magnitude too high due to systematic distance errors. In 1952, Walter Baade recognized two distinct Cepheid populations, doubling distances and halving H_0. Later, Allan Sandage showed that many of Hubble's brightest stars were actually H II regions or stellar clusters, so the distances were even larger, and he revised H_0 downward closer toward modern values.

78 Describe at least three "rungs" in the cosmic distance ladder.

The **cosmic distance ladder** is a succession of complementary methods used by astronomers to determine distances to celestial objects. Because no single technique spans all scales, each rung is calibrated by the more direct methods below it. See Fig. 7.1.

1. **Parallax** is accurate out to $3 \times 10^4\,\mathrm{lyr}$ or $10\,\mathrm{kpc}$. See Question 11.
2. Classical **Cepheids** are standard candles out to $10^8\,\mathrm{lyr}$ or $30\,\mathrm{Mpc}$. See Question 51. Cepheids pulsate through the κ-mechanism: in the partially ionized helium (He^+ to He^{++}) zone, compressing the gas increases the opacity, which traps heat and drives pulsation. The period-luminosity relation is roughly

$$M_V = -2.4(\log_{10} P - 1) - 4. \tag{7.5}$$

   By measuring the pulsating period, astronomers can infer the star's intrinsic luminosity; combined with the apparent magnitude, we can compute distances via the distance modulus (Eq. 2.4).
3. The **Tip of the RGB** is effective out to up to $\sim 50\,\mathrm{Mpc}$. See Question 58.
4. **Type Ia SNe** are standardizable candles out to $10^{10}\,\mathrm{lyr}$ or $3\,\mathrm{Gpc}$ ($z > 1$). See Questions 50 and 56. These are thermonuclear explosions of CO white dwarfs near the Chandrasekhar limit. ecause the exploding progenitor masses and fuel supplies are relatively uniform, they reach a nearly standard peak luminosity ($M_B \approx -19.3$). Small intrinsic variations in peak luminosity are empirically calibrated, but differences in progenitor systems can still introduce systematic uncertainties.

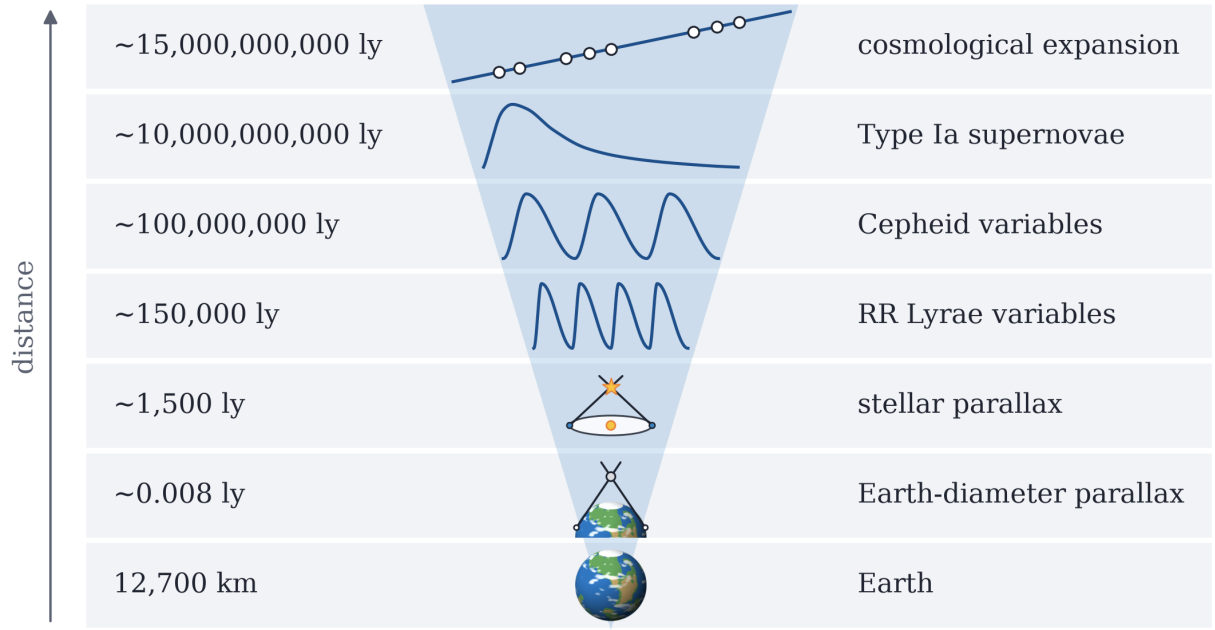


Figure 7.1: The cosmic distance ladder.

79 What were some of the first observations suggesting the need for dark matter?

Dark matter is an invisible form of matter that does not interact with EM radiation or baryonic matter. Dark matter constitutes ~85% of the total mass in the universe, or ~27% of the total mass-energy budget. The earliest observations suggesting the need for dark matter were based on the fact that things were moving too fast to be held together by the luminous matter that we can see.

1. **Coma Cluster Velocity Dispersion**: In 1933, Fritz Zwicky applied the virial theorem ($2\langle K\rangle + \langle U\rangle = 0$, Question 4) to the measured line-of-sight velocity dispersion ($\sigma_v \approx 1000\,\mathrm{km/s}$) of galaxies in the Coma Cluster,

$$M_{\mathrm{vir}} \sim \frac{R_{\mathrm{vir}}\sigma_v^2}{G}, \tag{7.6}$$

   showing that the galaxies were moving too fast for the cluster to be bound based on the inferred mass from the luminous matter. The resulting dynamical mass-to-light ratio ($M/L \gtrsim 100 - 500\,M_\odot/L_\odot$) exceeded the luminous contribution by many orders of magnitude. Zwicky coined the term "dunkle Materie" (German for dark matter) to suggest that there is missing undetectable mass that keeps the cluster bound.

2. **Galaxy Rotation Curves**: Observations dating back to the 1940s and popularized by Vera Rubin in the 1970s shows that stars at the outer edges of spiral galaxies orbit much faster than Newtonian dynamics suggest. For circular orbits, $v^2(r) = GM(<r)/r$. If most of the mass were in visible stars, i.e., an exponentially decaying stellar light profile, the rotation speed would fall off in a Keplerian manner, $v \propto r^{-1/2}$. Instead, a flat curve would imply that

$$M(<r) \propto r \Rightarrow \rho \propto r^{-2}, \tag{7.7}$$

   which is an isothermal halo extending beyond the visible light. They noticed this in the Milky Way,

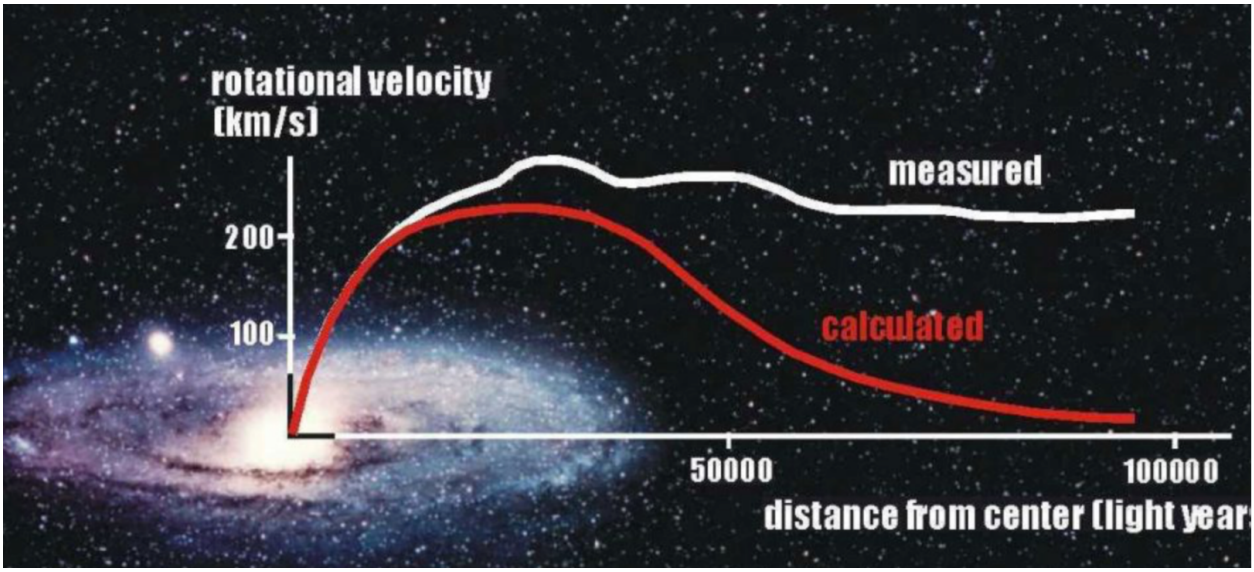


Figure 7.2: Rotation curve diagram.

where the velocity curves starts to be flat beyond the optical disk, implying that there is much more mass we cannot directly observe. See Question 69 and Fig. 7.2.

3. **Bullet Cluster**: This cluster, shown in Fig. 7.3, was formed after the violent collision of two large clusters of galaxies moving at high velocities. The red is the hot gas imaged by our observatories, and the blue is where most of the matter was found by gravitational lensing (see Question 96). The blue is non-luminous matter and did not experience the same drag as the hot gas, implying that it is invisible matter that does not interact with the luminous matter. This was a more modern confirmation of collisionless dark matter over modified gravity.

80 What are the main arguments in support of dark energy?

Dark energy is a proposed form of "energy" or component with negative pressure that drives the accelerating expansion of the universe. It is estimated that dark energy dominates the universe at ∼70% of the total mass-energy budget.

1. **Expansion via SNe**: It was found in 1998, using calibrated Type Ia SNe, that SNe at $z = 0.5$ are about 0.25 mag fainter (about 25 % dimmer in flux) than they would be in a decelerating Universe (an Einstein-de Sitter Universe, with $\Omega_m = 1$ and $\Omega_\Lambda = 0$). This showed that older, more distant standard candles are fainter, meaning that expansion has sped up since then. The luminosity distance to an object being larger at fixed redshift implies that the expansion was slower in the past and accelerated recently. Thus, the deceleration parameter is negative (implying acceleration), $q_0 \equiv -\ddot{a}a/\dot{a}^2\big|_{z=0} = \Omega_m/2 - \Omega_\Lambda \sim -0.55$, with $\Omega_m \approx 0.3$ and $\Omega_\Lambda \approx 0.7$. See Fig. 7.4.

2. **CMB Power Spectrum**: The CMB power spectrum implies that $\Omega_{\rm tot} \sim 1$. This is because the first acoustic peak is around $\ell = 220$, which is a measure of the maximum sound wave that could have formed before the CMB was emitted, which corresponds to the angular size of the sound horizon ($\theta_s = r_s/D_A$) at the surface of last scattering. The physical sound horizon r_s is fixed by early-Universe plasma physics, so the observed angular position ($\ell_1 \approx \pi/\theta_s \approx 220$) measures the spatial curvature, because D_A depends strongly on spatial geometry. This showed that the Universe is spatially flat,

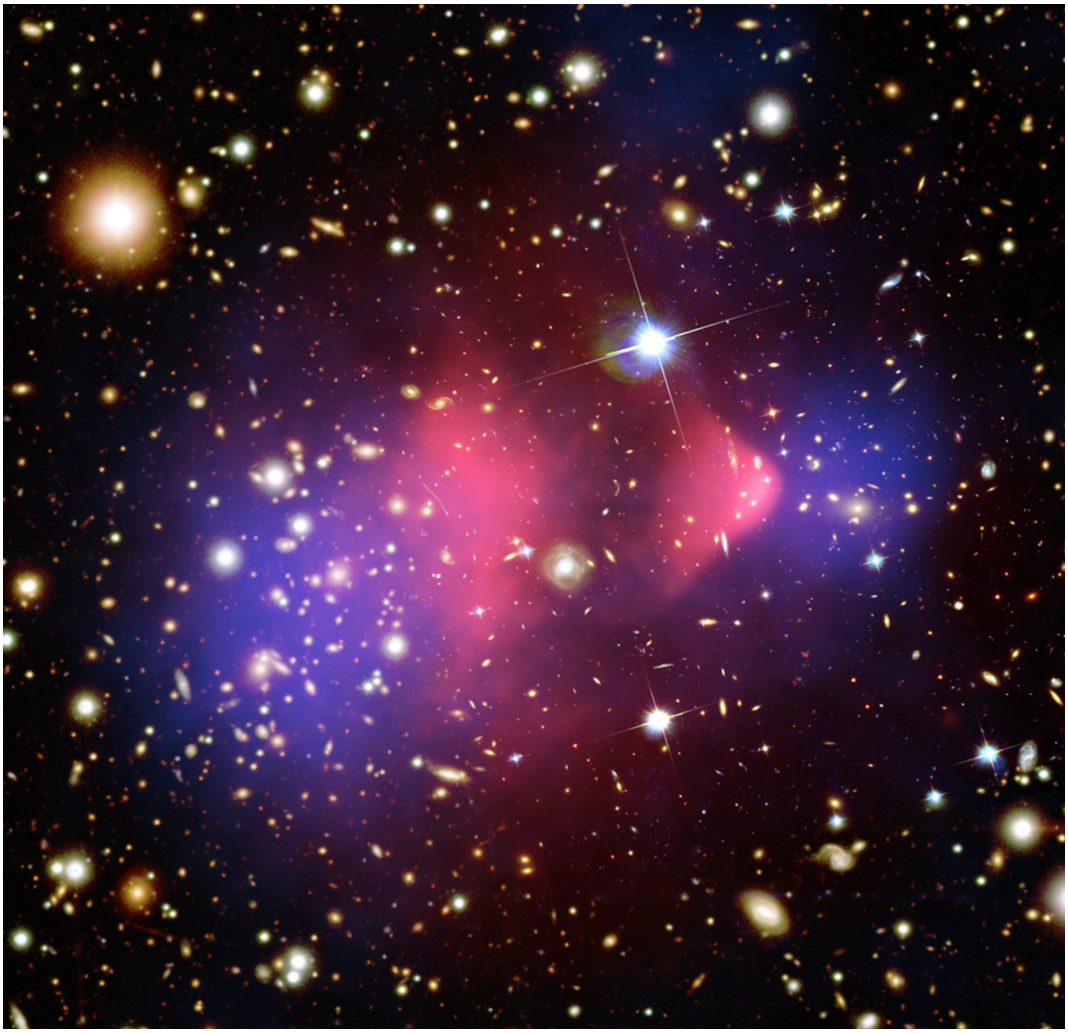


Figure 7.3: The Bullet Cluster.

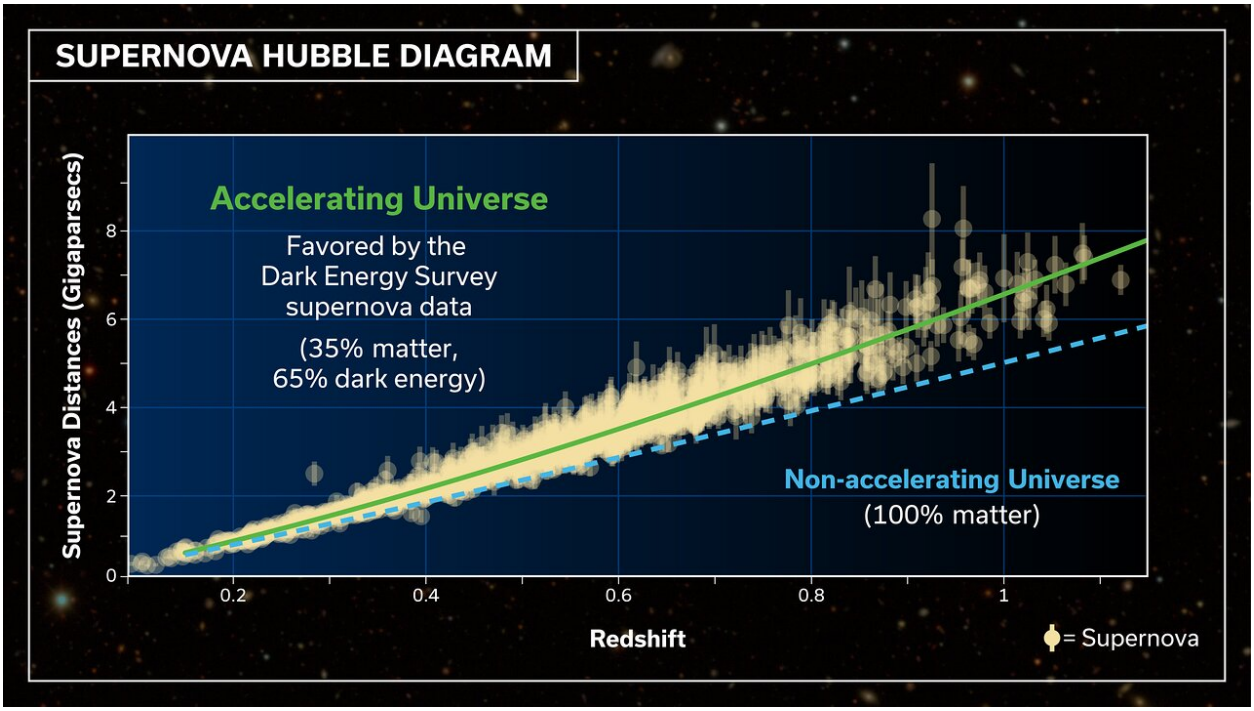


Figure 7.4: Supernova Hubble diagram.

$\Omega_k \approx 0 \Rightarrow \Omega_{\text{tot}} \approx 1$. Galaxy clusters and galaxy surveys find that $\Omega_m \sim 0.3$, so that the other 70 % has to be something else.

3. **Large-Scale Structure and Baryon Acoustic Oscillations**: The BAO peak acts as a standard ruler over time, caused by sound waves in the early Universe that froze at the drag epoch (when baryons decouple from photons, shortly after recombination) and imprinted on the matter distribution as a preferential separation of 150 Mpc. The BAO maps both the angular diameter distance $D_A(z)$ and the expansion rate $H(z)$ across cosmic time, so combining BAO with CMB data breaks degeneracies and proves accelerated expansion independently of SNe.

4. **The Cosmic Age Problem**: In a matter-only flat Universe ($\Omega_m = 1$), the age of the universe is $t_0 = \frac{2}{3} H_0^{-1} \approx 9 - 10\,\text{Gyr}$, which is younger than the oldest globular cluster stars ($\sim 12 - 13\,\text{Gyr}$, see Question 60). Adding Λ increases the expansion age to $t_0 \approx 13.8\,\text{Gyr}$, resolving this tension.

81 What is recombination and when did it occur?

Recombination was the period of time in the hot early Universe when the thermal plasma cooled sufficiently for protons and electrons to combine for the first time to create neutral atoms. Before recombination, the Universe was fully ionized, and the free electrons made it opaque, because Thomson scattering between photons and free electrons kept radiation and baryonic matter tightly coupled. The moment of recombination is the time when the number density of ions was equal to that of neutral atoms; note that the Saha equation describes the ratio of ions to baryons in thermal equilibrium. Considering only the ionization of hydrogen, the Saha equation gives the fractional ionization $X \equiv n_e/n_{\text{H}} = n_p/n_{\text{H}}$ (where $n_{\text{H}} = n_p + n_{\text{H}^0}$ is the total hydrogen number density) as

$$\frac{X^2}{1-X} = \frac{1}{n_{\text{H}}} \left(\frac{2\pi m_e kT}{h^2} \right)^{3/2} e^{-13.6\,\text{eV}/kT}, \tag{7.8}$$

where 13.6 eV is the ground-state binding energy of hydrogen. Using the scaling relations $T(z) = T_0(1+z)$ and $n(z) = n(1+z)^3$, we find that the point at which $X(z) \equiv n_p/n_{\rm H} = 0.5$ was at $z \approx 1370$, which corresponds to $T \approx 3700\,{\rm K} \approx 0.32\,{\rm eV}$. Although the binding energy is 13. eV, recombination is delayed until $kT \ll 13.6\,{\rm eV}$ because the large photon-to-baryon ratio supplies enough photons in the high-energy Wien tail of the Planck distribution to maintain ionization until the temperature drops substantially.

There are several closely linked events following this point that define the transition to a transparent Universe.

1. **Photon Decoupling ($z \approx 1100$):** The free electron density drops low enough that the Thomson scattering rate falls below the expansion rate, decoupling radiation from matter.

2. **Surface of Last Scattering ($z \approx 1090$, $t \approx 380{,}000\,{\rm yr}$):** The optical depth to Thomson scattering drops through unity ($\tau \approx 1$). Photons scatter for the final time and begin freely streaming through space. The photons that started freely streaming at this **surface of last scattering** are what we now know as the CMB.

3. **Freeze-out ($z \sim 800$):** The expansion rate surpasses the recombination rate, leaving a small residual ionized fraction of $X \approx 10^{-4}$ that persists until reionization by the first stars.

82 What is cosmic reionization and when did it occur?

Cosmic reionization was the period when neutral gas between galaxies was ionized again by UV light from the first luminous objects. This happened after the **dark ages**, which were after recombination (Question 81). There were two stages of reionization:

1. Hydrogen reionization occurred around $z \approx 6-10$, driven by early star-forming galaxies whose photons have energies above 13.6 eV. This was patchy, as ionized bubbles formed around sources. We can think about these sort of like cosmic Stromgren spheres that grew around early galaxies until they overlapped globally.

2. Helium II reionization (He II to He III) occurred around $z \approx 3-4$, driven by quasars. This happened later because a higher ionization energy is required, about 54.4 eV. Neutral helium has an ionization potential of 24.6 eV, so it was reionized alongside hydrogen.

A primary piece of evidence for reionization is the **Gunn-Peterson trough**, which is a feature found in high-redshift quasar spectra, where there is complete absorption blueward of Lyα at $z \gtrsim 6$, which indicates the tail end of hydrogen reionization. This is because neutral hydrogen has a very large absorption cross-section for Lyα photons, so any trace of neutral hydrogen is enough to block nearly all transmitted light. At redshifts below this limit, we see spiky transmission, the Lyα forest (see Question 87), since now photons can freely transmit between discrete gas clouds.

83 What assumptions are required to derive the Friedmann-Robertson-Walker metric?

The **Friedmann-Robertson-Walker (FRW) metric** is the most general spacetime metric describing a Universe that satisfies the Cosmological Principle. It is derived from purely geometric symmetry arguments, independently of Einstein's field equations.

The main assumption for the derivation is the Cosmological Principle, the idea that on sufficiently large scales ($\gtrsim 100\,{\rm Mpc}$), the spatial distribution of matter and energy in the Universe is homogeneous and

isotropic (see Question 85). The metric is

$$ds^2 = -c^2 d\tau^2 = -c^2 dt^2 + \overbrace{a(t)^2}^{a(z)=\frac{a_0}{1+z}} \underbrace{\left[\frac{dr^2}{1-kr^2} + r^2 d\Omega^2\right]}_{d\Sigma^2}, \tag{7.9}$$

where $a(t)$ is the dynamic **scale factor** relating proper distance to comoving coordinate distance and $k = \{+1, 0, -1\}$ corresponds to a closed, flat, and open Universe, respectively.

Inserting the FRW metric and an ideal fluid stress-energy tensor into Einstein's field equations ($G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$) yields the **Friedmann equations** governing the time evolution of $a(t)$:

$$H^2(a) \equiv \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3} \tag{7.10}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda c^2}{3}. \tag{7.11}$$

We can rewrite Eq. 7.10 as

$$H^2(a) = \frac{8\pi G}{3}\overbrace{(\rho_m + \rho_r + \rho_k + \rho_\Lambda)}^{\equiv \rho(a)}, \tag{7.12}$$

where we regroup $\rho = \rho_m + \rho_r$ (the total fluid energy density, i.e., matter and radiation) and define the curvature energy density and vacuum energy density as

$$\rho_k \equiv -\frac{3kc^2}{8\pi G a^2} \propto a^{-2}, \quad \rho_\Lambda \equiv \frac{\Lambda c^2}{8\pi G} = \text{constant}. \tag{7.13}$$

The **critical density** is the mass density at some z with flat curvature,

$$\rho_c(t) = \frac{3H(t)^2}{8\pi G} \approx 10^{-29}\,\text{g cm}^{-3}. \tag{7.14}$$

We define the dimensionless density parameters as $\Omega_i(t) \equiv \rho_i(t)/\rho_c(t)$, and vombining Eq. 7.10 with the continuity equation $\dot{\rho} + 3\frac{\dot{a}}{a}\left(\rho + \frac{p}{c^2}\right) = 0$ gives us

$$H^2(z) = H_0^2\left[\underbrace{\Omega_{m,0}(1+z)^3}_{\text{matter}} + \underbrace{\Omega_{r,0}(1+z)^4}_{\text{radiation}} + \Omega_{k,0}(1+z)^2 + \Omega_{\Lambda,0}\right]. \tag{7.15}$$

From this, the redshift of matter–radiation equality (z_{eq}) satisfies:

$$\Omega_{r,0}(1+z_{\text{eq}})^4 = \Omega_{m,0}(1+z_{\text{eq}})^3 \Rightarrow z_{\text{eq}} = \left(\frac{\Omega_{m,0}}{\Omega_{r,0}} - 1\right) \simeq \left(\frac{0.3}{10^{-4}} - 1\right) \simeq 3000. \tag{7.16}$$

84 What is the IGM and CGM, and how do we detect and study these components of the Universe?

See Figure 7.5. The **intergalactic medium (IGM)** and the **circumgalactic medium (CGM)** together contain the vast majority of the baryonic mass in the Universe. Both are are very diffuse, with $n_{\text{H}} \sim 10^{-6} - 10^{-2}\,\text{cm}^{-3}$, so their surface brightness is extremely faint, making them difficult to detect in emission. Thus, we study them primarily in absorption against bright background sources (quasars, etc.), dispersion measures of FRBs, specialized low-surface-brightness imaging (recall Pieter van Dokkum's colloquium and the MOTHRA telescope, which is designed to detect low surface brightness emission, with the goal of

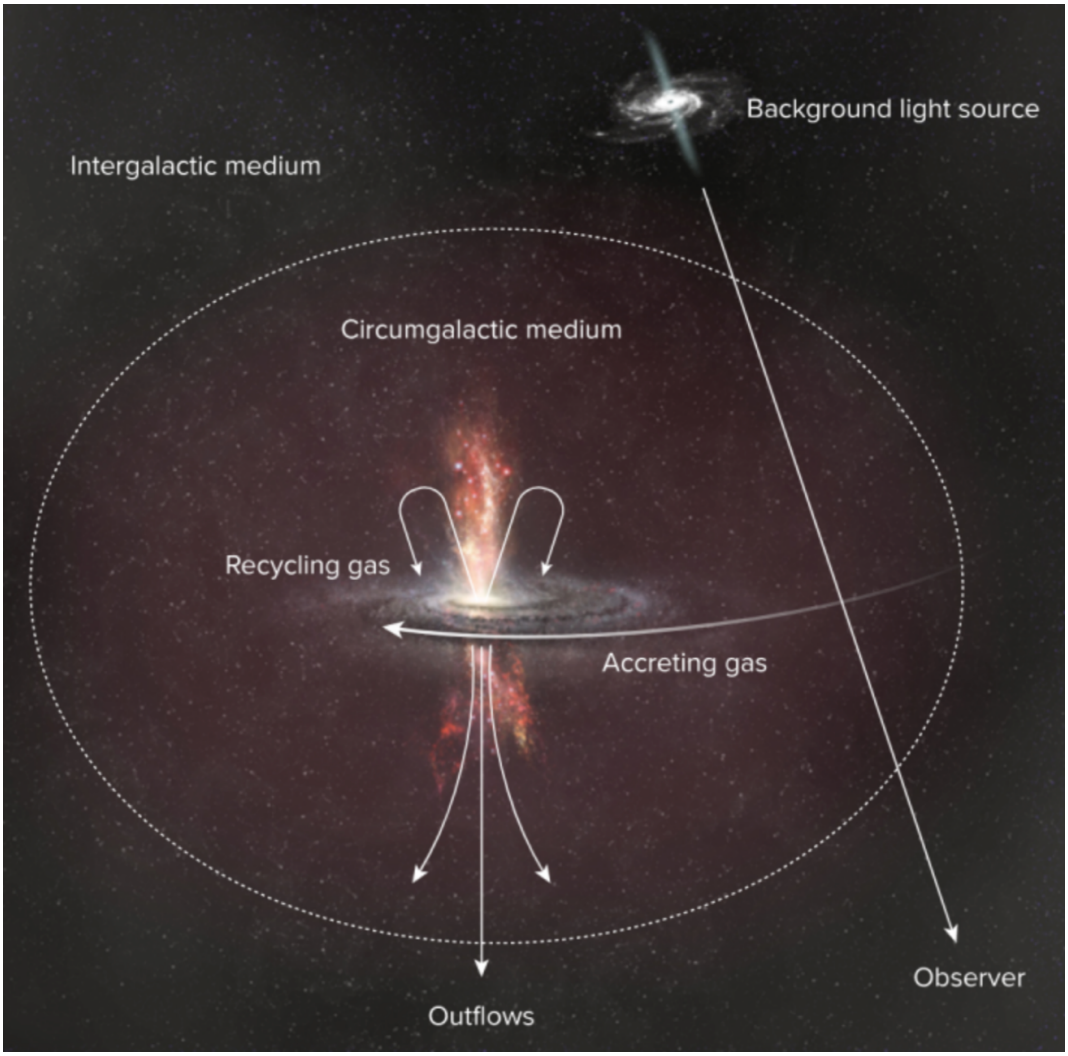


Figure 7.5: The IGM and the CGM.

mapping out the CGM and brighter pockets of the IGM).

84.1 The Intergalactic Medium

The IGM occupies the vast cosmic web filaments and voids between galaxies, outside their dark matter halos. It is composed of nearly pristine, highly ionized hydrogen and helium with trace metallicity enriched by galactic outflows. The IGM consists of a diffuse, photoionized warm phase ($T \sim 10^4\,\mathrm{K}$, photoheated by the UV background) and the shock-heated **Warm–Hot Intergalactic Medium** (WHIM; $T \sim 10^5$–$10^7\,\mathrm{K}$) along large-scale structure filaments. The photoionized IGM is mapped in exquisite detail via the neutral hydrogen resonant transitions in the **Lyman-α forest** (see Question 87) toward distant quasars. The hotter WHIM is probed via soft X-ray and far-UV absorption lines of highly ionized heavy elements.

The IGM was initially proposed as a solution to the missing baryon problem, where the amount of baryonic matter we infer from galaxies and the CGM do not match predictions from BBN and observed CMB anisotropies. Liam used FRBs to solve the missing baryon problem: the dispersion measure characterizes the frequency-dependent time delay observed in FRBs, which relates to the total column density of free electrons along the line of sight and can be used to show that $\sim$ 75–80% of cosmic baryons reside in the diffuse IGM/WHIM.

84.2 The Circumgalactic Medium

The CGM is the gas in a galaxy's dark matter halo outside of the disk and its ISM but within the virial radius. It mediates the **baryon cycle**, consisting of inflows of pristine and recycled gas feeding star formation (accretion from cosmic filaments) and outflows of metal-enriched material driven by SNe and AGN.

The CGM also has many phases: a cool/photoionized phase at $T \sim 10^4\,|rmK$ and a warm collisionalyl ionized phase at $T \sim 10^5\,\mathrm{K}$, both traced in UV/optical absorption, and a hot virialized phase at $T \sim 10^6 - 10^7\,\mathrm{K}$, which is collisionally ionized gas in hydrostatic equilibrium within massive halos and is traced by soft X-ray absorption and the thermal Sunyaev–Zel'dovich (tSZ) effect.

In general, the CGM can be studied by taking looking for absorption against background sources such as quasars. From the absorption lines, we can learn the chemical composition of the CGM, which is relevant for galaxy evolution. MOTHRA and integral field spectrographs such as MUSE now can directly map CGM emission (such as from Lyα) around individual galaxies.

85 What is the difference between homogeneous and isotropic? Describe a Universe that is one but not the other.

A **homogeneous** Universe is one that is uniform everywhere; it is invariant under spatial translations, and all points in space have local physical properties (e.g., density, pressure, curvature) identical. In an **isotropic** Universe, there is no preferential sense of direction; it is invariant under spatial rotations (importantly, about a given point).

A Universe consisting entirely of magnets (i.e., a Universe filled with a uniform, constant magnetic field pointing along the z-axis) would be homogeneous but not isotropic. A Universe with a radial, spherically symmetric density distribution would be isotropic (about the center, because an observer at that center sees the same density in every angular direction) but not homogeneous. There is no Universe that is isotropic everywhere but not homogeneous; in other words, a space that is isotropic about every point is necessarily homogeneous.

Because observation of the CMB shows near-perfect isotropy ($\Delta T/T \sim 10^{-5}$) from our location, and the Copernican principle posits that Earth does not occupy a privileged center of the Universe, we infer that the Universe is isotropic everywhere, and therefore both homogeneous and isotropic on cosmological scales ($\gtrsim 100\,\mathrm{Mpc}$)

86 Which elements were present in the Universe before the first stars? Where did they come from? How do we measure the abundances of these "primordial" elements?

Before the formation of Population III stars (Question 61), the Universe contained only light isotopes produced during BBN. The primordial abundances were

- ^{1}H: $\sim 75\,\%$ by mass, or $\sim 92\,\%$ by number.
- ^{4}He: $\sim 25\,\%$ by mass, or $\sim 8\,\%$ by number.
- Deuterium (^{2}H or D): D/H $\sim 2.5 \times 10^{-5}$ by number.
- ^{3}He: ^{3}He/H 10^{-5} by number.
- ^{7}Li: ^{7}Li/H 1.6×10^{-10} by number.
- Trace amounts of tritium (^{3}H, which decays into ^{3}He with $t_{1/2} \approx 12.3\,\mathrm{yr}$) and beryllium (^{7}Be, which electron-captured into ^{7}Be).

There were no heavier elements because there are not stable nuclei at mass numbers 5 and 8, and the expansion quickly dropped the temperature and density before three-body reactions (like the triple-α process) could occur.

Before the first second of BBN $t \sim 1\,\mathrm{s}$, weak interactions maintained statistical equilibrium between protons and neutrons, but at free-out, the expansion rate of the universe exceeded the weak interaction rate, locking the neutron-to-proton ratio at $\sim 1/6$ (using mass ratio and free-out temperature). Free neutrons began to β-decay, dropping n/p to $\sim 1/7$. Although nuclear binding favored forming ^{4}He, the first step required making deuterium, but because the photon-to-baryon ratio was high, deuterium was continuously photodissociated until the universe cooled $t \sim 200\,\mathrm{s}$. Once deuterium was able to survive, reactions quickly funneled almost all available neutrons into the tightly bound ^{4}He nucleus, with the resulting mass fraction set by the available neutrons:

$$Y_p \approx \frac{4n_{\mathrm{He}}}{n_n + n_p} = \frac{2n_n}{n_n + n_p} = \frac{2(n/p)}{1 + (n/p)} \approx \frac{2(1/7)}{1 + (1/7)} = 0.25. \tag{7.17}$$

To measure the primordial abundances of these elements, we must look at chemically unevolved environments. The best signature is the D/H ratio, because stars only destroy deuterium. By looking at high-redshift ($z \sim 2-3$) damped Lyα systems (DLAs), which have extremely high neutral hydrogen column densities, and measuring the shift between the H and D Lyman series, we can derive the primordial abundance value. Often, we also do things like plot Y (the ^{4}He mass fraction) against metallicity from ionized gas and extrapolate back to $Z \to 0$. But in general, measuring primordial abundances is hard, because stars destroy and produce many of these elements.

87 What is the Lyman-α forest and why is it significant to cosmology?

The **Lyman-α** forest is a dense series of redshifted Lyα absorption lines in the optical/UV spectra of distant quasars, as in Fig. 7.6. The Lyα absorption occurs when light from background sources passes through clouds of diffuse neutral hydrogen in the IGM (see Question 84), which absorb the Lyα transition ($n = 2 \leftrightarrow 1$ transition in hydrogen, at a rest wavelength of 1216 Å).

Continuum light from quasars passes through the IGM along a line sight, interacting with patches of neutral hydrogen at many redshifts. Each path of neutral hydrogen at redshift z_a absorbs at wavelength photons that redshift into its local rest-frame Lyα transition, $\lambda_a = \lambda_0(1 + z_a)$ (where $\lambda_0 = 1216\,\text{Å}$), which is blueward of the quasar's own Lyα emission (which is often a strong emission line, but not necessarily).

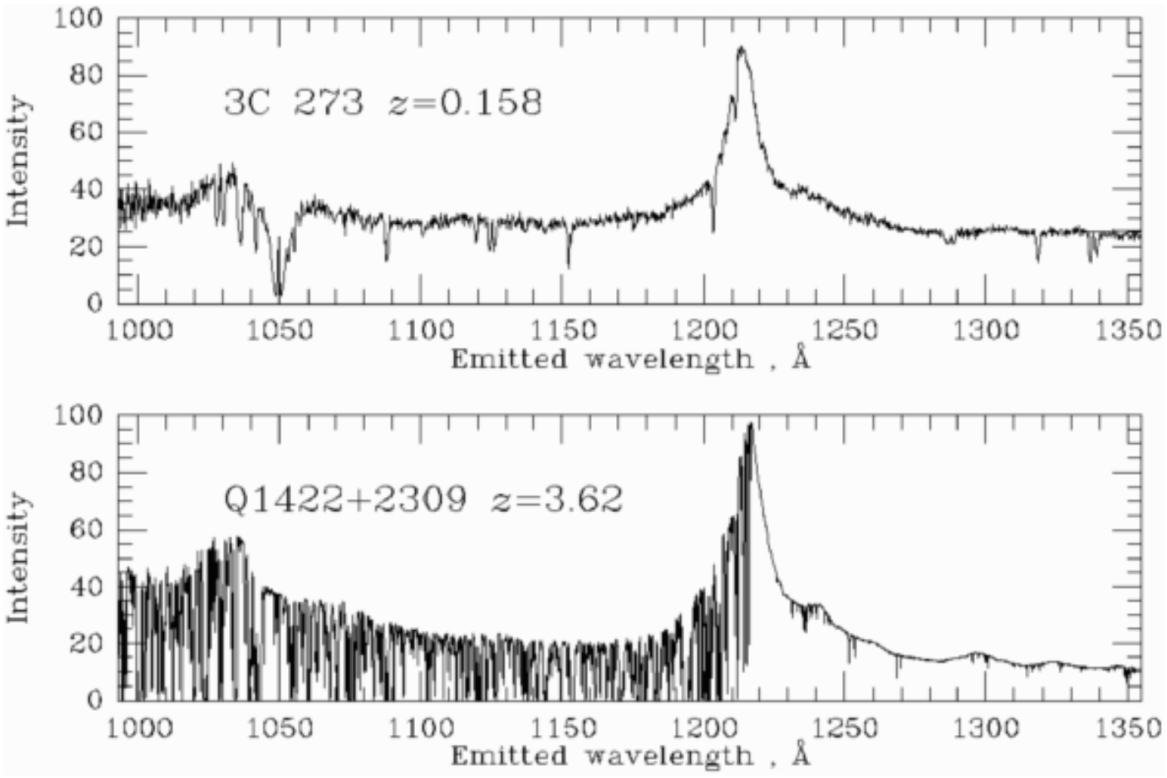


Figure 7.6: Lyman-α forest.

When we convert a spectrum into the quasar's rest frame, such that the emitted Lyα line is at λ_0, we see absorption lines at

$$\lambda_{\text{rest, qso}} = \frac{\lambda_a}{1+z_{\text{qso}}} = \lambda_0 \left(\frac{1+z_{\text{a}}}{1+z_{\text{qso}}} \right). \tag{7.18}$$

Since the intervening absorber has $z_a < z_{\text{qso}}$, the coefficient is less than unity, so that we see a series of absorption lines at shorter wavelengths than the emitted rest-frame Lyα wavelength. So, each line of sight to a quasar becomes a one-dimensional map of the neutral hydrogen density field of the IGM.

This is significant in astronomy because it can probe the small-scale power spectrum, particularly scales around $\sim 1\,\text{Mpc}$, which is much smaller than galaxy surveys. These small scales constrain WDM, the sum of neutrino masses, etc. It has also provided the highest BAO redshift measurements ($z \sim 2-3$) through cross-correlations across multiple lines of sight. And, it gives information about cosmic reionization, because its optical depth constrains the photoionization rate of the UV background and because the appearance of the Gunn-Peterson trough marks the end of reionization at $z \approx 5.5-6$ (Question 82). Note that the Lyα forest does not trace completely neutral gas clouds; it traces tiny residual fluctuations of $\sim 0.01\,\%$ neutral hydrogen. If the IGM ionization fraction were lower, we would see a trough rather than a forest of lines.

Chapter 8

High-Energy Astrophysics & Compact Objects

88 What is the strongest evidence for the existence of black holes? Name at least one other observation supporting the existence of black holes.

Historically, the first compelling evidence came from **X-ray binaries**, binary systems in which one companion accretes gas from the other and heats it to $T \gtrsim 10^6\,\mathrm{K}$, because only ultra-compact objects (neutron stars or black holes) possess gravitational potentials deep enough to drive thermal emission in the X-ray regime.

Consider the radiation energy of an accreting compact object. A lower limit for the temperature will assume the object in the X-ray binary system radiates as a black body, so that

$$T_b = (L_{\mathrm{edd}}/4\pi R_\star^2 \sigma)^{1/4}. \tag{8.1}$$

An upper limit for the temperature will assume that the mass being accreted turns its gravitational potential energy entirely into thermal energy, and that the accretion is done in electron-proton pairs, so that

$$\frac{GM(m_e + m_p)}{R_\star} \approx \frac{GMm_p}{R_\star} = 3kT_{\mathrm{therm}}. \tag{8.2}$$

If we solve for these temperatures for a canonical $1.4\,M_\odot$ neutron star with radius $R_\star \approx 10\,\mathrm{km}$, we find that

$$T_b \approx 10^7\,\mathrm{K}, \quad T_{\mathrm{therm}} \approx 5.5 \times 10^{11}\,\mathrm{K}. \tag{8.3}$$

Converting these temperatures into a photon energy scale (using roughly $h\nu \approx kT$, we get that

$$1\,\mathrm{keV} < E < 50\,\mathrm{MeV}, \tag{8.4}$$

which is comfortably in the X-ray range. Notice that in both Eqs. 8.1 and 8.2, the temperature scales inversely with radius, i.e., energy scales with compactness. Repeating these calculations for a white dwarf ($R_\star \sim 10^4\,\mathrm{km}$) shows that the photon energy will be primarily in the UV range.

The neutron star mass limit (the **TOV limit**, as in Question 90, which is less certain than the analogous Chandrasekhar limit for white dwarfs due partially to a unknown equation of state) of $\sim 2.2 - 3\,M_\odot$ can be used to claim that certain binaries must have a black hole, because the mass of the accretor, estimated from radial velocity and astrometric measurements, exceeds the TOV limit. For example, Cygnus X-1, a galactic X-ray source, has a compact object with $M \approx 20\,M_\odot$.

Other observations supporting the existence of black holes include:

- **Sgr A* orbits**: Decades of near-IR astrometry (e.g., of the star S2) track Keplerian orbits around the center of the Milky Way (the object Sgr A*), requiring that there be a central dark mass of $\sim 4\times10^6\,M_\odot$ confined within a periastron distance of $\lesssim 120\,\mathrm{AU}$. This rules out non-black-hole clusters, suggesting that Sgr A* is a supermassive black hole (SMBH).

- **Gravitational waves**: Detections from LIGO, VIRGO, and KAGRA match waveforms predicted by GR for compact objects above the neutron star mass limit; such binary black hole coalescences (e.g., GW150914) show the inspiral, merger, and distinctive quasi-normal mode ringdown matching Einstein's field equations.

- **Direct imaging**: The Event Horizon Telescope has resolved the black hole shadow and asymmetric emission ring of Sgr A* and M87*, matching GR predictions for null geodesics around a Kerr black hole.

- **Quasars, extreme nuclear transients, relativistic jets**: all of these sources produce electromagnetic signatures whose luminosity is so high ($\gtrsim 10^{46}\,\mathrm{erg\ s^{-1}}$) that there are no reasonable explanations for their liberation mechanisms that do not include gravitational potential energy conversion via accretion onto SMBHs.

89 What is the Schwarzschild Radius? Write down the relevant equation. What does this radius represent?

The **Schwarzshchild Radius** (R_S) is the characteristic radius defining the event horizon of a static, spherically symmetric, non-spinning, uncharged mass distribution in GR (described by the Schwarzschild metric). If a body of mass M is compressed entirely within this radius, no force can halt gravitational collapse, resulting in a black hole. It is given by

$$R_\mathrm{S} = 2GM/c^2. \tag{8.5}$$

Geometrically, R_S represents the location of the **event horizon**, which acts as a causal boundary. Inside $r < R_\mathrm{S}$, the radial coordinate r becomes timelike, meaning that all future-directed timelike and null worldlines inevitably terminate at the central coordinate singularity ($r = 0$).

The Schwarzschild metric is

$$ds^2 = -\left(1 - \frac{R_\mathrm{S}}{r}\right) c^2 dt^2 + \left(1 - \frac{R_\mathrm{S}}{r}\right)^{-1} dr^2 + r^2 d\Omega^2. \tag{8.6}$$

For an asymptotic, static external observer ($dr = d\theta = d\phi = 0$), the relation between proper time τ and coordinate time t for a stationary clock at radius r is:

$$d\tau = \sqrt{-ds^2/c^2} = \sqrt{1 - \frac{R_\mathrm{S}}{r}}\, dt. \tag{8.7}$$

As $r \to R_\mathrm{S}$, $g_{00} \to 0$ and $d\tau/dt \to 0$, leading to infinite gravitational time dilation and redshift ($z \to \infty$). Consequently, an external observer never observes an infalling object physically cross R_S; instead, the object appears to asymptotically decelerate while rapidly dimming exponentially into undetectable wavelengths. (Note that $r = R_\mathrm{S}$ is a coordinate singularity, not a physical curvature singularity; an infalling observer crosses the horizon in finite proper time.)

The Schwarzschild Radius of a human ($M \approx 70\,\mathrm{kg}$) is of order $R_\mathrm{S} \sim 10^{-23}\,\mathrm{cm}$, which is much smaller than the nucleus of an atom. For the Sun ($M_\odot \approx 2\times10^{30}\,\mathrm{kg}$, it is $R_\mathrm{S} \sim 3\,\mathrm{km}$. For Sgr A* ($M \approx 4.15\times10^6\,M_\odot$), it is $R_\mathrm{S} \approx 1.2\times10^7\,\mathrm{km} \approx 0.08\,\mathrm{AU}$. If the black hole possesses angular momentum (Kerr metric) or electric charge, the event horizon is at a radius less than R_S.

90 What are white dwarfs and neutron stars, and how do they form?

90.1 White Dwarfs

White dwarfs are the compact remnants of low- to intermediate-mass stars, roughly $0.5-8\,M_\odot$. Throughout their evolution, these stars undergo shell and core burning phases, but their cores never reach the temperatures required to ignite carbon (or neon, in higher-mass cases). At some point, their compactness gets so high that the degeneracy pressure of electrons takes over. The boundary of this compactness is when the distance between atoms is less than their de Broglie wavelengths, $\lambda = h/p$, and p is much lower for electrons than protons. At this point, the Pauli exclusion principle kicks in, dictating that no two electrons can occupy the same quantum state, establishing a non-thermal **electron degeneracy pressure** that halts further contraction and supports hydrostatic equilibrium.

The radius of a white dwarf scales as $R_{\rm WD} \propto M_{\rm WD}^{-1/3}$. In other words, MV is constant. This means that the more massive a white dwarf, the smaller it is. As mass increases, relativistic effects become significant, softening the equation of state until the degeneracy pressure can no longer withstand self-gravity. This defines the **Chandrasekhar limit**, where the degeneracy pressure can no longer support the star, and it will collapse (see Questions 50, 56, and 91).

White dwarfs are less than $1.4\,M_\odot$, the Chandrasekhar limit, and roughly the size of Earth. They are hot and blue when they form, at the lower left of the HR diagram. Because they have exhausted all nuclear fuel and cannot undergo gravitational contraction, they radiate strictly from residual thermal energy. Their thin, non-degenerate outer envelopes act as thermal insulators, causing them to cool exceptionally slowly over tens of billions of years; over time, the internal ion plasma crystallizes into a solid lattice.

90.2 Neutron Stars

Neutron stars are the compact stellar remnants of stars with mass $\gtrsim 8\,M_\odot$. As massive stars burn, their core unlocks temperatures and densities high enough to burn heavier and heavier elements. Eventually, their core reaches the iron group, where fusion absorbs energy rather than releases it (is endothermic, rather than exothermic). Once the core exceeds its effective Chandrasekhar mass, it collapses in near-free fall, accelerated by nuclear photodisintegration and neutronization,

$$e^- + p \to n + \nu_e. \tag{8.8}$$

At a certain point, this collapse abruptly halts, and neutron degeneracy pressure supports hydrostatic equilibrium. The layers of matter surrounding the core fall onto it and produce a shockwave, which we observe as a core-collapse (Type II/Ib/Ic) SN (see Question 50).

The equation of state for neutron stars (see Question 56) is less certain, and models show that their radii remain quite consistent ($\sim 10-15\,\mathrm{km}$) across most of their mass range ($\sim 1.2-2.0\,M_\odot$). The maximum allowable mass of a non-rotating neutron star, the **TOV limit**, is estimated to be about $2.0-2.3\,M_\odot$, beyond which the core collapses into a stellar-mass black hole.

Young neutron stars typically conserve much of their progenitor's angular momentum and magnetic flux. Pulsars are rapidly spinning, highly magnetized neutron stars that produce radio emission as a beam from their poles (see Question 62). As they radiate magnetic dipole radiation and accelerate particle winds, pulsars steadily lose rotational kinetic energy, leading to a measurable spin-down ($\dot{P} > 0$) over time that can be used to estimate their characteristic ages and magnetic field strengths. Magnetars are neutron stars with extremely strong magnetic fields, of order 10 to 100 times that of pulsars. Their magnetic field decay powers X-ray and gamma-ray emission.

91 What is the Chandrasekhar limit? Describe its relevance to astrophysics and cosmology.

The **Chandrasekhar limit** is the upper mass limit of white dwarfs, roughly equal to $1.4\,M_\odot$. This limit arises because increasing density forces degenerate electrons to relativistic speeds, which softens the equation of state from the non-relativistic relation ($P \propto \rho^{5/3}$) to the ultra-relativistic relation ($P \propto \rho^{4/3}$), at which point hydrostatic equilibrium becomes marginally unstable and self-graavity overcomes electron degeneracy pressure above this mass (see Question 56).

In CO white dwarfs accreting mass toward the Chandrasekhar limit, compressional heating ignites carbon fusion before total collapse can occur. Because degenerate gas pressure is independent of temperature, the core cannot expand to cool, triggering a thermonuclear runaway explosion that completely disrupts the star, leaving no remnant (a Type Ia SN, see Question 50). Progenitor models traditionally fall into two classes: the **single-degenerate** channel (accretion from a non-degenerate binary companion) and the **double-degenerate** channel (the merger of two white dwarfs driven by gravitational radiation loss). Modern models also recognize sub-Chandrasekhar mass detonations (e.g., double-detonation mechanisms) as significant contributors.

There is also accretion-induced collapse (AIC), where oxygen-neon-magnesium white dwarfs approaching the Chandrasekhar limit undergo rapid electron capture, which removes degeneracy support before thermonuclear runaway unbinds the star, collapsing the white dwarf directly into a neutron star.

Because Type Ia supernovae originate from white dwarfs with a characteristic mass near $M_{\rm Ch}$, their peak intrinsic luminosities are remarkably homogeneous ($M_B \approx -19.3$), powered primarily by the radioactive decay chain $^{56}\mathrm{Ni} \to {}^{56}\mathrm{Co} \to {}^{56}\mathrm{Fe}$. While slightly variable in practice due to different channels for exceeding the Chandrasekhar limit, variations in chemical composition, or asymmetries, they serve as precise **standardizable candles** using empirical calibrations (see Question 78). Dstance measurements to high-redshift Type Ia supernovae revealed that distant supernovae were dimmer than expected in a matter-dominated decelerating universe. This provided the first direct observational evidence for the accelerated expansion of the universe.

92 What is an active galactic nucleus (AGN) and how do we identify them observationally?

Active galactic nuclei (AGN) are galactic nuclei for which a significant fraction of the luminosity comes from a non-stellar origin. AGN are thought to be powered by the accretion of matter by supermassive black holes ($M_{\rm SMBH} \sim 10^6 - 10^{10}\,M_\odot$) at the centers of galaxies.

Observationally, AGN have:

- Large, variable luminosities from a very compact region (a few pc). Extreme variability on short timescales implies that the origin of the light must be small, since $R < c\Delta t \sim 7\,\mathrm{AU}$ for observationally inferred timescales of hours, i.e., $\sim 3600\,\mathrm{s}$.
- Relativistic jets of ionized plasma, launched along the SMBH spin axis, produce non-thermal synchrotron radiation observable in visible, radio, and X-ray wavelengths.
- Broadband non-thermal SED spanning radio to hard X-rays and gamma-rays, featuring optical/UV emission from the accretion disk, X-ray power-law corona emission, and IR re-radiation by dusty tor. The spectra are distinctly non-stellar.
- Spectra show both broad and narrow lines. The broad line region has Dopper broadening of $\gtrsim 10^4\,\mathrm{km/s}$ and is thought to be smaller (more concentrated around the nucleus) than the narrow line region.

There are many types of AGN; classification depends largely on the orientation of an obscuring dusty torus relative to our line of sight (see Question 93), as well as jet power:

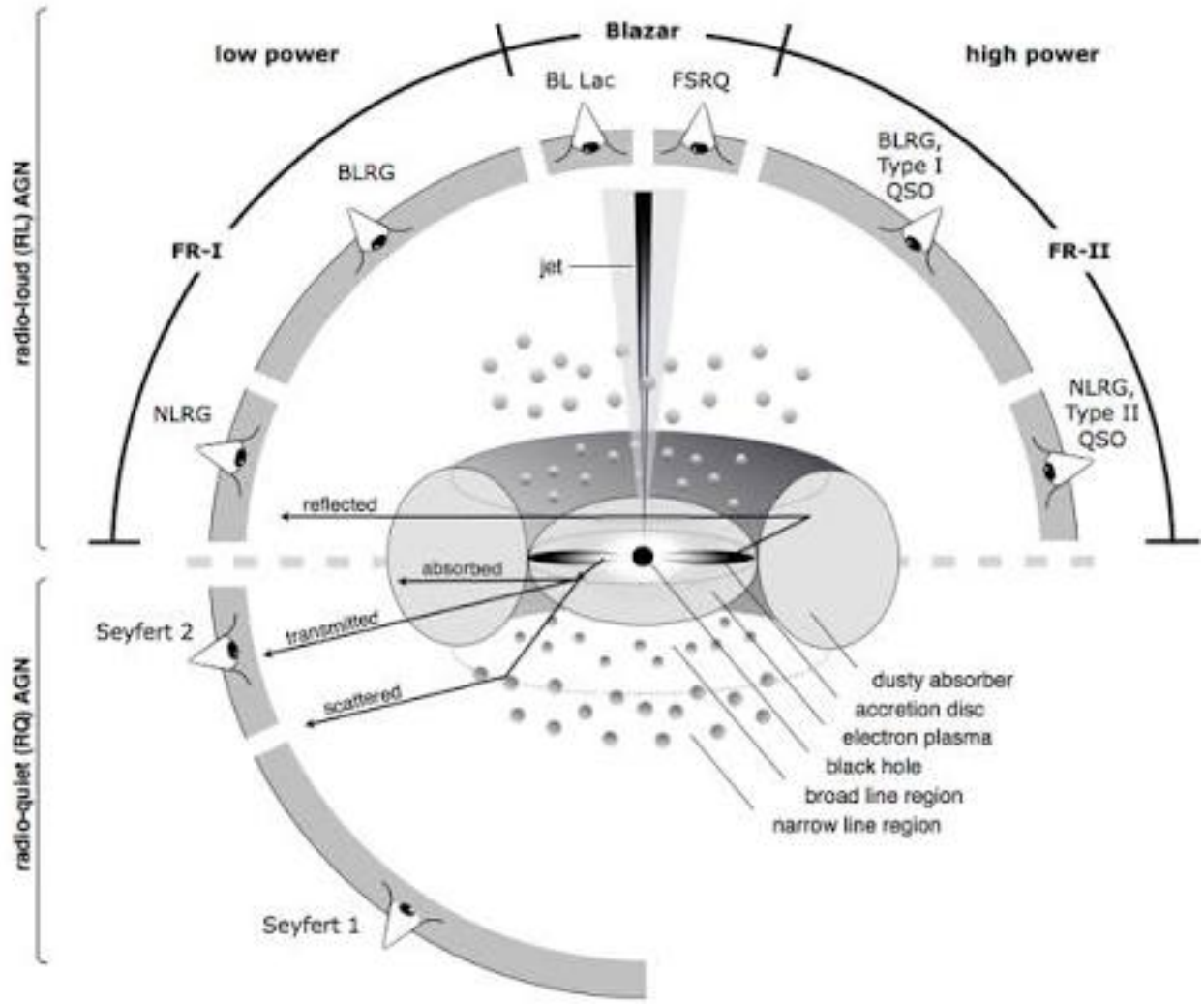


Figure 8.1: The unified model of AGN.

- **Quasars (QSOs, see Question 98)** have the highest luminosity ($L_{\text{bol}} \gtrsim 10^{46}\,\text{erg/s}$), overpowering the rest of the galaxy's emission, observed predominantly at higher redshift ($z \gtrsim 1$).

- **Seyferts** are less luminous than quasars, with $L_{\text{bol}} \lesssim 10^{45}\,\text{erg/s}$, and are typically hosted in nearby spiral galaxies.

- **Blazars**, including BL Lacs and Flat-spectrum radio quasars (FSRQs), have relativistic jets pointed directly along the observer's line of sight, producing beamed continuum emission with high variability, polarization, and weak or absent emission lines.

- **Type I AGN** have broad and narrow lines, while **Type II AGN** have only narrow lines.

93 What is the "unified model" of AGN, and what observations was it developed to unify?

The **unified model of AGN** states that the diverse observational classes of AGN represent the same intrinsic physical system viewed at different inclination angles relative to an axisymmetric obscuring structure. The fundamental components of an AGN are (see Fig. 8.1):

- The **SMBH** at the center of the system, of order $M_{\rm BH} \sim 10^6 - 10^{10}\,M_\odot$, which sets the central gravitational potential at scales $r_{\rm g} = GM/c^2$.
- The **accretion disk**, extending out $r \sim 10^{-4} - 10^{-2}\,{\rm pc}$, is the site of UV and optical emission from viscous dissipation. A hot ($T_e \sim 10^8 - 10^9\,{\rm K}$) electron **corona** inverse-Compton scatters the photons into a power-law X-ray continuum.
- The **broad line region (BLR)**, extending out $r \sim 0.01 - 0.1\,{\rm pc}$), consists of dense, fast-moving clouds ($n_e \gtrsim 10^9\,{\rm cm^{-3}}$, $v \sim 10^3 - 10^4\,{\rm km/s}$) orbiting close to the SMBH, emitting Doppler-broadened permitted lines.
- The **dusty torus**, extending out $r \sim 0.1 - 10\,{\rm pc}$, is an optically and geometrically thick structure of gas and dust, blocking emission from the inner core and reprocesses it into IR.
- The **narrow line region (NLR)**, extending out $r \sim 10 - 10^3\,{\rm pc}$, consists of low-density, extended gas ($n_e \sim 10^3 - 10^6\,{\rm cm^{-3}}$, $v \sim 200 - 500\,{\rm km/s}$) photoionized by the central engine, producing narrow permitted and forbidden lines.
- **Relativistic jets** in **Radio-loud AGN** are highly collimated plasma flows launched along the rotation axis of the SMBH, emitting non-thermal synchrotron radiation and inverse-Compton gamma-rays. Ionized gas traveling from the disk along the magnetic fields lines to the jet is thought to be the mechanism of jet launching.

The model was formulated primarily to reconcile the spectroscopic dichotomy between Type I (broad + narrow lines) and Type II (narrow lines only) AGN. The different viewing angles correspond to different AGN classifications.

- Looking directly at the relativistic jet are blazars (BL Lacs and FSRQs). Relativistic Doppler boosting dominates the continuum, producing rapid broadband variability, high polarization, and suppressed or dominated line features.
- Looking at intermediate angles gives a good view of the accretion disk and the BLR (as well as the extended NLR), so this corresponds to Type I QSOs or Seyferts.
- Looking edge-on makes it such that the accretion disk and BLR is obscured by the dusty torus, leaving observers with only narrow lines from the NLR, corresponding to Type II QSOs or Seyferts.
- Radio quiet AGN just do not have jets, which are the power source for synchrotron radio emission.

94 Describe the Eddington Limit and one application

The **Eddington Limit** is the maximum radiative luminosity an object in hydrostatic equilibrium can sustain when the outward radiative force balances the inward gravitational pull.

Consider a spherical shell (first, of fully ionized hydrogen plasma) with radius r and mass dm around a central source with mass M (e.g., an accreting black hole or a star). The inward gravitational force acting on the shell is

$$F_{\rm grav} = \frac{GM\,dm}{r^2}. \tag{8.9}$$

The central source emits a bolometric luminosity L, producing a radiative flux $F = L/4\pi r^2$. Photons transfer momentum to electrons via Thomson scattering, with cross section $\sigma_{\rm T}$. Electrostatic attraction tightly couples electrons to protons ($dm \approx N_p m_p$), so that the effective opacity per unit mass is $\kappa = \sigma_{\rm T}/m_p$. The outward radiative force imparted by momentum transfer to the shell is

$$F_{\rm rad} = \frac{F \cdot (\kappa dm)}{c} = \frac{L}{c} \cdot \frac{\kappa dm}{4\pi r^2}. \tag{8.10}$$

Equating these two gives

$$L_{\rm Edd} = \frac{4\pi GMc}{\kappa} = \frac{4\pi GMm_p c}{\sigma_{\rm T}}. \tag{8.11}$$

The exact value depends on the composition of the gas, but for ionized hydrogen, this yields

$$L_{\rm Edd} \approx 1.26 \times 10^{38} \left(\frac{M}{M_\odot}\right) \,{\rm erg\,s^{-1}} \approx 3.3 \times 10^4 \left(\frac{M}{M_\odot}\right) L_\odot \tag{8.12}$$

Beyond the Eddington luminosity, the radiation pressure will expel mass. In accretion, this sets a limit on the accretion rate. In stars, this sets a limit on stellar luminosity.

For an accreting black hole with radiative efficiency η (i.e., $L = \eta \dot{M} c^2$, where typically $\eta \sim 0.1$ for a thin disk), the Eddington luminosity sets the characteristic maximum mass accretion rate:

$$\dot{M}_{\rm Edd} = \frac{L_{\rm Edd}}{\eta c^2} = \frac{4\pi GM}{\eta c \kappa}. \tag{8.13}$$

Assuming continuous accretion at this rate ($\dot{M} \propto M$), black hole growth follows an exponential growth law. Explaining how $M \sim 10^9\, M_\odot$ quasars exist at high redshift ($z \gtrsim 6-7$) when the universe was only $\sim 800\,$Myr old requires near-continuous Eddington (or episodic super-Eddington) growth from initial stellar seeds.

It is possible to achieve super-Eddington accretion when the spherical and steady-state assumptions fail. Asymmetries (e.g., accretion from a disk and radiation from the poles), clumpiness (radiation leaks through low-density channels around high-density clumps), or beaming (collimated radiation causing apparent excessive luminosity to an observer) all can lead to super-Eddington accretion.

95 What are gamma-ray bursts? What are the two classes of astrophysical phenomena believed to give rise to them?

First discovered serendipitously in the late 1960s by the Vela satellites (designed to monitor compliance with the nuclear test ban treaty), their cosmological nature and isotropic sky distribution were confirmed in the 1990s by the BATSE detector aboard CGRO, which recorded an observed rate of ~ 1 burst/day across the sky.

Gamma-ray bursts (GRBs) are a class of transients consisting of a shower of gamma-rays with durations spanning $T_{90} \sim 10^{-2} - 10^3\,$s, where T_{90} is the time over which $90\,\%$ of the burst's fluence is detected, exhibiting exponential decay and afterglows in X-ray, optical, and radio frequencies. They were first discovered serendipitously in the 1960s by the Vela military spacecraft, which was looking for evidence of Soviet nuclear non-compliance. Their astrophysical nature was later confirmed and found to be at a rate of about one event per day across the sky. There is a clear bimodal distribution in T_{90} duration (at about $2\,$s) and spectral hardness (the ratio of high-energy to low-energy counts).

Long-soft GRBs last longer than two seconds and have a softer prompt photon spectra ($E_{\rm peak} \sim 100 - 300\,$keV. They are believed to come from SNe: a compact object and a debris disk form, a relativistic jet is launched with Lorentz factors $\gtrsim 100$, which produces gamma-rays via internal shocks, with the afterglow produced by the external forward shock through the ISM. The primary mechanism is through the collapsar model, where very massive stars undergo core collapse to a black hole with a debris disk, associated with

broad-lined Type Ic SNe. Another secondary mechanism is the supranova model, where a rapidly spinning neutron star is formed from core collapse, which exceeds the TOV limit due to its spin; as the NS spins down, it collapses to a black hole with the remnant electric field from the spin, causing a jet to launch.

Short-hard GRBs are less than two seconds and have harder prompt spectra ($E_{\rm peak} \gtrsim 400\,{\rm keV}$). They are believed to come from kilonovae (NS + NS or BH + NS mergers), which were observationally confirmed by GW170817. The energetics of this event and its very small crossing time made it one of the most persuasive sources of short GRBs, because there were multi-messenger detections of gravitational waves, a short GRB, and a kilonova.

96 What is a "gravitational lens," and what can we learn from observing them?

Gravitational lensing is the deflection of light rays by the gravitational field of a massive intervening object, the **gravitational lens**. In the thin-lens approximation, the deflection angle for an impact parameter b relative to a point mass M is $\hat{\alpha} = \frac{4GM}{c^2 b}$. Gravitational lenses cause **magnification** because of differential bending, because rays closer to the source get bent more than those farther from the source. Because gravitational lensing preserves surface brightness (flux per solid angle), any magnification of the solid angle subtended by the source results in a net **amplification** of flux. There are several regimes of gravitational lensing.

96.1 Strong lensing

Strong lensing is the regime in which we can observe distortions in a source (background) object, which occurs when the lens consists of a dense mass concentration. The deflection is large enough to produce multiple images, arcs, or an **Einstein ring** if the source, lens, and observer are closely aligned.

Strong lenses can be used to amplify distant objects in their backgrounds, with magnification factors possibly exceeding $10 - 100\times$, allowing observations of faint, high-redshift galaxies ($z \gtrsim 6 - 10$), individual lensed stars at cosmological distances, and faint quasar host galaxies. We can also use ray-tracing methods to reconstruct the total (baryonic and dark matter) mass profiles of the lenses without relying on dynamical assumptions or virialization. In merging systems such as the Bullet Cluster, the gravitational potential reconstructed from lensing aligns with the collisionless galaxies rather than the collisional gas (emitting in X-ray), demonstrating that the dominant mass component is collisionless dark matter. And, when a background variable source (e.g., a quasar or supernova) is strongly lensed, differing geometric path lengths create measurable arrival-time differences between images (Shapiro delay), which provides an independent geometric measurement of the Hubble constant.

96.2 Weak lensing

Background galaxies are lensed by the integrated foreground along the line of sight; this **weak lensing** produces a very small shape distortion (at the 0.1 to 1 % level) called shear, alongside a size and flux magnification called convergence. These distortions cannot be individually resolved, because individual galaxies themselves have intrinsically random ellipticities ("shape noise"), but by correlating the shapes of millions of background galaxies, we can detect coherent alignments caused by foreground structures. This allows us to reconstruct the three-dimensional matter power spectrum, constrain the amplitude of matter fluctuations (often parametrized as $S_8 \equiv \sigma_8(\Omega_m/0.3)^{0.5}$), and prove the equation of state of dark energy $w(z)$.

Apart from background galaxies, the CMB is also weakly lensed; this is observable as non-Gaussian mode coupling in the CMB power spectrum. While in galaxy weak lensing surveys, we can split source galaxies into redshift bins to allow for three-dimensional structure reconstructions, for CMB lensing, all mass along the line of sight out to the CMB redshift ($z \approx 1100$) is integrated, so this is not possible. This can help constrain the integrated matter distribution, neutrino mass sum, and spatial curvature.

96.3 Microlensing

When the deflection is too small to be resolved, the effect is observed solely as a time-variable magnification of the background source as the lens traverses the line of sight. This is **microlensing**, because the deflection is of order microarcseconds. This small effect can be used to detect exoplanets by measuring deviations or blips in the amplification over time of a lensing star passing by a background source (see Question 31).

97 What was the solar neutrino problem and how was it resolved? How do we detect neutrinos?

Beginning in the late 1960s, the number of neutrinos from the Sun collected in detectors (specifically the Homestake chlorine experiment) was only $\sim 1/3$ of the theoretical expectation from the Standard Solar Model. This discrepancy was the **solar neutrino problem**.

Because neutrinos interact solely via the weak force and gravity, their cross-section with baryonic matter is extremely small ($\sigma \sim 10^{-44}\,\mathrm{cm}^2$). Detectors must therefore be massive and situated deep underground to shield against cosmic-ray backgrounds. Primary detection mechanisms include:

1. **Radiochemical Capture:** Early detectors used inverse β-decay on target nuclei, such as:

$$\nu_e + {}^{37}\mathrm{Cl} \to {}^{37}\mathrm{Ar} + e^- \quad \text{(Homestake)}, \tag{8.14}$$

   or $\nu_e + {}^{71}\mathrm{Ga} \to {}^{71}\mathrm{Ge} + e^-$ (GALLEX/SAGE). The radioactive product isotopes are periodically flushed and counted via radioactive decay.

2. **Elastic Scattering via Cherenkov Radiation:** In ultra-pure water or ice detectors (e.g., Super-Kamiokande, IceCube), neutrinos elastically scatter atomic electrons:

$$\nu_x + e^- \to \nu_x + e^-. \tag{8.15}$$

   The relativistic recoil electrons exceed the phase velocity of light in the medium (c/n), generating cones of directional Cherenkov radiation collected by photomultiplier tube (PMT) arrays.

3. **Deuterium Disassociation (Heavy Water):** Using heavy water (D_2O), the Sudbury Neutrino Observatory (SNO) isolated two crucial channels:

   - **Charged-Current (CC):** $\nu_e + d \to p + p + e^-$ (sensitive exclusively to ν_e).
   - **Neutral-Current (NC):** $\nu_x + d \to p + n + \nu_x$ (equally sensitive to all three flavorsr).

The resolution came from the understanding that neutrinos possess non-zero mass and undergo flavor transitions. Neutrinos exist as flavor eigenstates (ν_e, ν_μ, ν_τ), which are superpositions of distinct mass eigenstates (ν_1, ν_2, ν_3). They are created as a definite flavor eigenstate; as they propagate, each mass eigenstate travels as a wave with a slightly different frequency due to different masses; and the probability that a neutrino will be measured as an electron, muon, or tau neutrino oscillates, depending on the distance traveled divided by the neutrino's energy. For higher-energy solar neutrinos, these vacuum oscillations are enhanced by electrons in the solar interior through the **Mikheyev-Smirnov-Wolfenstein effect**, which converts electron neutrinos into muon and tau neutrinos before exiting the Sun.

The p-p chain (see Question 55) produces electron neutrinos, and early detectors were only sensitive to these electron neutrinos. The problem was definitively resolved in 2001–2002 by the Sudbury Neutrino Observatory, which was sensitive both to the electron neutrino flux and the total neutrino flux, the combination of which matched the Standard Solar Model.

98 Describe the historical significance of quasars (QSOs) for the development of modern cosmology.

In the 1950s and 1960s, QSOs were identified as quasi-stellar radio sources, because they looked like stars but had strange absorption and emission lines. In 1963, Maarten Schmidt recognized that some of the broad lines resembled the Balmer series, just at higher redshifts than they were used to; in particular, he identified 3C 273 at a redshift of $z = 0.158$ and, soon after, 3C 48 at $z = 0.367$. This established that quasars were located at cosmological distances, requiring extreme energy outputs ($L \sim 10^{46}\,\mathrm{erg/s}$), which led to the realization that accretion onto SMBHs powers AGN (Donald Lynden-Bell in 1969).

Quasars were significant for the development of modern cosmology in many ways:

- Quasar counts revealed strong redshift evolution, because their spatial density peaked sharply around $z \sim 2$–3 ("cosmic noon", see Question 70) and dropped at $z \to 0$. This (alongside the discovery of the CMB in 1965) refuted the Steady State hypothesis (which, at the time, was in competition with Big Bang cosmology), which required the universe to be statistically homogeneous across all cosmic epochs.

- As high-luminosity continuum sources, quasars enabled absorption-line spectroscopy of intervening structures. In particular, this allowed for Lyα forest studies and detection of the Gunn-Peterson trough (see Question 87).

- That quasar activity peaked at $z \sim 2$–3 shows that SMBH growth mirrored cosmic star formation history (Question 70), establishing AGN feedback as a dominant mechanism governing galaxy quenching.

- The 1979 discovery of the doubly imaged quasar Q0957+561 provided the first observational confirmation of strong gravitational lensing, opening an independent geometric pathway to measure the Hubble constant (H_0) via time-delay cosmography.

99 What are sources/phenomena that give rise to gravitational waves? Why are they strong sources of GWs?

See Question 18. Gravitational waves are metric perturbations generated by accelerated mass distributions with a time-varying mass quadrupole moment.

In the linearized quadrupole approximation, the strain amplitude h and radiated GW luminosity L_{GW} scale with the second and third time derivatives of the reduced quadrupole moment tensor I_{ij}:

$$h_{ij} \sim \frac{2G}{c^4 r}\ddot{I}_{ij}, \qquad L_{\mathrm{GW}} = \frac{G}{5c^5}\langle \dddot{I}_{ij}\, \dddot{I}_{ij}\rangle. \tag{8.16}$$

Because the fundamental coupling constant is suppressed by the speed of light to the fifth power ($G/c^5 \sim 10^{-60}\,\mathrm{erg^{-1}s}$), laboratory-scale or ordinary celestial motions produce entirely negligible radiation.

For a self-gravitating binary system of mass M, separation R, and orbital velocity $v^2 \sim GM/R$, the quadrupole moment scales as $I \sim MR^2$, giving $\dddot{I} \sim Mv^3/R$. The radiated power therefore scales as:

$$L_{\mathrm{GW}} \sim \frac{c^5}{G}\left(\frac{R_{\mathrm{s}}}{R}\right)^2\left(\frac{v}{c}\right)^6, \tag{8.17}$$

where $R_{\mathrm{s}} = 2GM/c^2$ is the Schwarzschild radius and $c^5/G \approx 3.6 \times 10^{59}\,\mathrm{erg\,s^{-1}}$ is the maximum Dyson luminosity. Strong GW emission therefore fundamentally requires extreme compactness ($R \sim R_{\mathrm{S}}$), relativistic velocities ($v \sim c$), and large mass and asymmetry).

Examples of astrophysical sources include:

- **Compact object mergers**: solar mass ($1 - 100\,M_\odot$) neutron stars and/or black holes inspiraling and merging emit gravitational waves. This is what LIGO/VIRGO/KAGRA are sensitive to, with frequencies of order $10 - 10^3\,\mathrm{Hz}$.
- **Supermassive black hole binaries**: mergers between SMBH binaries ($M \sim 10^4 - 10^7\,M_\odot$) are what LISA will detect with its longer baselines (frequencies of order $10^{-4} - 10^{-1}\,\mathrm{Hz}$), along with extreme mass ratio black hole binaries (where a stellar-mass compact object spirals into a SMBH).
- **SMBH background**: the cosmic ensemble of slowly inspiraling SMBH binaries ($M \sim 10^8 - 10^{10}\,M_\odot$) generates a stochastic nanohertz background ($10^{-9}\,\mathrm{Hz} - 10^{-7}\,\mathrm{Hz}$), detecting via correlated timing residuals in pulsar timing arrays such as NANOGrav.
- **Non-axisymmetric rotating stars):** rapidly spinning neutron stars with, for example, tiny "mountains" sustained by magnetic or elastic stresses, emit continuous, monochromatic gravitational radiation at twice their spin frequency.
- **Asymmetric core-collapse supernova**: normal SNe are ideally perfectly spherically symmetric, so they do not produce gravitational waves, but if there are asymmetries, that would lead to short bursts at kilohertz frequencies.
- **Primordial/stochastic background:** Quantum fluctuations of the tensor metric during cosmological inflation produce a primordial stochastic GW background. Unlike electromagnetic radiation (which cannot penetrate the Universe prior to recombination at $z \approx 1100$), GWs decouple near the Planck scale, offering an unshielded probe of the early universe. This background imprints characteristic curly B-mode polarization patterns in the Cosmic Microwave Background (CMB).
- The **primordial gravitational wave background**: there should be a gravitational wave background (like the CMB) from the early Universe, produced by quantum fluctuations of the tensor metric during inflation, which would allow us to prove before recombination, because gravitational waves, unlike EM waves, decouple from matter and radiation almost immediately. B-mode polarization in the CMB would be a signature of such gravitational waves and would allow us to constrain inflation.

100 What are fast radio bursts (FRBs)? Describe two proposed physical explanations for the phenomena.

Fast radio bursts (FRBs) are radio transients that typically last from a microseconds to tens of milliseconds, and possibly up to a couple of seconds. They are very bright and span a large range of radio frequencies. Some are repeaters, and there has been one discovered that repeats every 16 days. That some repeat implies that there is probably not just one source of FRBs.

The time-resolved spectra of FRBs exhibit a characteristic frequency-dependent time delay, where higher-frequency radio waves arrive earlier than lower-frequency waves, following $\Delta t \propto \nu^{-2}$. The time delay is directly related to the **dispersion measure (DM)**, which is the integrated column density of free electrons along the line of sight. Observed DM values significantly exceed the contribution from the MW ISM, confirming an extragalactic origin.

Two proposed explanations for FRBs are

- **Magnetars**, which have a large energy reservoir in their magnetic field. The plasma region from which FRBs may be emitted is small, which is important for short signals. Magnetic reconnection events, for example, can trigger Alfvén waves and relativistic plasma outflows; this cloud of plasma can act as a coherent emitter. The two models are far-field synchrotron maser emission (far from the magnetar's surface) or near-field coherent curvature radiation (near the magnetar's surface). The magnetar hypothesis was observationally validated in April 2020 by the simultaneous detection of an FRB-luminosity radio burst and hard X-ray flare from the Galactic magnetar SGR 1935+2154.

- **Coalescing compact objects (BHs or NSs)**: a large burst of emission caused by the merging of compact objects could also explain such radio signals. At coalescence, the collapse of a neutron star into a black hole disrupts the magnetosphere, launching a burst of coherent radio emission. Because the progenitor is destroyed, this mechanism would account only for the non-repeating population and would be a natural candidate for gravitational wave associations.